

Case #: 19-ZN-2004#2

Review Cycle: 2

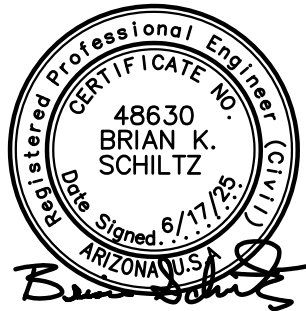
Reviewed By: GA

Date: 07/02/2025

Status: Accepted

**Preliminary Drainage Report
For
Finish Line Auto Club – Perimeter Center
Princess Drive and Loop 101
Scottsdale, Arizona 85255**

19-ZN-2004#2

COS Plan Check #: ~~1059-PA-2024~~**June 2025**

**Prepared by:
Brian Schiltz
Hunter Engineering, Inc.
10446 N. 74th Street. Suite 140
Scottsdale, AZ 85258**

H.E. PROJECT NO. LGEC332

Table of Contents

1.0	Introduction.....	1
1.1	FEMA Section	1
2.0	Existing Condition.....	1
3.0	Off-site Drainage Management.....	2
4.0	On-site Drainage Management.....	2
5.0	Storm Water Storage and First Flush Mitigation	2
6.0	Lowest Floor Elevations	3
7.0	Environmental Requirements	4
8.0	Conclusion.....	4
9.0	References	4

Appendix

Vicinity and FEMA Map	Appendix A
Drainage Calculations	Appendix B
Reference Information.....	Appendix C
Exhibits.....	Appendix D
Master Drainage Report for Scottsdale Perimeter Center	Appendix E
Conditions, Covenants and Restrictions with Design Guidelines	Appendix F



1.0 Introduction

This preliminary design report has been prepared under a contract from LGE Corporation, developer of the Finish Line Auto Club – Perimeter Center project. The purpose of this report is to provide a preliminary drainage analysis, required by the City of Scottsdale, to support this development. Preparation of this report has been done according to the procedures detailed in the City of Scottsdale's *Design Standards and Policy Manual* (Reference 1).

This development project is located along the west side of N. Perimeter Drive and along the east side of N. Pacesetter Way, between E. Pacesetter Way and E. Princess Drive, within the City of Scottsdale, Maricopa County, Arizona. The site is specifically located in the south half of Section 36, Township 4 North, Range 4 East of the Gila and Salt River Base and Meridian. Figure 1, in Appendix A, illustrates the location of the project site in relation to the City of Scottsdale street system. Access to the site is provided from N. Perimeter Drive and N. Pacesetter Way.

The proposed project site is currently a developed parcel within the Perimeter Center master development. The existing parcel is bound by existing commercial developments to the north and south, N. Perimeter Drive to the east and N. Pacesetter Way to the west.

The development proposes the construction of five new commercial buildings and one new industrial building. Site improvements will include construction of driveway entrances, parking lots, sidewalk/hardscape, landscape areas, and supporting infrastructure including new storm water drainage systems, water, sewer and fire line services. The overall project site is approximately 10.394± acres. Exhibit EXH 1, located in Appendix A, illustrates the proposed drainage improvements for the project.

1.1 FEMA Section

The current FEMA Flood Insurance Rate Map (FIRM) for this area, map number 04013C1320 L (Revision date October 16, 2013) shows the entire project site is in a flood hazard Zone AO with depths of 1ft and velocities of 3 ft/sec. Zone AO is defined as, "Flood depths of 1 to 3 feet (usually sheet flow on sloping terrain); average depths determined. For area of alluvial fan flooding, velocities also determined". See Appendix A for FEMA FIRMette map.

2.0 Existing Condition

The existing site consists of a single commercial building and associated parking lots with parking canopies. The site is part of the overall Scottsdale Perimeter Center master development. A city approved Master Drainage Report was prepared in 1989 for the Perimeter Center Development. See the *Master Drainage Report for Scottsdale Perimeter Center* located in Appendix E. That report discusses the existing site channels along the roadways within the development designed to route onsite stormwater flows to the ultimate downstream outfall, designated at the Scottsdale Tournament Players Course (TPC). See Exhibit EXH 2, located in Appendix D, for existing topography of the site.

3.0 Off-site Drainage Management

The only improvements to the adjacent roads will be to add driveways where required to access the site. The flow generated from the existing streets will remain undisturbed and match historical patterns. Pipe culverts will be required for the proposed driveways along the existing N. Pacesetter Way drainage channel, see EXH 1 in Appendix D for locations.

4.0 On-site Drainage Management

The onsite drainage will be conveyed into two separate stormwater retention systems, west and east. The west system will include the new industrial building and the east system will include the five new commercial buildings.

The west system will be within drainage areas DA1 and DA2 on EXH 1 in Appendix D. These tributary areas will drain to respective underground retention tanks. The east system will include Buildings A through E, within drainage areas DA3, DA4, and BA1. These tributary areas will consist of underground retention tanks and a 1-foot deep retention basin, respectively. Stormwater will be conveyed via overland flows from the structures and drives to catch basins and storm drains which will outfall to the underground retention tanks. The underground retention tanks will bleed off via drywells within 36 hours, as required by City of Scottsdale requirements. The 1-foot deep retention basin will drain via natural percolation. See storage capacity in Appendix B. All proposed underground retention tanks will be enveloped within a public drainage easement per DSMP Chapter 4, see Appendix C for easement requirements.

5.0 Storm Water Storage and First Flush Mitigation

City of Scottsdale requires the pre vs post runoff condition for runoff management, see Appendix B for the pre vs post analysis. This method requires stormwater runoff and peak discharge rates will not increase due to post development as per Chapter 4 of the City of Scottsdale DSPM 2018. There is an existing retention basin on the east side of the existing development. The current site topography was used to determine that the existing basin provides approximately 4,720 cf of volume. The proposed development will provide 20,729 cf. Therefore, the estimated runoff exiting the site after the first flush will decrease by approximately 16,009 cf.

Per the master drainage report and the design guidelines of the Perimeter Center, in Appendix E and F, respectively, onsite retention is not required, however the site is required to account for the first flush per City of Scottsdale standards. Since this site is part of the Scottsdale Perimeter Center master development only the first-flush event will be retained onsite as onsite storm water from the master development is designed to flow into the existing roadway channels to the Scottsdale TPC golf course outfall to the south.

The project will mitigate first flush requirements by retaining the first flush volume. The required volume is assessed as follows:

$$V_{R(FF)} = C * P/12 * A$$

Where:

$V_{R(FF)}$ = Required First Flush volume in cubic feet

P = rainfall depth (0.5)

A = Disturbed area = 10.08 acres

C = Runoff coefficient 1.00 for first flush

$$V_{R(FF)} = 1.0 * (0.5/12) * 10.08 * 43,560 = 18,295 \text{ cu-ft.}$$

The site is proposing onsite retention which can retain a total of 20,729 cu-ft of stormwater, which is more than the 18,295 cu-ft required for first flush. As mentioned in Section 4.0, all retention facilities will bleed off within the required 36 hours. See drainage calculations in Appendix B.

6.0 Lowest Floor Elevations

As discussed in Section 1.1 the entire site is located within FEMA Flood Zone AO. Therefore, the Lowest Floor NAVD 88 Datum must be at least 2' above the Highest Adjacent Grade (HAG) for any part of the building. This is to be based upon the historic site topography. Since the site is currently developed the recent topographic survey no longer represents the historic site topography. Therefore, the topography from the plans for the master perimeter center development was utilized to determine the HAG for the buildings.

The digital CAD files for the master development from 1989 were not available so the plan pdf's were scaled and inserted into the project site to be used as the historic topographic elevations. The master development plans were based on the City of Scottsdale NGVD29 datum. Therefore, the 1989 plan contour elevations need to be raised by 1.864' to match the current City of Scottsdale NAVD88 datum. The 1.864' was obtain by using the benchmark used and finding the difference by using UGS.NOAA.GOV to obtain the difference in elevation, see Appendix C for findings. To be conservative the proposed LF88 elevation is more than 2' above the highest HAG based on the adjusted 1989 topo elevations See the 2025 and 1989 Highest Adjacent Grade Elevation Exhibits located in Appendix D.

The existing site topography and locations of the proposed buildings presented challenges for the site grading design. Therefore, a majority of the proposed buildings will incorporate stepped finished floors to avoid extreme pavement slopes within the site. The stepped finished floors are in compliance with the FEMA Flood Zone AO requirements.

Building	LF88	HAG(1989)
Industrial Building	1580.50	1578.29
Building A	1583.00	1580.38
Building B	1581.70	1576.66
Building C	1581.70	1576.10
Building D	1582.85	1578.64
Building E	1584.40	1582.31

Note: *Elevation certification for each structure is required based on finished construction as a condition of obtaining the certification of occupancy.*

The existing extreme outfall for this site is approximately 1569.12 at the southwest corner of the site along N. Pacesetter Way.

7.0 Environmental Requirements

As the site will be disturbing more than one acre, an NOI and SWPPP plan and report will need to be submitted with the final plan submittal as condition of plan approval.

8.0 Conclusion

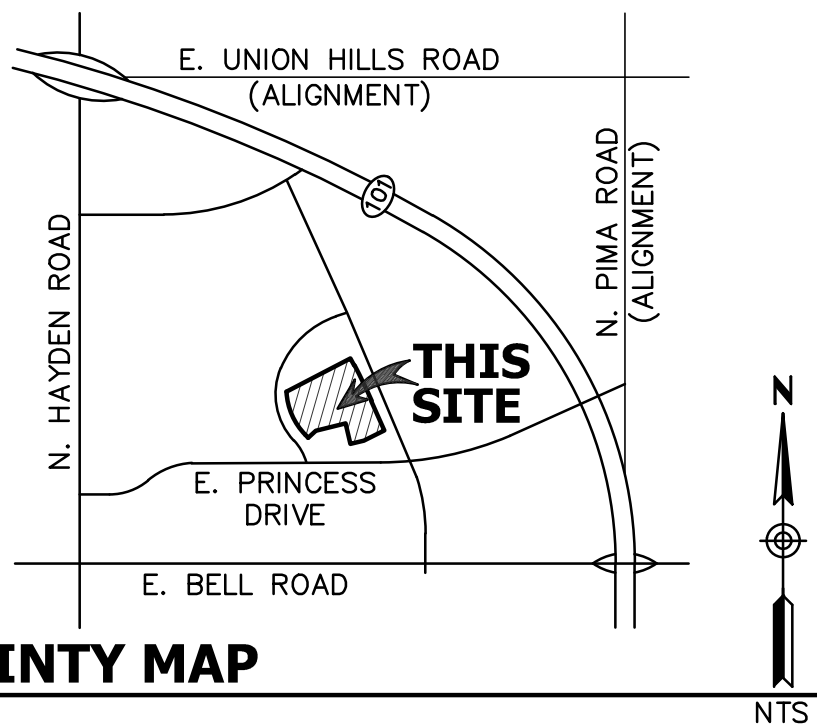
Based on the results of this study, it can be concluded that:

- The site will retain the development retention requirements for first flush.
- The proposed LF88 elevations are more than 2' above the adjacent Highest Adjacent Grade per the FEMA Flood Zone AO requirements.
- Underground retention tanks meet COS DSPM requirements.
- Surface percolation and drywells will be used to dispose of accumulated stormwater within 36-hours.

9.0 References

- 1) City of Scottsdale Design Standards & Policies Manual, January 2018.
- 2) Drainage Design Manual for Maricopa County, Arizona, Hydrology, December 2018.
- 3) Drainage Design Manual for Maricopa County, Arizona, Hydraulics, December 2018.
- 4) Master Drainage Report for Scottsdale Perimeter Center, February 1989.
- 5) Conditions, Covenants and Restrictions with Design Guidelines for The Perimeter Center Scottsdale, December 1990.

Appendix A
Vicinity and FEMA Map

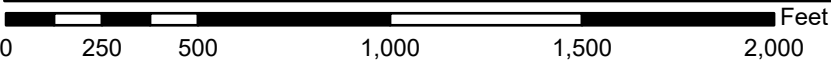


**VICINITY MAP
FIGURE 1**

National Flood Hazard Layer FIRMMette



111°54'18"W 33°39'3"N



1:6,000

111°53'41"W 33°38'33"N

Basemap Imagery Source: USGS National Map 2023

Legend

SEE FIS REPORT FOR DETAILED LEGEND AND INDEX MAP FOR FIRM PANEL LAYOUT

SPECIAL FLOOD HAZARD AREAS		Without Base Flood Elevation (BFE) Zone A, V, A99
		With BFE or Depth Zone AE, AO, AH, VE, AR
		Regulatory Floodway
OTHER AREAS OF FLOOD HAZARD		0.2% Annual Chance Flood Hazard, Areas of 1% annual chance flood with average depth less than one foot or with drainage areas of less than one square mile Zone X
		Future Conditions 1% Annual Chance Flood Hazard Zone X
		Area with Reduced Flood Risk due to Levee. See Notes. Zone X
		Area with Flood Risk due to Levee Zone D
OTHER AREAS		NO SCREEN Area of Minimal Flood Hazard Zone X
		Effective LOMRs
		Area of Undetermined Flood Hazard Zone D
GENERAL STRUCTURES		Channel, Culvert, or Storm Sewer
		Levee, Dike, or Floodwall
OTHER FEATURES		20.2 Cross Sections with 1% Annual Chance Water Surface Elevation
		17.5 Cross Sections with 1% Annual Chance Water Surface Elevation
		Coastal Transect
		Base Flood Elevation Line (BFE)
		Limit of Study
		Jurisdiction Boundary
		Coastal Transect Baseline
MAP PANELS		Digital Data Available
		No Digital Data Available
		Unmapped



The pin displayed on the map is an approximate point selected by the user and does not represent an authoritative property location.

This map complies with FEMA's standards for the use of digital flood maps if it is not void as described below. The basemap shown complies with FEMA's basemap accuracy standards

The flood hazard information is derived directly from the authoritative NFHL web services provided by FEMA. This map was exported on 4/4/2025 at 3:21 PM and does not reflect changes or amendments subsequent to this date and time. The NFHL and effective information may change or become superseded by new data over time.

This map image is void if the one or more of the following map elements do not appear: basemap imagery, flood zone labels, legend, scale bar, map creation date, community identifiers, FIRM panel number, and FIRM effective date. Map images for unmapped and unmodernized areas cannot be used for regulatory purposes.

Appendix B
Drainage Calculations

Retention Basin Calculations

$$Vr = C * (D/12) * A * 43,560, C=1, D=0.5$$

Design Storm: First Flush

C = 1.00 First Flush
D = 0.50 inches (First Flush)
A = 10.08 Site acres
Vr = 18,295 cubic feet

Underground Retention 1

Required

Location	Type	Area (ac)	C' Coefficient	Depth	Required (cf)
DA1	First Flush	2.76	1.00	0.50	5,009
		2.76			5,009

Provided

Length	Diameter (ft)	Area (sf)	Volume (cf)
256	5	19.63	5,027

5,027 cf Prov

5,009 cf Req

17 cf Excess

Underground Retention 2

Required

Location	Type	Area (ac)	C' Coefficient	Depth	Required (cf)
DA2	First Flush	0.78	1.00	0.50	1,416
		0.78			1,416

Provided

Length	Diameter (ft)	Area (sf)	Volume (cf)
115	4	12.57	1,445

1,445 cf Prov

1,416 cf Req

29 cf Excess

Underground Retention 3

Required

Location	Type	Area (ac)	C' Coefficient	Depth	Required (cf)
DA3	First Flush	2.78	1.00	0.50	5,046
		2.78			5,046

Provided

Length	Diameter (ft)	Area (sf)	Volume (cf)
101	8	50.27	5,077

5,077 cf Prov

5,046 cf Req

31 cf Excess

Underground Retention 4

Required

Location	Type	Area (ac)	C' Coefficient	Depth	Required (cf)
DA4	First Flush	3.42	1.00	0.50	6,207
		3.42			6,207

Provided

Length	Diameter (ft)	Area (sf)	Volume (cf)
124	8	50.27	6,233

6,233 cf Prov

6,207 cf Req

26 cf Excess

Retention Basin 1

Required

Location	Type	Area (ac)	C' Coefficient	Depth	Required (cf)
BA1	First Flush	0.34	1.00	0.50	617
		0.34			617

Provided

Elevation	Area (sf)	Avg. Area (sf)	Depth (ft)	Volume (cf)	Σ Volume (cf)
1,572.5	3,470				
		2,948	1.00	2,948	2,948
1,571.5	2,425				

2,948 cf Prov

617 cf Req

2,330 cf Excess

Total Retention
20,729 cf Prov
18,295 cf Req
2,434 cf Excess

Percolation Calculations

Project: LGEC332
Date: 6/16/2025

Calc'd By: AG
Chck'd By: BKS

Drywell Percolation Calculations

Design Percolation Rate **0.10** cfs

Basin ID	Number of Drywells	Drywell Perc. Rate (cfs)	Drywell Perc. Rate (cfh)	Basin Volume (cf)	Dry-Up Time (hr)
UGRT 1	1	0.1	360	5,009	14
UGRT 2	1	0.1	360	1,416	4
UGRT 3	1	0.1	360	5,046	14
UGRT 4	1	0.1	360	6,207	17

Pre vs Post Volume Provided

Project: LGEC332
Date: 6/16/2025

Calc'd By: AG
Chck'd By: BKS

Pre vs Post

$V_{P(Pre)}$ = 4,720 cf
 $V_{P(Post)}$ = 20,729 cf
Post-Pre = 16,009 cf

Appendix C
Reference Information



NOAA Atlas 14, Volume 1, Version 5
Location name: Scottsdale, Arizona, USA*
Latitude: 33.6465°, Longitude: -111.8999°
Elevation: 1576 ft**
* source: ESRI Maps
** source: USGS



POINT PRECIPITATION FREQUENCY ESTIMATES

Sanja Perica, Sarah Dietz, Sarah Heim, Lillian Hiner, Kazungu Maitaria, Deborah Martin, Sandra Pavlovic, Ishani Roy, Carl Trypaluk, Dale Unruh, Fenglin Yan, Michael Yekta, Tan Zhao, Geoffrey Bonnin, Daniel Brewer, Li-Chuan Chen, Tye Parzybok, John Yarchoan

NOAA, National Weather Service, Silver Spring, Maryland

[PF tabular](#) | [PF graphical](#) | [Maps & aeriels](#)

PF tabular

PDS-based point precipitation frequency estimates with 90% confidence intervals (in inches) ¹										
Duration	Average recurrence interval (years)									
	1	2	5	10	25	50	100	200	500	1000
5-min	0.197 (0.164-0.242)	0.258 (0.216-0.316)	0.348 (0.288-0.425)	0.417 (0.343-0.508)	0.510 (0.413-0.618)	0.580 (0.465-0.699)	0.653 (0.514-0.785)	0.726 (0.562-0.871)	0.824 (0.621-0.991)	0.899 (0.664-1.08)
10-min	0.300 (0.250-0.368)	0.393 (0.328-0.481)	0.529 (0.438-0.647)	0.634 (0.522-0.772)	0.776 (0.629-0.941)	0.884 (0.707-1.06)	0.994 (0.782-1.20)	1.10 (0.855-1.33)	1.25 (0.946-1.51)	1.37 (1.01-1.65)
15-min	0.372 (0.309-0.457)	0.487 (0.407-0.597)	0.656 (0.543-0.802)	0.786 (0.648-0.958)	0.962 (0.779-1.17)	1.10 (0.877-1.32)	1.23 (0.969-1.48)	1.37 (1.06-1.64)	1.56 (1.17-1.87)	1.70 (1.25-2.04)
30-min	0.501 (0.416-0.615)	0.655 (0.547-0.803)	0.883 (0.731-1.08)	1.06 (0.872-1.29)	1.30 (1.05-1.57)	1.48 (1.18-1.78)	1.66 (1.30-2.00)	1.84 (1.43-2.21)	2.09 (1.58-2.52)	2.28 (1.69-2.75)
60-min	0.620 (0.515-0.761)	0.811 (0.677-0.994)	1.09 (0.905-1.34)	1.31 (1.08-1.60)	1.60 (1.30-1.94)	1.82 (1.46-2.20)	2.05 (1.62-2.47)	2.28 (1.77-2.74)	2.59 (1.95-3.12)	2.83 (2.09-3.40)
2-hr	0.724 (0.609-0.868)	0.936 (0.791-1.13)	1.25 (1.05-1.49)	1.48 (1.23-1.77)	1.81 (1.48-2.15)	2.05 (1.66-2.43)	2.30 (1.84-2.72)	2.56 (2.00-3.02)	2.90 (2.22-3.42)	3.16 (2.37-3.75)
3-hr	0.798 (0.672-0.978)	1.02 (0.865-1.26)	1.34 (1.12-1.63)	1.58 (1.32-1.93)	1.93 (1.58-2.33)	2.21 (1.78-2.65)	2.49 (1.98-2.99)	2.80 (2.18-3.34)	3.20 (2.42-3.84)	3.54 (2.61-4.23)
6-hr	0.961 (0.826-1.14)	1.21 (1.04-1.44)	1.55 (1.32-1.83)	1.82 (1.54-2.14)	2.18 (1.82-2.55)	2.46 (2.03-2.88)	2.76 (2.24-3.21)	3.06 (2.44-3.57)	3.46 (2.69-4.04)	3.78 (2.87-4.42)
12-hr	1.09 (0.939-1.28)	1.37 (1.18-1.61)	1.73 (1.48-2.02)	2.01 (1.72-2.35)	2.39 (2.02-2.79)	2.68 (2.24-3.12)	2.99 (2.45-3.46)	3.29 (2.67-3.82)	3.70 (2.92-4.31)	4.01 (3.11-4.70)
24-hr	1.28 (1.12-1.48)	1.62 (1.42-1.88)	2.09 (1.83-2.42)	2.47 (2.15-2.86)	3.00 (2.59-3.46)	3.42 (2.92-3.93)	3.85 (3.26-4.44)	4.31 (3.60-4.96)	4.94 (4.06-5.71)	5.45 (4.42-6.32)
2-day	1.39 (1.21-1.60)	1.77 (1.54-2.04)	2.32 (2.01-2.67)	2.76 (2.38-3.17)	3.37 (2.89-3.87)	3.85 (3.27-4.43)	4.37 (3.68-5.03)	4.90 (4.09-5.67)	5.65 (4.63-6.56)	6.25 (5.05-7.29)
3-day	1.49 (1.31-1.71)	1.91 (1.67-2.19)	2.51 (2.20-2.88)	3.00 (2.62-3.43)	3.70 (3.20-4.22)	4.26 (3.65-4.86)	4.85 (4.13-5.56)	5.48 (4.61-6.30)	6.37 (5.28-7.34)	7.09 (5.80-8.21)
4-day	1.60 (1.41-1.82)	2.04 (1.80-2.33)	2.71 (2.39-3.08)	3.25 (2.85-3.70)	4.03 (3.51-4.58)	4.66 (4.03-5.30)	5.34 (4.58-6.08)	6.06 (5.14-6.93)	7.09 (5.92-8.13)	7.93 (6.54-9.14)
7-day	1.80 (1.58-2.07)	2.31 (2.03-2.65)	3.07 (2.69-3.51)	3.69 (3.22-4.22)	4.58 (3.97-5.22)	5.30 (4.56-6.05)	6.07 (5.18-6.94)	6.90 (5.83-7.92)	8.08 (6.72-9.31)	9.05 (7.43-10.5)
10-day	1.96 (1.73-2.24)	2.52 (2.22-2.88)	3.34 (2.93-3.81)	4.00 (3.50-4.56)	4.95 (4.30-5.62)	5.71 (4.93-6.49)	6.53 (5.58-7.44)	7.40 (6.27-8.45)	8.63 (7.20-9.90)	9.63 (7.94-11.1)
20-day	2.44 (2.15-2.78)	3.14 (2.77-3.57)	4.16 (3.66-4.72)	4.94 (4.33-5.60)	6.00 (5.24-6.80)	6.82 (5.93-7.74)	7.67 (6.62-8.72)	8.54 (7.32-9.74)	9.73 (8.25-11.2)	10.7 (8.94-12.3)
30-day	2.86 (2.52-3.26)	3.69 (3.26-4.20)	4.89 (4.30-5.53)	5.80 (5.09-6.55)	7.03 (6.14-7.95)	7.99 (6.94-9.03)	8.97 (7.75-10.2)	9.98 (8.56-11.3)	11.4 (9.64-12.9)	12.4 (10.5-14.2)
45-day	3.35 (2.97-3.80)	4.33 (3.84-4.90)	5.73 (5.06-6.47)	6.77 (5.97-7.66)	8.17 (7.16-9.23)	9.23 (8.06-10.4)	10.3 (8.95-11.7)	11.4 (9.84-13.0)	12.9 (11.0-14.8)	14.1 (11.9-16.2)
60-day	3.74 (3.32-4.22)	4.84 (4.29-5.45)	6.38 (5.65-7.18)	7.51 (6.63-8.46)	9.00 (7.92-10.1)	10.1 (8.86-11.4)	11.3 (9.80-12.7)	12.4 (10.7-14.0)	13.9 (11.9-15.8)	15.0 (12.8-17.2)

¹ Precipitation frequency (PF) estimates in this table are based on frequency analysis of partial duration series (PDS).

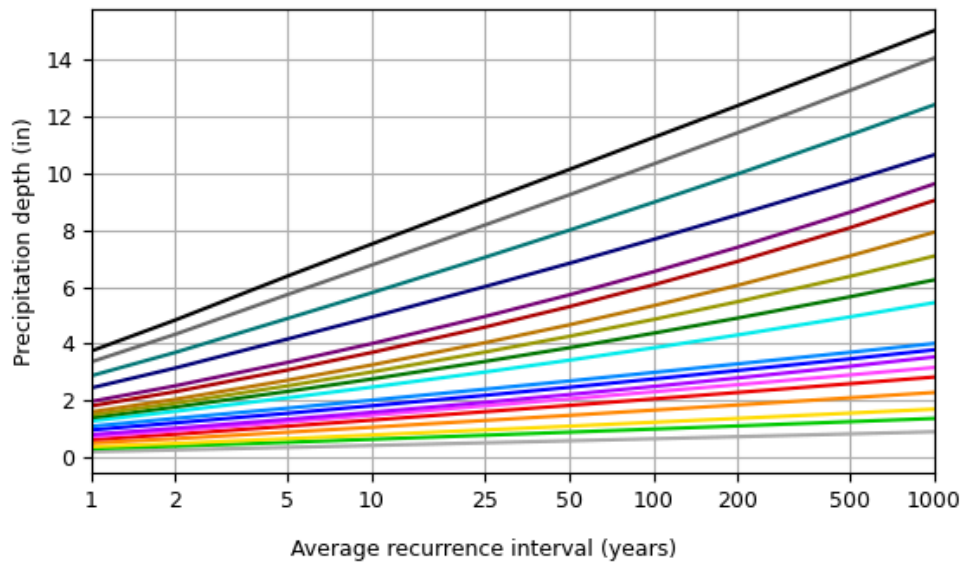
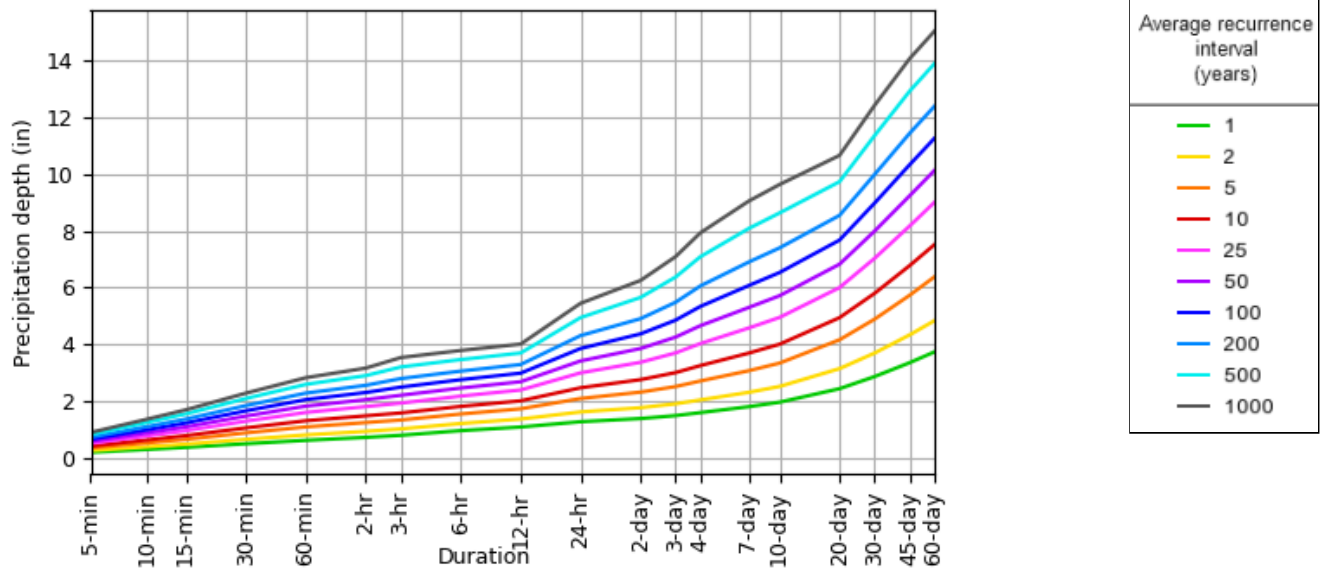
Numbers in parenthesis are PF estimates at lower and upper bounds of the 90% confidence interval. The probability that precipitation frequency estimates (for a given duration and average recurrence interval) will be greater than the upper bound (or less than the lower bound) is 5%. Estimates at upper bounds are not checked against probable maximum precipitation (PMP) estimates and may be higher than currently valid PMP values.

Please refer to NOAA Atlas 14 document for more information.

[Back to Top](#)

PF graphical

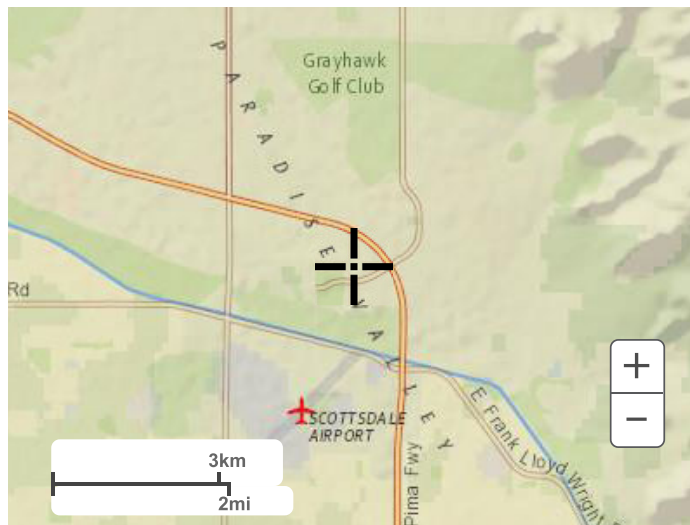
PDS-based depth-duration-frequency (DDF) curves
Latitude: 33.6465°, Longitude: -111.8999°



[Back to Top](#)

Maps & aerials

Small scale terrain



Large scale terrain



Large scale map



Large scale aerial



[Back to Top](#)

[US Department of Commerce](#)
[National Oceanic and Atmospheric Administration](#)
[National Weather Service](#)
[National Water Center](#)
1325 East West Highway
Silver Spring, MD 20910
Questions?: HDSC.Questions@noaa.gov

[Disclaimer](#)



NOAA Atlas 14, Volume 1, Version 5
Location name: Scottsdale, Arizona, USA*
Latitude: 33.6465°, Longitude: -111.8999°
Elevation: 1576 ft**
* source: ESRI Maps
** source: USGS



POINT PRECIPITATION FREQUENCY ESTIMATES

Sanja Perica, Sarah Dietz, Sarah Heim, Lillian Hiner, Kazungu Maitaria, Deborah Martin, Sandra Pavlovic, Ishani Roy, Carl Trypaluk, Dale Unruh, Fenglin Yan, Michael Yekta, Tan Zhao, Geoffrey Bonnin, Daniel Brewer, Li-Chuan Chen, Tye Parzybok, John Yarchoan

NOAA, National Weather Service, Silver Spring, Maryland

[PF tabular](#) | [PF graphical](#) | [Maps & aeriels](#)

PF tabular

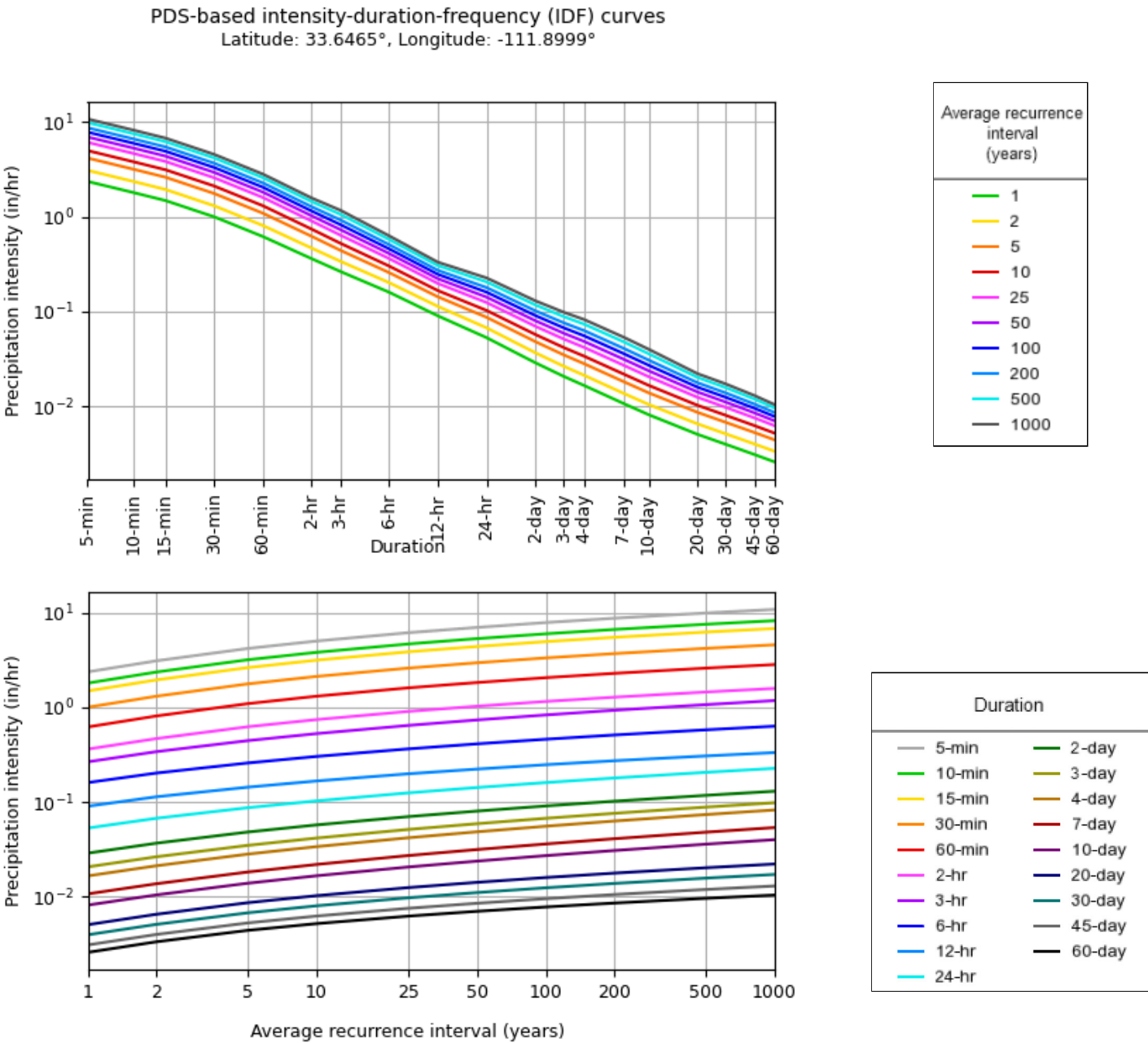
PDS-based point precipitation frequency estimates with 90% confidence intervals (in inches/hour) ¹										
Duration	Average recurrence interval (years)									
	1	2	5	10	25	50	100	200	500	1000
5-min	2.36 (1.97-2.90)	3.10 (2.59-3.79)	4.18 (3.46-5.10)	5.00 (4.12-6.10)	6.12 (4.96-7.42)	6.96 (5.58-8.39)	7.84 (6.17-9.42)	8.71 (6.74-10.5)	9.89 (7.45-11.9)	10.8 (7.97-13.0)
10-min	1.80 (1.50-2.21)	2.36 (1.97-2.89)	3.17 (2.63-3.88)	3.80 (3.13-4.63)	4.66 (3.77-5.65)	5.30 (4.24-6.38)	5.96 (4.69-7.17)	6.63 (5.13-7.96)	7.52 (5.68-9.04)	8.21 (6.07-9.88)
15-min	1.49 (1.24-1.83)	1.95 (1.63-2.39)	2.62 (2.17-3.21)	3.14 (2.59-3.83)	3.85 (3.12-4.66)	4.38 (3.51-5.28)	4.93 (3.88-5.93)	5.48 (4.24-6.58)	6.22 (4.69-7.48)	6.78 (5.01-8.17)
30-min	1.00 (0.832-1.23)	1.31 (1.09-1.61)	1.77 (1.46-2.16)	2.12 (1.74-2.58)	2.59 (2.10-3.14)	2.95 (2.36-3.55)	3.32 (2.61-3.99)	3.69 (2.86-4.43)	4.19 (3.16-5.03)	4.57 (3.38-5.50)
60-min	0.620 (0.515-0.761)	0.811 (0.677-0.994)	1.09 (0.905-1.34)	1.31 (1.08-1.60)	1.60 (1.30-1.94)	1.82 (1.46-2.20)	2.05 (1.62-2.47)	2.28 (1.77-2.74)	2.59 (1.95-3.12)	2.83 (2.09-3.40)
2-hr	0.362 (0.304-0.434)	0.468 (0.395-0.563)	0.623 (0.523-0.745)	0.742 (0.615-0.885)	0.904 (0.742-1.07)	1.03 (0.831-1.21)	1.15 (0.918-1.36)	1.28 (1.00-1.51)	1.45 (1.11-1.71)	1.58 (1.18-1.87)
3-hr	0.265 (0.223-0.325)	0.340 (0.288-0.418)	0.444 (0.373-0.544)	0.527 (0.438-0.641)	0.643 (0.526-0.776)	0.735 (0.594-0.882)	0.830 (0.658-0.996)	0.930 (0.725-1.11)	1.07 (0.807-1.28)	1.18 (0.870-1.41)
6-hr	0.160 (0.137-0.190)	0.202 (0.174-0.241)	0.258 (0.220-0.305)	0.303 (0.256-0.356)	0.363 (0.303-0.426)	0.411 (0.338-0.480)	0.460 (0.373-0.536)	0.510 (0.406-0.596)	0.578 (0.448-0.675)	0.632 (0.478-0.738)
12-hr	0.090 (0.077-0.106)	0.113 (0.098-0.133)	0.143 (0.123-0.168)	0.166 (0.142-0.195)	0.198 (0.167-0.231)	0.222 (0.185-0.258)	0.247 (0.203-0.287)	0.273 (0.221-0.316)	0.306 (0.242-0.357)	0.332 (0.258-0.390)
24-hr	0.053 (0.046-0.061)	0.067 (0.059-0.078)	0.087 (0.076-0.101)	0.102 (0.089-0.118)	0.124 (0.107-0.144)	0.142 (0.121-0.163)	0.160 (0.135-0.185)	0.179 (0.150-0.206)	0.205 (0.169-0.237)	0.226 (0.184-0.263)
2-day	0.028 (0.025-0.033)	0.036 (0.032-0.042)	0.048 (0.041-0.055)	0.057 (0.049-0.066)	0.070 (0.060-0.080)	0.080 (0.068-0.092)	0.090 (0.076-0.104)	0.102 (0.085-0.118)	0.117 (0.096-0.136)	0.130 (0.105-0.151)
3-day	0.020 (0.018-0.023)	0.026 (0.023-0.030)	0.034 (0.030-0.039)	0.041 (0.036-0.047)	0.051 (0.044-0.058)	0.059 (0.050-0.067)	0.067 (0.057-0.077)	0.076 (0.064-0.087)	0.088 (0.073-0.101)	0.098 (0.080-0.114)
4-day	0.016 (0.014-0.018)	0.021 (0.018-0.024)	0.028 (0.024-0.032)	0.033 (0.029-0.038)	0.041 (0.036-0.047)	0.048 (0.042-0.055)	0.055 (0.047-0.063)	0.063 (0.053-0.072)	0.073 (0.061-0.084)	0.082 (0.068-0.095)
7-day	0.010 (0.009-0.012)	0.013 (0.012-0.015)	0.018 (0.016-0.020)	0.021 (0.019-0.025)	0.027 (0.023-0.031)	0.031 (0.027-0.035)	0.036 (0.030-0.041)	0.041 (0.034-0.047)	0.048 (0.040-0.055)	0.053 (0.044-0.062)
10-day	0.008 (0.007-0.009)	0.010 (0.009-0.011)	0.013 (0.012-0.015)	0.016 (0.014-0.018)	0.020 (0.017-0.023)	0.023 (0.020-0.027)	0.027 (0.023-0.030)	0.030 (0.026-0.035)	0.035 (0.030-0.041)	0.040 (0.033-0.046)
20-day	0.005 (0.004-0.005)	0.006 (0.005-0.007)	0.008 (0.007-0.009)	0.010 (0.009-0.011)	0.012 (0.010-0.014)	0.014 (0.012-0.016)	0.015 (0.013-0.018)	0.017 (0.015-0.020)	0.020 (0.017-0.023)	0.022 (0.018-0.025)
30-day	0.003 (0.003-0.004)	0.005 (0.004-0.005)	0.006 (0.005-0.007)	0.008 (0.007-0.009)	0.009 (0.008-0.011)	0.011 (0.009-0.012)	0.012 (0.010-0.014)	0.013 (0.011-0.015)	0.015 (0.013-0.017)	0.017 (0.014-0.019)
45-day	0.003 (0.002-0.003)	0.004 (0.003-0.004)	0.005 (0.004-0.005)	0.006 (0.005-0.007)	0.007 (0.006-0.008)	0.008 (0.007-0.009)	0.009 (0.008-0.010)	0.010 (0.009-0.012)	0.011 (0.010-0.013)	0.013 (0.011-0.014)
60-day	0.002 (0.002-0.002)	0.003 (0.002-0.003)	0.004 (0.003-0.004)	0.005 (0.004-0.005)	0.006 (0.005-0.007)	0.007 (0.006-0.007)	0.007 (0.006-0.008)	0.008 (0.007-0.009)	0.009 (0.008-0.011)	0.010 (0.008-0.011)

¹ Precipitation frequency (PF) estimates in this table are based on frequency analysis of partial duration series (PDS).

Numbers in parenthesis are PF estimates at lower and upper bounds of the 90% confidence interval. The probability that precipitation frequency estimates (for a given duration and average recurrence interval) will be greater than the upper bound (or less than the lower bound) is 5%. Estimates at upper bounds are not checked against probable maximum precipitation (PMP) estimates and may be higher than currently valid PMP values.

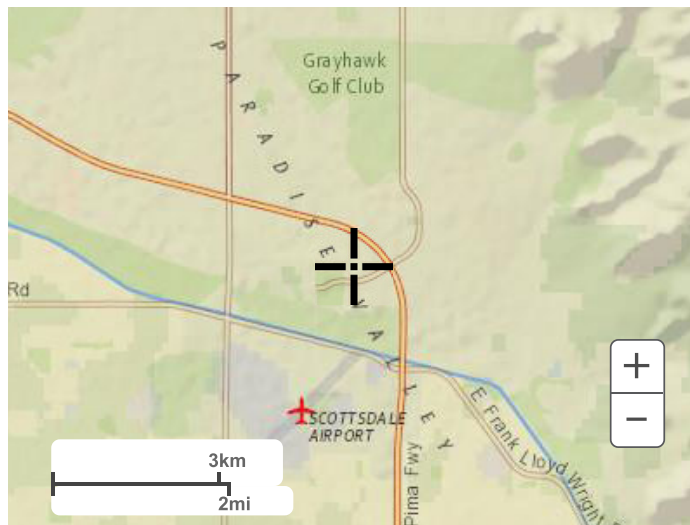
Please refer to NOAA Atlas 14 document for more information.

PF graphical



Maps & aerials

Small scale terrain



Large scale terrain



Large scale map



Large scale aerial



[Back to Top](#)

[US Department of Commerce](#)
[National Oceanic and Atmospheric Administration](#)
[National Weather Service](#)
[National Water Center](#)
1325 East West Highway
Silver Spring, MD 20910
Questions?: HDSC.Questions@noaa.gov

[Disclaimer](#)



Submit

Enter lat-lon in decimal degrees

Lat 33.6473313389

Lon -111.8906256861

or degrees-minutes-seconds

Lat N 33-38-50.39282

Lon W 111-53-26.25247

or drag map marker to a location of interest

Input reference frame
(historically called 'horizontal datum')

NAD83(2011)

Don't see a reference frame in the list? Click [here](#) to learn more.

Orthometric Height

1,595.253

Units of height

International Feet

Input geopotential datum
(historically called 'vertical datum')

NAVD88

Output reference frame
(historically called 'horizontal datum')

NAD83(2011)

Output geopotential datum
(historically called 'vertical datum')

NGVD29

SPC zone
AZ C-0202

Click blue bar(s) to expand/collapse

Transformed Coordinate

Input Coordinate		Output Coordinate		Total Change + Uncertainty	
Latitude	N33° 38' 50.39282" N333850.39282 33.6473313389	Latitude	N33° 38' 50.39282" N333850.39282 33.6473313389	Latitude	0.00000' ±0.000000" (0.000 m ±0.0000 m)*
Longitude	E248° 06' 33.74753" W1115326.25247 -111.8906256861	Longitude	E248° 06' 33.74753" W1115326.25247 -111.8906256861	Longitude	0.00000' ±0.000000" (0.000 m ±0.0000 m)*
Ellipsoid Height (ift)	Not given	Ellipsoid Height (ift)	Not given	Ellipsoid Height	Not given
Orthometric Height (ift)	1595.253	Orthometric Height (ift)	1593.389	Orthometric Height	-1.864 ift ±0.069 ift
Reference Frame	NAD83(2011)	Reference Frame	NAD83(2011)		
Geopotential	NAVD88	Geopotential	NGVD29		

permanent body of water with appropriate health, safety, and water quality measures.

13. Drain times for full storage facilities should be maximized to ensure the effectiveness of the facilities. Drain times should generally be from 12 to 24 hours.
14. Drain times for partial storage facilities shall be designed to maximize the attenuation of peak discharge rates from the facilities.
15. For stormwater storage basins designed as detention basins and being used to address first flush stormwater quality requirement, the invert of the outflow pipe shall be located typically 3 inches above the basin floor.
16. The outflow discharge from basins may be regulated with use of orifice plates placed over the entrance of the outlet pipe. The orifice size should be a minimum of 6 inches in diameter unless the clogging potential of the orifice is mitigated by other means in the design of the outlet works.
17. Stormwater storage may occur in a parking lot if the following conditions are met:
 - a. The first 50% of the total required storage volume is provided in an above-ground stormwater storage basin or underground storage tank, if approved;
 - b. No more than 50% of the total required storage volume is provided in the parking lot;
 - c. The depth of water does not exceed six inches at any location in the parking lot;
 - d. Interference with pedestrian traffic is minimized where the parking lot acts as a storage facility; and
 - e. A public drainage easement will not be required by the city over the limits of 100-year inundation for areas of storage within parking lots only.

UNDERGROUND STORMWATER STORAGE POLICY

4-1.202

A. POLICY

This policy supplements the SRC requirements for all stormwater storage. Underground stormwater storage involves constructing underground tanks, pipes, or vaults that accept stormwater runoff by means of inlets and storm drain (SD) pipes. The city approves underground storage only after rigorous analysis of storage system location, specifications, access, operation and maintenance, liability, and signage.

B. PROJECTS QUALIFYING FOR UNDERGROUND STORMWATER STORAGE

1. Project must meet the following criteria:
 - a. The project must be located within an industrial, commercial, or multi-family development.
 - b. Residential development may be considered on a case-by case basis subject to the following considerations:
 - i. The development shall have a minimum of 20 lots.
 - ii. The proposed storage facilities do not include contributing watersheds that includes natural desert or mitigation of are not subject to sedimentation

- iii. The viability of the property maintenance organization as discussed below is acceptable to the City.
2. The project must have a viable property maintenance organization or other maintenance mechanism to assume the continued maintenance, repair, and the potential replacement of the underground stormwater storage system and to protect the public's interest. The viability of the property maintenance organization or other mechanism shall be approved by the city on a case by case basis based on an assessment of the long-term ability of the property maintenance organization or other mechanism to maintain, repair and potentially replace the system in conjunction with the review of development review and preliminary plat submittals for the project.

C. GENERAL CRITERIA FOR UNDERGROUND STORMWATER STORAGE SYSTEM DESIGN

1. Underground stormwater storage systems must demonstrate protection of public health, safety, and welfare as established by the SRC and related policies.
2. Storage systems must not be located under structures, parking garages, or significant landscaping such as trees or sizable cactus that would preclude access to or replacement of the facilities.
3. The owner must dedicate a public drainage easement to the city which meets the standards for all drainage easements.
4. Design must address:
 - a. Water quality protection measures to protect underground and surface water resources to meet applicable water quality standards.
 - b. Vector control within storage system.
 - c. Redundancy in case of storage system failure, with attention to the possibility of structure or street flooding, sediment accumulation, or storm events that are greater than the 100-year, 2-hour event.
 - d. Initial suspended sediment load removal.
 - e. At least a 75-year life of entire system, including the lining and coating of the underground storage tank.
 - f. Drainage by gravity. Pumped systems will only be considered if no other reasonable alternative exists with dry wells as a preferred alternative if drainage by gravity is not feasible.

D. SPECIFIC CRITERIA FOR UNDERGROUND STORMWATER STORAGE DESIGN

1. Outfall—underground storage systems must have some sort of outfall, such as gravity drains or pumps.
2. Pipes—underground storage system pipes must have a smooth interior floor. The city's Maricopa Association of Governments (MAG) supplemental standard detail 2554 shall be used to meet the smooth interior floor requirement for the use of corrugated metal pipes in underground stormwater storage facilities.
3. Installation—excavation, bedding, and backfill procedures and materials must be in accordance with MAG standards.
4. Access—a minimum of two access points must be provided for each underground storage system unless waived in advance by stormwater staff to enable inspections and access for removal of accumulated sediment and

debris. Access must be in accordance with MAG standards and be placed to maximize the ability to maintain the underground system.

E. CRITERIA FOR OPERATIONS, MAINTENANCE AND LIABILITY

1. Operations and maintenance generally—owner must provide:
 - a. Maintenance staff with expertise in operating, inspecting, and maintaining an underground stormwater storage system;
 - b. An Operations and Maintenance Manual on site for the system that includes:
 - i. a schedule for inspections and maintenance, and
 - ii. provisions for emergency operations due to power failure, pump failure, and clogged outlet structures;
 - iii. A log of the inspections and required maintenance services.
2. Inspections and maintenance required—In addition to maintenance required by the SRC and other applicable requirements, owner shall:
 - a. Inspect system after each storm event of 0.6 inch or more, and semiannually, preferably before summer and winter rains.
 - b. Remove accumulated trash and debris from inlet and outlet structures as needed to ensure free flow of stormwater.
 - c. Inspect all other elements of the drainage system (pipes, geotextiles, and stone) and repair/replace elements as needed for the storage system to operate at peak efficiency.
3. Signage—before receiving a C of O, the owner must install signs at each end of the underground storage tank that read “Notice—Underground Stormwater Storage Tank.” The size, color, and locations of signs are subject to city staff approval.
4. Ownership Responsibility Statement – The owner shall provide and sign a statement of responsibility for the system stating and acknowledging the owner is responsible for the maintenance, repair, and potential replacement of the system. Prior to final plan approval, the owner must provide a signed and notarized document to this effect, in a form satisfactory to the City Attorney, for recordation by the city in the Maricopa County Recorder’s Office.

STORMWATER STORAGE WAIVERS

4-1.203

A. WAIVER OF STORMWATER STORAGE REQUIREMENTS

See SRC, Chapter 37, for waiver criteria. A waiver approval does not relieve the developer of liability if flood damage occurs resulting from the waiver.

B. WAIVER OF FIRST FLUSH

Generally, there is no waiver for stormwater storage volume required to hold runoff from the first one-half inch of precipitation. However, the owner may provide a smaller basin and/or alternative stormwater controls, to the satisfaction of the Stormwater Quality Coordinator.

C. WAIVER PROCESS

To apply for a stormwater storage waiver, the developer shall complete and submit a stormwater storage waiver application in conjunction with the preliminary drainage report as part of the development review or preliminary plat or similar case submittal for the project.

Appendix D
Exhibits

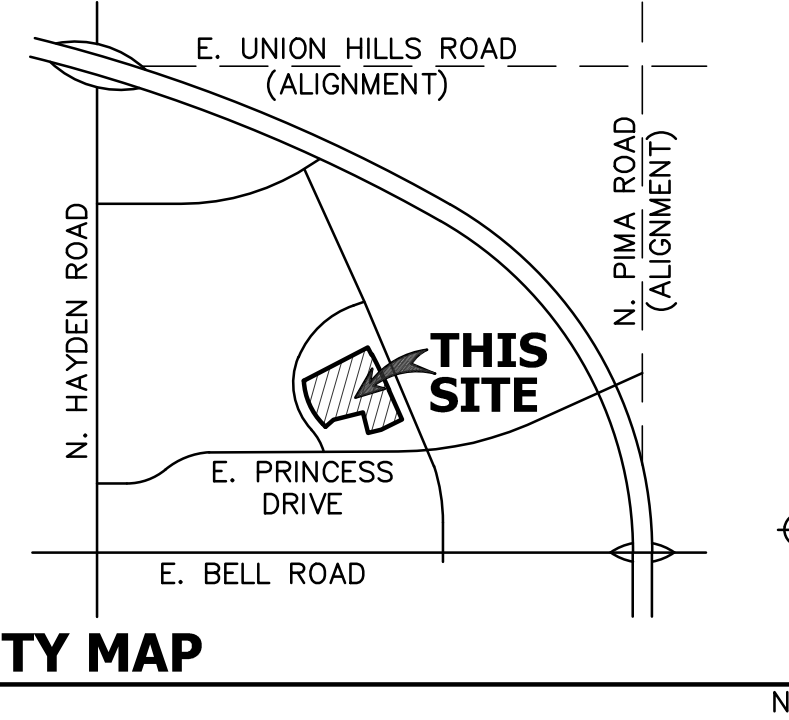
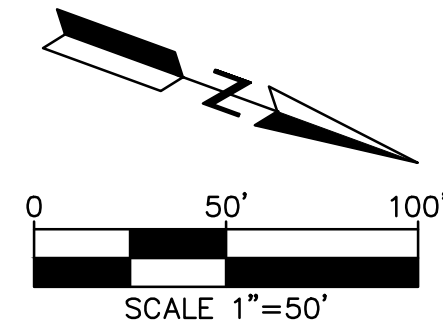
FLOOD PLAIN DESIGNATION

COMMUNITY NUMBER	PANEL # PANEL DATE	SUFFIX	DATE OF FIRM (INDEX DATE)	FIRM ZONE	BASE FLOOD ELEVATION (IN AO ZONE USE DEPTH)
04013C	1320 10-16-13	L	2/8/2024	ZONE AO	1FT DEPTH AND VELOCITIES OF 3 FT/SEC

ENGINEER'S CERTIFICATION: THE LOWEST FLOOR ELEVATION(S) AND/OR FLOOD PROOFING ELEVATION(S) ON THIS PLAN ARE SUFFICIENTLY HIGH TO PROVIDE PROTECTION FROM FLOODING CAUSED BY A ONE-HUNDRED YEAR STORM, AND ARE IN ACCORDANCE WITH CITY OF SCOTTSDALE REVISED CODE, CHAPTER 37 - FLOODWAYS & FLOOD PLAINS ORDINANCE.

PRELIMINARY DRAINAGE EXHIBIT FOR FINISH LINE AUTO CLUB PRINCESS DRIVE AND LOOP 101 SCOTTSDALE, ARIZONA

LOT 1 OF FNBA CENTER, AS RECORDED IN BOOK 844 OF MAPS, PAGE 47 OF MARICOPA COUNTY, ARIZONA, LOCATED WITHIN THE SOUTH HALF OF SECTION 36, TOWNSHIP 4 NORTH, RANGE 4 EAST, OF THE GILA AND SALT RIVER MERIDIAN, MARICOPA COUNTY, ARIZONA.



VICINITY MAP

APN:

PROJECT APNS

DRAINAGE STATEMENT

THE PROPOSED UNDERGROUND STORMWATER STORAGE TANKS (USST'S) SHALL MEET THE THE CITY'S USST POLICY IN SECTION 4-1.202 OF THE CITY'S DESIGN STANDARDS AND POLICY MANUAL (DSPM).

OWNER/ DEVELOPER

LGE DESIGN BUILD

1200 N 52ND ST
PHOENIX, ARIZONA 85008
PHONE: (480) 966-4001
CONTACT: BLAKE WELLS
EMAIL: BWELLS@LGEDESIGNGROUP.COM

ARCHITECT

LGE DESIGN GROUP

1200 N 52ND ST
PHOENIX, ARIZONA 85008
PHONE: (480) 966-4001
CONTACT: CARLOS ELIAS
EMAIL: CELIAS@LGEDESIGNGROUP.COM

CIVIL ENGINEER

HUNTER ENGINEERING, INC.

10446 N. 74TH STREET, SUITE 140
SCOTTSDALE, ARIZONA 85258
PHONE: (480) 991-3985
FAX: (480) 991-3986
CONTACT: BRIAN SCHILTZ, P.E.
EMAIL: BSCHILTZ@HUNTERENGINEERINGPC.COM

LEGEND

	PROPOSED STORM LINE
	EXISTING STORM LINE
	PROPOSED CATCH BASIN
	PROPOSED FLOW ARROW
	EXISTING FLOW ARROW
	DRAINAGE AREA BOUNDARY
CB# 0.00 AC	CATCH BASIN
CP# 0.00 AC	CONCENTRATION POINT
BA# 0.00 AC	RETENTION BASIN AREA

RETENTION CALCULATIONS

ONSITE AREA: 10.08 ACRES

RETENTION REQUIRED:

$$V_R = (C) * (P/12) * (A) \quad C=1.00 \quad P=0.5 \quad (\text{FIRST FLUSH})$$
$$V_R (\text{FIRST FLUSH}) = (1.00) * (0.5/12) * (10.08) * (43,560) = 18,295 \text{ CF}$$

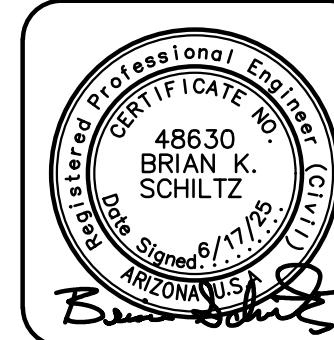
TOTAL REQUIRED = 18,295 CF

RETENTION TABLE				
BASIN LOCATION	VOLUME REQ'D (CF)	VOLUME PROV'D (CF)	EXCESS (CF)	DRYWELLS
UGRT 1	5,009	5,027	17	1
UGRT 2	1,416	1,445	29	1
UGRT 3	5,046	5,077	31	1
UGRT 4	6,207	6,233	26	1
BASIN 1	617	2,948	2,222	0
TOTAL	18,295	20,729	2,434	4

NO.	DATE	REVISION	BY

PURPOSE:
1ST DR SUBMITTALDESIGN BY: BKS
DRAWN BY: MM
CHECKED BY: BKS

HUNTER
ENGINEERING
CIVIL AND SURVEY
10446 NORTH 74TH STREET
SUITE 140
SCOTTSDALE, AZ 85258
P 480 991 3985



PRELIMINARY DRAINAGE EXHIBIT 'A'
FOR
FINISH LINE AUTO CLUB
PRINCESS DRIVE AND LOOP 101
SCOTTSDALE, ARIZONA

CONTACT AROUND AT LEAST 2 FEET
AWAY FROM THE EXISTING FOUNDATION
AR ZONAS811
CALL 811 OR CLIX ARIZONAS811.COM

THESE PLANS ARE
NOT APPROVED FOR
CONSTRUCTION
WITHOUT AN
APPROVED SIGNATURE
FROM THE GOVERNING
MUNICIPALITY.

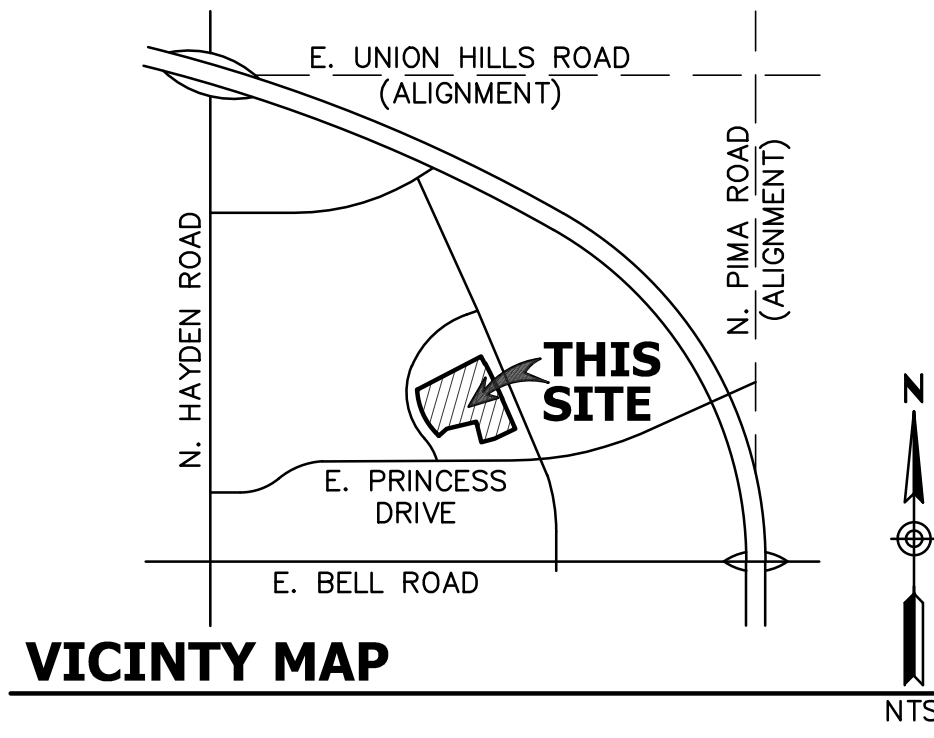
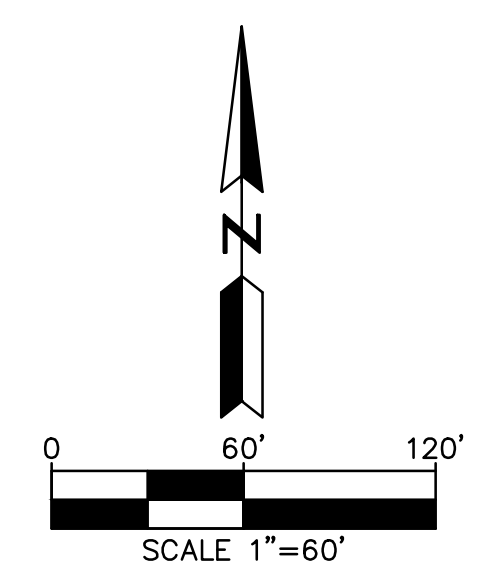
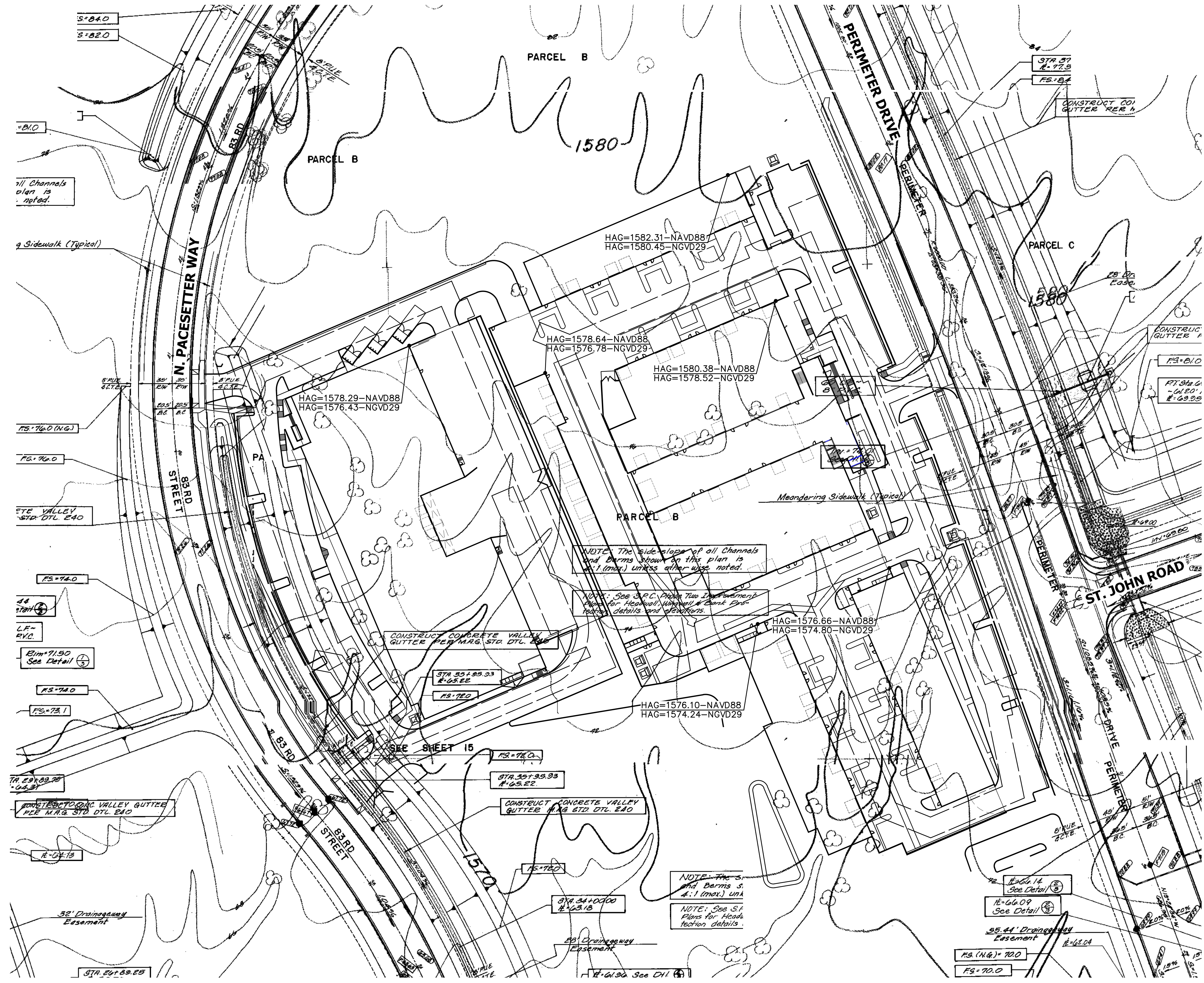
PROJECT NAME:
FINISH LINE
AUTO CLUB

HE NO.: LGECC332
SCALE: 1"=50'

SHEET:
EXH 1

1989 - HAG AND LAG EXHBIT
FOR
FINISH LINE AUTO CLUB
PRINCESS DRIVE AND LOOP 101
SCOTTSDALE, ARIZONA

LOT 1 OF FNBA CENTER, AS RECORDED IN BOOK 844 OF MAPS, PAGE 47 OF MARICOPA COUNTY, ARIZONA, LOCATED WITHIN THE SOUTH HALF OF SECTION 36, TOWNSHIP 4 NORTH, RANGE 4 EAST, OF THE GILA AND SALT RIVER MERIDIAN, MARICOPA COUNTY, ARIZONA.



VICINITY MAP

OWNER/ DEVELOPER
LGE DESIGN BUILD
1200 N 52ND ST
PHOENIX, ARIZONA 85008
PHONE: (480) 966-4001
CONTACT: BLAKE WELLS
EMAIL: BWELLS@LGEDESIGNGROUP.COM

ARCHITECT
LGE DESIGN GROUP
1200 N 52ND ST
PHOENIX, ARIZONA 85008
PHONE: (480) 966-4001
CONTACT: CARLOS ELIAS
EMAIL: CELIAS@LGEDESIGNGROUP.COM

CIVIL ENGINEER
HUNTER ENGINEERING, INC.
10446 N. 74TH STREET, SUITE 140
SCOTTSDALE, ARIZONA 85258
PHONE: (480) 991-3985
FAX: (480) 991-3986
CONTACT: BRIAN SCHILTZ, P.E.
EMAIL: BSCHILTZ@HUNTERENGINEERINGPC.COM

LEGEND

--- PROPERTY LINE

95.86 1989 CONTOURS NGVD29 AND NAVD88 DATUM

PLAN NOTE

THIS PLAN SHOWS THE EXISTING CONTOURS BASED UPON A RECENT 2023 TOPOGRAPHIC SURVEY. THE 2023 CONTOURS ARE BASED UPON THE NAVD88 DATUM. SEE THE 1989 HIGHEST ADJACENT GRADE EXHIBIT FOR THE 1989 CONTOURS BASED UPON THE NGVD29 DATUM.

THE DATUM FORMULA IS NAVD88=NGVD29+1.86'.

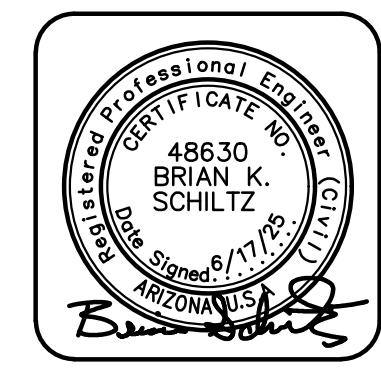
THE HIGHEST ADJACENT GRADES (HAG) AND LOWEST ADJACENT GRADES (LAG) FOR EACH BUILDING WERE DETERMINED BASED ON THE HIGHER OF THE HISTORIC AND EXISTING NAVD88 CONTOUR ELEVATIONS.

NO.	DATE	REVISION	BY

PURPOSE:
1ST DR SUBMITTAL

DESIGN BY: BKS
DRAWN BY: MM
CHECKED BY: BKS

HUNTER ENGINEERING
CIVIL AND SURVEY
10446 NORTH 74TH STREET
SUITE 140
SCOTTSDALE, AZ 85258
P 480 991 3985



1989 - HAG AND LAG EXHBIT
FOR
FINISH LINE AUTO CLUB
PRINCESS DRIVE AND LOOP 101
SCOTTSDALE, ARIZONA

CONTACT AROUND BIT AT LEAST 2 FEET
WORKING DAYS BEFORE BEGIN EXCAVATION

AR ZONAS11
CALL 811 OR CLICK ARZONAS11.COM

THESE PLANS ARE NOT APPROVED FOR CONSTRUCTION WITHOUT AN APPROVED SIGNATURE FROM THE GOVERNING MUNICIPALITY.

PROJECT NAME:
FINISH LINE AUTO CLUB

HE NO.: LGE0332
SCALE: 1"=60'

SHEET:
EXH 2

Appendix E

Master Drainage Report for Scottsdale Perimeter Center

MASTER DRAINAGE REPORT
FOR
SCOTTSDALE PERIMETER CENTER
C.W.W. JOB NO. 870514-1

PREPARED FOR:
WESTCOR COMPANY
11411 NORTH TATUM BOULEVARD
PHOENIX, ARIZONA 85028

SUBMITTED BY:
COLLAR, WILLIAMS & WHITE ENGINEERING
2702 NORTH 44TH STREET, SUITE 100-A
PHOENIX, ARIZONA 85008

JANUARY, 1988
REVISED JUNE, 1988
REVISED JULY, 1988
REVISED JANUARY, 1989

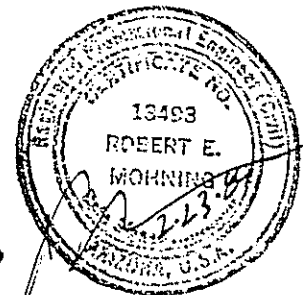
ACCEPTED MASTER PLAN

Water and Wastewater

Drainage

Circulation

MASTER PLANNING UNIT
COMMUNITY DEVELOPMENT DEPARTMENT
CITY OF SCOTTSDALE



SCOTTSDALE PERIMETER CENTER MASTER DRAINAGE REPORT

Table of Contents

	<u>Page</u>
I. Introduction.....	1
A. Location.....	1
B. Related Drainage Studies.....	1
II. Off-site Drainage.....	2
III. On-site Drainage.....	3
A. Existing Drainage Conditions.....	3
B. Developed Drainage Conditions.....	4
1. Storm Water Storage Facilities.....	4
2. Peak Discharges.....	5
3. Channels.....	5
4. Culverts.....	7
IV. Phasing of Development.....	8
V. Summary.....	9

Figure 1 - Vicinity Map

Exhibit 1 - Proposed Drainage Plan

Exhibit 2 - Proposed Downstream Drainage Plan

Appendix I - Peak Discharge Assumptions, HEC-1 Input
Data and Output Summaries (2, 10, 25
and 100 Year Storms)

42 pages

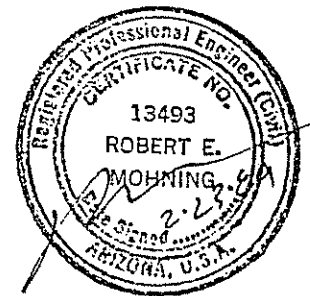
Appendix II - Channel Calculations

1 page

Appendix III - Culvert Calculations

3 pages

Appendix IV - Letter for Downstream Drainage



I. INTRODUCTION

A. Location

Scottsdale Perimeter Center is a planned industrial and commercial development, located within the corporate limits of Scottsdale, Arizona. The area comprising the Scottsdale Perimeter Center is a part of the former State Trust Lands, north of the Bureau of Reclamation's Central Arizona Project Canal.

The site lies west of the proposed Outer Loop Freeway, between Union Hills Drive and Bell Road and east of 83rd Street. Scottsdale Perimeter Center encompasses about 240 acres (gross). See Figure 1 - Vicinity Map. Exhibit 1 shows Union Hills Drive, Pima Road and Bell Road extended to Hayden Road from 83rd Street, although the area west of 83rd Street is not a part of this site.

Westcor purchased Parcels 16, 17, 18, 19 and 20, a portion of the above described State Trust Lands. Parcels 16-20 have been split due to the internal street configuration, and have been renumbered in this report with letters A through K.

The surrounding properties are developing rapidly. In the past two years, the City of Scottsdale Tournament Players Club Golf Course (TPC) and the Princess Hotel have been constructed; and the Eagle Apartments are currently under construction. Based upon current trends, it is anticipated that development in the immediate area will occur at faster rates than most other portions of Maricopa County.

B. Related Drainage Studies

Collar, Williams & White Engineering (CWW) completed the "Preliminary Drainage Report for Forest City Scottsdale Company" in August, 1985, which included about 1400 acres of State Trust Lands. The subject site was included in the preliminary report. Revisions to the report were made in February and May of 1986. Simons, Li and Associates, Inc. (SLA) completed the "Drainage Analysis for State Trust Lands - Outer Loop Highway - Scottsdale Road to Pima Road" in June 1987.

SLA essentially utilized the channelization network proposed in the preliminary report by CWW, with the addition of a detention basin north of the Outer Loop. Freeway between Hayden Road and Pima Road. This basin and the off-site drainage are discussed in further detail in the next section. The SLA report has been approved by the City of Scottsdale.

II. OFF-SITE DRAINAGE

The drainage basins contributing runoff to the subject site have been extensively analyzed and evaluated in the aforementioned drainage reports. Therefore, the off-site drainage will not be discussed in detail for this report. A brief summary of the off-site watersheds from the SLA report, however, is provided below.

The contributing offsite drainage areas affecting the site extend beyond Reata Pass, approximately 8 miles northeast of the subject site. According to SLA, about 1100 acre-feet of runoff volume is generated from the off-site watersheds during the 100 year storm. This volume is proposed to be routed to a large detention basin via upstream training dikes. The proposed location of the detention basin is adjacent to the northern edge of the Outer Loop Freeway, between Hayden Road and Pima Road. The detention basin is proposed to be drained by two outlets; one near Hayden Road and the other near the future Union Hills Road alignment. The peak discharge released from the Hayden Road spillway is 1796 cfs, while the 100 year peak discharge at Union Hills is proposed to be 722 cfs. It is the Union Hills outlet that affects the subject site. Although this drainage concept and corresponding peak discharge rates and locations have been approved by the City of Scottsdale, it should be noted that the Arizona State Land Department and the Arizona Department of Transportation have yet to reach a similar final decision.

In conclusion, Scottsdale Perimeter Center will be subject to an ultimate off-site peak discharge of 722 cfs from the upstream detention basin, plus 300 $\pm$ cfs from an adjacent State Land parcel at Pima Road (northeast of the Outer Loop). The off-site drainage is proposed to be routed in a future concrete-lined channel (subject to the Design Review Board approval) that will parallel the southerly and westerly property boundary of the Outer Loop Freeway from Union Hills to Bell Road, and then south to the TPC. Coordination with the City of Scottsdale and affected property owners will be necessary to design the routing of the 722 cfs from Bell Road to the TPC.

The future channel location and cross-section for routing the above described off-site flows are shown on Exhibit 1.

The construction of the channel, including the roadway culverts, and the detention basin must occur concurrently, or before the construction of, the Outer Loop.

III. ON-SITE DRAINAGE

A. Existing Drainage Conditions

As above, the on-site drainage basins have been previously analyzed. Therefore, only a brief description is presented below.

The site is presently undeveloped and slopes gently to the southwest with an average slope of 1.5 percent. The area is characterized by native desert plant species, with an estimated cover density of 20 percent. Several defined dry washes occur on the site. However, due to the high sediment load during runoff flows, the channels frequently shift depending upon the intensity and duration of the rainfall event. Many of the smaller washes transform from channelized flow to sheet flow, and then back to channelized flow downstream.

According to an earlier Simons, Li and Associates drainage report, entitled "Hydrology Report, Outer Loop Highway, North of the CAP Aqueduct", April 1987, the historical drainage across Bell Road, between Hayden Road and Pima Road, is 6879 cfs. The SLA hydrologic analysis showed that the historic channels are not capable of carrying flows greater than the 2 year event. Larger events are therefore carried in wide strips adjacent to the historic channels with shallow depths and considered sheet flow.

According to the aforementioned CWW drainage report, three (3) concentration points were identified at Bell alignment, between Hayden and Pima Roads. The location of the concentration points and the peak discharges, per the report, are:

Location	100 Year Peak Discharge (cfs)
83rd Street and Bell Road	1643
86th Street and Bell Road	6784
700 ± feet west of Pima Road	193

The magnitudes of the peak discharge values from both reports are similar. However, the location(s) are uncertain. It is obvious, though, that no well defined channel exists between Hayden and Pima Roads to contain the above peak discharges.

B. Developed Drainage Conditions

Scottsdale Perimeter Center encompasses 11 parcels. All parcels are zoned for either industrial or commercial use. A channelization network is proposed to route the on-site runoff safely and effectively through the site. The channels will convey the on-site generated runoff to the TPC.

Since development will probably occur prior to the construction of the detention basin north of the Outer Loop, temporary measures will be required to protect the structures from the off-site flows, as described earlier. Temporary measures could consist of elevating the finished floors, constructing berms, walls and channels, excavating retention basins, or a combination of the above.

Any of the above measures will need to be evaluated and implemented concurrently with the design of any individual site development with evaluation of any change in water surface elevation which would affect downstream properties.

1. Storm Water Storage Facilities

Scottsdale City Code Title 37, Section 37-42 (12)a. requires "as a minimum, all development will make provisions to store runoff from rainfall events up to and including the 100-year 2-hour duration event".

This requirement may be waived if per Section 37-42 (12)a. "The runoff has been included in a storage facility at another location". In the same section of the code, it is stated that if storage is waived, the development shall be required to contribute to the cost of drainage works on the basis of runoff contribution. Section 37-42 allows exceptions to the regulations for "areas covered by master drainage plans, providing that the standards established by the code are met".

The Perimeter Center is an integral part of the State Land Department Core South Master Plan which requires large retention basins to be constructed along the Outer Loop Highway. These basins are specifically intended to mitigate the need for on-site storage within the Core South project area.

Each of the respective parties are required to construct their portion of the drainage project.

The Perimeter Center lays the foundation to satisfy the provisions of the City Code Title 37 for storm water storage facilities.

2. Peak Discharges

Post development peak discharges (100 year, one hour storm) are listed below for each parcel.

Parcel	Area (acres)	Peak Discharge (cfs)			
		2 Year	10 Year	25 Year	100 Year
A	30.8	70	117	151	186
B	25.2	57	96	124	153
C	13.3	30	51	66	80
D	22.2	50	85	109	134
E	6.9	16	26	34	42
F	9.2	21	35	45	56
G	29.7	67	114	146	179
H	27.2	62	104	134	165
I	24.4	56	93	120	148
J	22.2	51	85	109	134
K	27.6	63	105	136	167

Exhibit 1 shows the routing of the above on-site peak discharges and the proposed channel network.

The U.S. Army Corps of Engineers HEC-1 model was used to calculate the on-site peak discharges. Appendix I contains the peak discharge assumptions, HEC-1 input data and the output summaries.

3. Channels

The channels are proposed to be grass-lined rather than natural earth for the following reasons:

- a. Grass-lined channels can sustain a velocity of 7 feet per second (fps) or higher before erosion and scour becomes a serious problem, whereas the velocity limit for earth-lining is about 4 to 5 fps.
- b. In order to maintain velocities near 7 fps, the approximate maximum channel bed slope allowed is 0.6 percent; a 0.25 percent slope would be needed to maintain a velocity of 5 fps. Since the natural landform slopes about 1.0 to 1.5 percent, more drop structures would be required to maintain flatter channel slopes.
- c. If the channel bed slope was to remain constant, then additional channel width would be necessary to decrease the channel depth, which in turn, would decrease the channel velocity.

The following analysis illustrates why grass-lined channels are more favorable for the subject site:

SHAPE	Trapezoidal	Trapezoidal	Trapezoidal
Lining	Grass	Earth	Earth
Sideslopes	4:1	4:1	4:1
Bottom Width	20'	20'	80'
Slope	0.6%	0.25%	0.6%
"n"	0.027	0.030	0.030
Velocity	7.1 fps	4.8 fps	5.0 fps
Design	681 cfs	681 cfs	681 cfs
Top Width	44 feet	52 feet	93 feet
1' Drop			
structures required at			
1.5% natural slope			
	1 per 112 feet	1 per 80 feet	1 per 112 feet

Therefore, it is apparent that the grass-lined channels are the superior alternative in terms of "acceptable" velocities, minimizing the number of drop structures, and reducing the widths of drainage easements.

To minimize the top width of the required channels and drainage easements, the vertical dimension of the drop structures is 1 foot. The channel depth is usually 4 feet. This depth includes one (1) foot of freeboard. This depth is consistent with the results of the SLA report. Also, 4H:1V sideslopes improve channel stability and maintenance.

Typical channel sections and drainage easement widths are shown on Exhibit 1. The calculations showing the velocities and discharges of the proposed channels are provided in Appendix II.

Two possible corridors for routing stormwater runoff at 83rd Street and Bell Road exist. The primary route i.e., routing runoff south at 83rd Street to the golf course, would require a drainage easement through private land. This route is consistent with historical flow patterns (see Section III A).

Alternative #1, i.e., routing runoff west to Hayden Road and then south to the golf course would also require a drainage easement on private land. Runoff, from this alternative would be combined with runoff from the north (adjacent Hayden Road). Refer to SLA report.

4. Culverts

Culverts are necessary to route the channelized flows under the major streets. Concrete box culverts are proposed for the 100 year peak discharges exceeding about 150 cfs. For locations where peak discharges are less than about 150 cfs, reinforced concrete pipes will probably be a more economical alternative.

Sixteen culvert locations are proposed. The locations are shown on Exhibit 1; the preliminary culvert sizes are listed below. The design sizes are based on an inlet control condition with an available headwater depth of 5 ± 0.5 feet, unless otherwise noted.

CULVERT NUMBER	LOCATION	SIZE	Q100(cfs)
1	W 1/4 Corner & Bell	Quad 10x4 CBC	1348
2	Loop & Pima South	Triple 8x3 CBC	632
3	Loop Road (west)		
4	& Pima East	Single 8x3 CBC	184
5	86th & Pima	Double 6x3 CBC	306
6	Loop Road (channel "F" & "G" to "H")	8x3 CBC	186
7	Loop Road (channel "D" to "G") & 86th	30" RCP	51
8	Loop Road (channel "E" to "F")	30" RCP	57
9	Loop Road (channels "R" & "S" to "V")	Triple 10x4 CBC	1026
10	Loop Road (channels "P" & "Q" to "R") and 86th	10x3 CBC	268
11	Loop Road (channel "O" to "P")	30" RCP	53
12	Loop Road (channels "M" & "N" to "Q") and 86th	Double 36" RCP	95
13	86th & Bell	6x3 CBC	167
14	Hayden & Bell	Six - 10x5 CBC*	3321 (max. est.)
15	Pima & Outer Loop	10x5 CBC**	700
16	Outer Loop (channel "B" to "C")	10x5 CBC**	1022
	Bell & Outer Loop	10x5 CBC**	1022

* Concrete box culvert based on 6 feet of available headwater depth. This preliminary size of culvert is required for alternative #2 only.

**Based on open channel flow calculations

The final culvert sizes and locations will be determined during the design phase of the roadway infrastructure.

Appendix III contains the culvert calculation worksheets.

IV. PHASING OF DEVELOPMENT

Phase I consists solely of the FFCA site (encompassing a portion of Parcel "J"). The developer intends to construct the roadways to the parcel, which consist of Bell Road to 86th Street, and then 86th Street to the parcel, plus the adjacent Loop Road.

Any excess top soil during the construction phases will be stockpiled in the northeast corner of the site. The soil material will be spread out to eliminate the possibility of floodwater damming, ponding and/or diversion. The damage to any existing vegetation will be minimal, thus preserving the area's visual resources.

Only 20% of the Perimeter Center can be developed until the drainage infrastructure has been completed.

The timeframe for the complete buildout of Scottsdale Perimeter Center is anticipated to be several years. During the interim, preventative siltation measures will be necessary to minimize downstream sedimentation. A small perimeter berm around each parcel boundary is proposed to prevent sediment from entering the channel system.

It will not be possible to alter upstream offsite siltation effects until the proposed upstream detention system is in place.

During and after parcel development, the grading and drainage plan will include a design to "trap and collect" sediment from draining into the channel network.

Careful measures and coordination will be undertaken where drainage enters and crosses the City of Scottsdale golf course to prevent erosion and silting.

It is apparent to CWW that the existing drainage facilities to route runoff through the golf course are inadequate to contain the flows previously given. Therefore, we believe the City of Scottsdale needs to explore further channelization improvements on the TPC property.

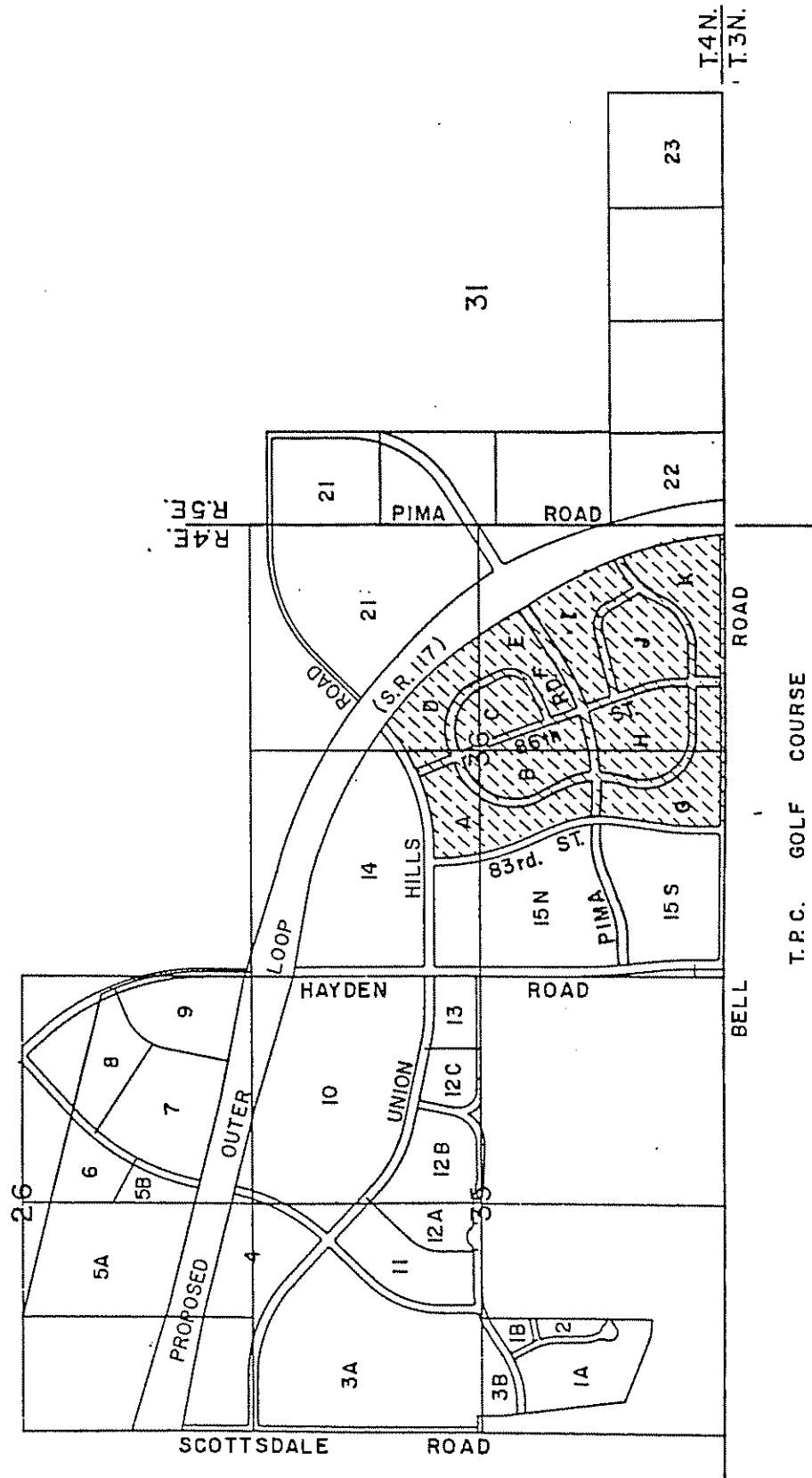
V. SUMMARY

The proposed master drainage plan for the Scottsdale Perimeter Center was prepared with close coordination with the City of Scottsdale staff. The master plan follows the recommendations suggested by the Simons, Li and Associates' report entitled "Drainage Analysis for State Trust Lands, Outer Loop Highway, Scottsdale Road to Pima Road", June 1987. This report has been approved by the City of Scottsdale.

The primary goal of this report is two-fold. First, to establish the drainage concept for the eleven parcels of the Scottsdale Perimeter Center; and secondly, to effectively route the off-site runoff and the on-site generated runoff through the development.

The off-site drainage concept as contained herein, i.e., detention basin with training dikes and the 100 year peak discharge bleed-off rate, are adopted in full from the Simons, Li and Associates' report.

The proposed on-site channel network, with the recommended channel cross sections, will satisfy the fully developed conditions of the site.



COLLAR, WILLIAMS & WHITE
ENGINEERING

FIGURE...
 PAGE

APPENDIX I - PEAK DISCHARGE ASSUMPTIONS, HEC-1 INPUT DATA, AND OUTPUT SUMMARIES

1. Per the SCS Aguila-Carefree Soil Survey, April, 1986, the existing soils are classified in the "B" Hydrologic Soil Group.
2. The existing vegetative cover density is 20%. This is consistent with previous drainage reports by CWW and others.
3. Per Figure 2-15, COS Section 2 Design Manual, a "B" soil and 20% cover yields a runoff curve number of 83.
4. During a telephone discussion with Bill Erickson, COS, on 12/22/87, it was agreed upon to calculate post peak discharges based on additional impervious area, rather than adjusting (increasing) the runoff curve number. This approach is consistent with new State and Federal guidelines. The percent of impervious areas with development, both industrial and commercial, is assumed to be 85.
5. Due to the small drainage parcels, all on-site parcels were assumed to have a time of concentration of 10 minutes. Although several of the parcels would have a time of concentration less than 10 minutes, by using either Kirpich's formula or a time of travel relationship in parking areas and/or street, a 10 minute minimum was assumed. This is consistent with ADOT drainage procedures.

This is an conservative assumption, as the developer will impose sediment control on each development site, thereby increasing the time of concentration.

6. The individual drainage area boundaries are shown on Exhibit 1. The post development sub-basin drainage areas, within the individual parcels, were depicted to conform to the existing topography and landform as much as possible.
7. The 100 year, 1 hour precipitation value of 2.46 inches was obtained from Fig. 2-6, COS Section 2 Design Manual. The 5 minute and 15 minute values, 0.72 inches and 1.41 inches, respectively, were obtained by multiplying the 1 hour value of 0.29 and 0.57. This follows ADOT guidelines, per the "Hydrologic Design for Highway Drainage, Arizona". Likewise, the 2 year, 1 hour value was 0.99 inches, the 10 year, 1 hour value was 1.62 inches, and the 25 year, 1 hour value was 2.05 inches.
8. Channel routing was utilized by the Kinematic Wave channel routing method.

LINE ID.....1.....2.....3.....4.....5.....6.....7.....8.....9.....10

*** FREE ***

```

1      ID      SCOTTSDALE'S PERIMETER CENTRE
2      ID
3      ID      POST DEVELOPMENT DRAINAGE
4      ID      2 YEAR - 1 HOUR
5      ID
6      ID      FILE NAME:PERIM
7      ID
8      IDIAGRAM
9      IT      1 28DEC88      1200      100
10     IO      5
11     KE      D1
12     BA      0.0133
13     PH              0.29      0.56      0.99
14     LS              85      85
15     UD      0.10
16     KE      ROUT01
17     RE      900      0.012      0.027      TRAP      8      4
18     KE      C1
19     BA      0.0088
20     KECONBINED
21     HC      2
22     KE      D2
23     BA      0.0147
24     KE      ROUT02
25     RE      630      0.010      0.027      TRAP      8      4
26     KE      C2
27     BA      0.0120
28     KECONBINED
29     HC      2
30     KECONBINED
31     HC      2
32     KE      ROUT01
33     RE      450      0.011      0.027      TRAP      8      4
34     KE      F1
35     BA      0.0076
36     KECONBINED
37     HC      2

```

LINE	ID	1	2	3	4	5	6	7	8	9	10
37	EX	B2									
38	BA	0.0067									
39	EX	E									
40	BA	0.0100									
41	RECOMBINED										
42	HC	2									
43	FI	ROUTE									
44	RE	700	0.009	0.027		TRAP	8		4		
45	EX	F2									
46	BA	0.0108									
47	RECOMBINED										
48	HC	2									
49	RECOMBINED										
50	HC	2									
51	FI	ROUTE2									
52	RE	700	0.007	0.027		TRAP	8		4		
53	EX	F2									
54	BA	0.0152									
55	RECOMBINED										
56	HC	2									
57	EX	F1									
58	BA	0.0242									
59	RECOMBINED										
60	HC	2									
61	EX	A1									
62	BA	0.0375									
63	EX	ROUTE1									
64	RE	350	0.008	0.027		TRAP	8		4		
65	EX	A2									
66	BA	0.0100									
67	RECOMBINED										
68	HC	2									
69	EX	ROUTE2									
70	RE	400	0.0045	0.027		TRAP	8		4		

[illegible]

SCHEMATIC DIAGRAM OF STREAM NETWORK

INPUT LINE	(V) ROUTING	(---) DIVERSION OR PUMP FLOW
NO.	(.) CONNECTOR	(:---) RETURN OF DIVERTED OR PUMPED FLOW
10	D1	
	V	
	V	
15	ROUTD1	
	.	
17	.	C1
	.	.
	.	.
19	COMBINED.....	
	.	
21	.	D2
	.	V
	.	V
23	ROUTD2	
	.	
	.	
25	.	C2
	.	.
	.	.
27	COMBINED.....	
	.	
	.	
29	COMBINED.....	
	V	
	V	
31	ROUTC1	
	.	
33	.	F1
	.	.
	.	.
35	COMBINED.....	
	.	
37	.	D3
	.	.
	.	.
39	.	E
	.	.
	.	.
41	COMBINED.....	
	V	
	V	
43	ROUTE	
	.	
	.	
45	.	F2
	.	.
	.	.
47	COMBINED.....	
	.	
	.	
49	COMBINED.....	
	V	
	V	

53	.	B2	.
55	COMBINED.....		
57	.	F1	.
59	COMBINED.....		
61	.	A1	.
	.	V	.
	.	7	.
63	ROUTE1		
65	.	A2	.
67	COMBINED.....		
	.	V	.
	.	V	.
69	ROUTE2		
71	COMBINED.....		
	.	V	.
	.	V	.
73	ROUTE1		
75	.	B1	.
77	COMBINED.....		
79	.	F2	.
81	.		F1
83	COMBINED.....		
	.	V	.
	.	V	.
85	ROUTE12		
87	.		B1
89	COMBINED.....		
91	.		F3
	.		V
	.		V
93	ROUTE13		

97	.	.	COMBINED.....
	.	.	
99	.	.	COMBINED.....
	.	.	V
	.	.	V
101	.	.	ROUTJ2
	.	.	
103	.	.	H2
	.	.	
105	.	.	COMBINED.....
	.	.	
107	.	.	COMBINED.....
	.	.	V
	.	.	V
109	.	.	ROUTH1
	.	.	
111	.	.	V
	.	.	V
	.	.	V
113	.	.	ROUTE
	.	.	
115	.	.	63
	.	.	
117	.	.	COMBINED.....
	.	.	
119	.	.	COMBINED.....
	.	.	
121	.	.	61
	.	.	V
	.	.	V
123	.	.	ROUTG1
	.	.	
125	.	.	62
	.	.	
127	.	.	COMBINED.....
	.	.	
129	.	.	COMBINED.....
	.	.	
131	.	.	152

(***) RUNOFF ALSO COMPUTED AT THIS LOCATION

FILE NAME: PERIOD

```

IT      HYDROGRAPH TIME DATA
        DMIN      1 MINUTES IN COMPUTATION INTERVAL
        IDATE      20DEC68 STARTING DATE
        ITIME      1200 STARTING TIME
        NO        100 NUMBER OF HYDROGRAPH ORDINATES
        HDATE      20DEC68 ENDING DATE
        HDTIME     1339 ENDING TIME

```

ENGLISH UNITS

[illegible]

[illegible]

RUNOFF SUMMARY
FLOW IN CUBIC FEET PER SECOND
TIME IN HOURS, AREA IN SQUARE MILES

OPERATION	STATION	PEAK FLOW	TIME OF PEAK	AVERAGE FLOW FOR MAXIMUM PERIOD			BASIN AREA	MAXIMUM STAGE	TIME OF MAX STAGE
				6-HOUR	24-HOUR	72-HOUR			
HYDROGRAPH AT	D1	19.	.62	4.	4.	4.	.01		
ROUTED TO	ROUTD1	19.	.65	4.	4.	4.	.01		
HYDROGRAPH AT	C1	13.	.62	3.	3.	3.	.01		
2 COMBINED AT	COMBINED	31.	.65	7.	7.	7.	.02		
HYDROGRAPH AT	D2	21.	.62	5.	5.	5.	.01		
ROUTED TO	ROUTD2	21.	.65	5.	5.	5.	.01		
HYDROGRAPH AT	C2	17.	.62	4.	4.	4.	.01		
2 COMBINED AT	COMBINED	38.	.65	9.	9.	9.	.03		
2 COMBINED AT	COMBINED	68.	.65	16.	16.	16.	.05		
ROUTED TO	ROUTC1	68.	.65	16.	16.	16.	.05		
HYDROGRAPH AT	F1	7.	.62	1.	1.	1.	.03		
2 COMBINED AT	COMBINED	74.	.65	18.	18.	18.	.05		
HYDROGRAPH AT	B3	10.	.62	2.	2.	2.	.01		
HYDROGRAPH AT	E	16.	.62	4.	4.	4.	.01		
2 COMBINED AT	COMBINED	27.	.62	6.	6.	6.	.02		
ROUTED TO	ROUTEF	25.	.65	6.	6.	6.	.02		
HYDROGRAPH AT	I1	16.	.62	4.	4.	4.	.01		
2 COMBINED AT	COMBINED	40.	.65	10.	10.	10.	.03		
2 COMBINED AT	COMBINED	114.	.65	27.	27.	27.	.08		
ROUTED TO	ROUTE2	112.	.67	27.	27.	27.	.08		
HYDROGRAPH AT	J2	22.	.62	5.	5.	5.	.02		
2 COMBINED AT	COMBINED	132.	.65	32.	32.	32.	.10		
HYDROGRAPH AT	K1	35.	.62	8.	8.	8.	.02		
2 COMBINED AT	COMBINED	165.	.65	40.	40.	40.	.12		
HYDROGRAPH AT	A1	54.	.62	13.	13.	13.	.04		
ROUTED TO	ROUTA1	54.	.65	13.	13.	13.	.04		
HYDROGRAPH AT	A2	16.	.62	4.	4.	4.	.01		
2 COMBINED AT	COMBINED	69.	.62	17.	16.	16.	.05		

2 COMBINED AT COMBINED	271.	.60	52.	57.	52.	.11
ROUTED TO ROUTE1	128.	.68	56.	56.	56.	.11
HYDROGRAPH AT H1	37.	.62	9.	9.	9.	.05
2 COMBINED AT COMBINED	258.	.67	65.	65.	65.	.19
HYDROGRAPH AT 12	31.	.62	7.	7.	7.	.12
HYDROGRAPH AT 11	5.	.62	1.	1.	1.	.00
2 COMBINED AT COMBINED	36.	.62	8.	8.	8.	.02
ROUTED TO ROUTE2	35.	.65	8.	8.	8.	.02
HYDROGRAPH AT J1	10.	.62	2.	2.	2.	.01
2 COMBINED AT COMBINED	44.	.63	10.	10.	10.	.05
HYDROGRAPH AT 15	20.	.62	5.	5.	5.	.01
ROUTED TO ROUTE3	10.	.70	5.	5.	5.	.01
HYDROGRAPH AT J2	41.	.62	9.	9.	9.	.07
2 COMBINED AT COMBINED	53.	.65	14.	14.	14.	.04
2 COMBINED AT COMBINED	97.	.63	24.	24.	24.	.07
ROUTED TO ROUTE2	97.	.67	24.	24.	24.	.00
HYDROGRAPH AT 02	25.	.62	6.	6.	6.	.02
2 COMBINED AT COMBINED	119.	.65	30.	30.	30.	.09
2 COMBINED AT COMBINED	376.	.60	95.	95.	95.	.20
ROUTED TO ROUTE1	271.	.68	95.	95.	95.	.20
HYDROGRAPH AT 1	63.	.62	15.	15.	15.	.04
ROUTED TO ROUTE	61.	.65	14.	14.	14.	.04
HYDROGRAPH AT 53	12.	.62	3.	3.	3.	.01
2 COMBINED AT COMBINED	73.	.63	17.	17.	17.	.05
2 COMBINED AT COMBINED	444.	.67	112.	112.	112.	.55
HYDROGRAPH AT 61	20.	.62	7.	7.	7.	.02
ROUTED TO ROUTE1	26.	.67	7.	7.	7.	.02
HYDROGRAPH AT 62	27.	.62	6.	6.	6.	.02
2 COMBINED AT COMBINED	51.	.65	15.	15.	15.	.04
2 COMBINED AT COMBINED	494.	.67	125.	125.	125.	.57
HYDROGRAPH AT 153	106.	.62	24.	24.	24.	.07

LINE ID.....1.....2.....3.....4.....5.....6.....7.....8.....9.....10

*** FREE ***

1	ID	SCOTTSDALE'S PERIMETER CENTRE							
2	ID								
3	ID	POST DEVELOPMENT DRAINAGE							
4	ID	10 YEAR - 1 HOUR							
5	ID								
6	ID	FILE NAME:PERM1							
7	ID								
	DIAGRAM								
8	IT	1	20DEC88	1200	100				
9	IO	5							
10	IF	D1							
11	BA	0.0133							
12	PH		0.47	0.92	1.62				
13	LS		83	83					
14	UD	0.10							
15	IF	ROUTD1							
16	RE	900	0.012	0.027		TRAP	8	4	
17	IF	C1							
18	BA	0.0000							
19	RECOMBINED								
20	HC	2							
21	IF	C2							
22	BA	0.0117							
23	IF	ROUTD2							
24	RE	650	0.010	0.027		TRAP	8	4	
25	IF	C2							
26	BA	0.0120							
27	RECOMBINED								
28	HC	2							
29	RECOMBINED								
30	HC	2							
31	IF	ROUTC1							
32	RE	450	0.011	0.027		TRAP	8	4	
33	IF	C1							
34	BA	0.0000							
35	RECOMBINED								
36	HC	2							

LINE	ID	1	2	3	4	5	6	7	8	9	10
37	EE	03									
38	EA	0.0067									
39	EE	5									
40	EA	0.0108									
41	EE	COMBINED									
42	EC	2									
43	EE	ROUTE									
44	EE	700	0.009	0.027		TRAP	8	4			
45	EE	12									
46	EA	0.0108									
47	EE	COMBINED									
48	EC	2									
49	EE	COMBINED									
50	EC	2									
51	EE	ROUTE2									
52	EE	00	0.007	0.027		TRAP	8	4			
53	EE	12									
54	EA	0.0152									
55	EE	COMBINED									
56	EC	7									
57	EE	11									
58	EA	0.0217									
59	EE	COMBINED									
60	EC	2									
61	EE	A1									
62	EA	0.0373									
63	EE	ROUTE1									
64	EE	350	0.008	0.027		TRAP	8	4			
65	EE	A2									
66	EA	0.0108									
67	EE	COMBINED									
68	EC	2									
69	EE	ROUTE2									
70	EE	400	0.0045	0.027		TRAP	8	4			

40

SCHEMATIC DIAGRAM OF STREAM NETWORK

INPUT LINE	(V) ROUTING	(---) DIVERSION OR PUMP FLOW
NO.	(.) CONNECTOR	(---) RETURN OF DIVERTED OR PUMPED FLOW
10	B1 V V	
15	ROUTD1 .	
17	. C1 .	
19	COMBINED..... .	
21	. B2 V V	
23	. ROUTD2 .	
25	. C2 .	
27	COMBINED..... .	
29	COMBINED..... V V	
31	ROUTC1 .	
33	. B1 .	
35	COMBINED..... .	
37	. B3 .	
39	. F .	
41	COMBINED..... V V	
43	. ROUTE .	
45	. F2 .	
47	COMBINED..... .	
49	COMBINED..... V /	

53	.	B2	
	.		
55	COMBINED.....		
	.		
57	.	B1	
	.		
59	COMBINED.....		
	.		
61	.	A1	
	.	V	
	.	V	
63	ROUTE1		
	.		
65	.	A2	
	.		
67	COMBINED.....		
	.	V	
	.	V	
69	ROUTE2		
	.		
71	COMBINED.....		
	.	V	
	.	V	
73	ROUTE1		
	.		
75	.	B1	
	.		
77	COMBINED.....		
	.		
79	.	I2	
	.		
81	.	I1	
	.		
83	COMBINED.....		
	.	V	
	.	V	
85	ROUTE2		
	.		
87	.	B1	
	.		
89	COMBINED.....		
	.		
91	.	I3	
	.	V	
	.	V	
93	ROUTE3		
	.		
	.		

97	.	.	COMBINED.....
99	.	.	COMBINED.....
	.	.	V
	.	.	V
101	.	.	ROUT77
103	.	.	H2
105	.	.	COMBINED.....
107	.	.	COMBINED.....
	.	.	V
	.	.	V
109	.	.	ROUT81
111	.	.	K
	.	.	V
	.	.	V
113	.	.	ROUTE
115	.	.	G3
117	.	.	COMBINED.....
119	.	.	COMBINED.....
121	.	.	G1
	.	.	V
	.	.	V
123	.	.	ROUT61
125	.	.	G2
127	.	.	COMBINED.....
129	.	.	COMBINED.....
131	.	.	158

(***) RUNOFF ALSO COMPUTED AT THIS LOCATION

POST DEVELOPMENT DRAINAGE
10 YEAR - 1 HOUR

FILE NAME: PERM

9 10 OUTPUT CONTROL VARIABLES

```

IPRNT      5  PRINT CONTROL
IPLOT      0  PLOT CONTROL
OSCAL      0.  HYDROGRAPH PLOT SCALE

```

HYDROGRAPH TIME DATA

MMIN	1	MINUTES IN COMPUTATION INTERVAL
IDATE	28DEC88	STARTING DATE
ITIME	1200	STARTING TIME
NO	100	NUMBER OF HYDROGRAPH ORDINATES
NDDATE	28DEC88	ENDING DATE
NDTIME	1339	ENDING TIME

COMPUTATION INTERVAL .02 HOURS
TOTAL TIME BASE 1.65 HOURS

1991 12 15

[illegible]

[illegible]

RUNOFF SUMMARY
FLOW IN CUBIC FEET PER SECOND
TIME IN HOURS, AREA IN SQUARE MILES

OPERATION	STATION	PEAK FLOW	TIME OF PEAK	AVERAGE FLOW FOR MAXIMUM PERIOD			BASIN AREA	MAXIMUM STAGE	TIME OF MAX STAGE
HYDROGRAPH AT	D1	33.	.62	8.	8.	8.	.01		
ROUTED TO	ROUTD1	32.	.65	8.	8.	8.	.01		
HYDROGRAPH AT	C1	22.	.62	5.	5.	5.	.01		
2 COMBINED AT	COMBINED	53.	.63	12.	12.	12.	.02		
HYDROGRAPH AT	D2	36.	.62	8.	8.	8.	.01		
ROUTED TO	ROUTD2	35.	.65	8.	8.	8.	.01		
HYDROGRAPH AT	C2	29.	.62	7.	7.	7.	.01		
2 COMBINED AT	COMBINED	64.	.63	15.	15.	15.	.03		
2 COMBINED AT	COMBINED	117.	.63	28.	28.	28.	.05		
ROUTED TO	ROUTC1	116.	.65	28.	28.	28.	.05		
HYDROGRAPH AT	F1	9.	.62	2.	2.	2.	.00		
2 COMBINED AT	COMBINED	124.	.65	30.	30.	30.	.05		
HYDROGRAPH AT	D3	16.	.62	4.	4.	4.	.01		
HYDROGRAPH AT	E	26.	.62	6.	6.	6.	.01		
2 COMBINED AT	COMBINED	43.	.62	10.	10.	10.	.02		
ROUTED TO	ROUTE	43.	.63	10.	10.	10.	.02		
HYDROGRAPH AT	I2	26.	.62	6.	6.	6.	.01		
2 COMBINED AT	COMBINED	69.	.63	16.	16.	16.	.03		
2 COMBINED AT	COMBINED	193.	.63	46.	46.	46.	.08		
ROUTED TO	ROUTE2	191.	.65	45.	45.	45.	.08		
HYDROGRAPH AT	H2	37.	.62	9.	9.	9.	.02		
2 COMBINED AT	COMBINED	326.	.65	54.	54.	54.	.10		
HYDROGRAPH AT	H1	59.	.62	14.	14.	14.	.02		
2 COMBINED AT	COMBINED	782.	.65	68.	68.	68.	.12		
HYDROGRAPH AT	A1	91.	.62	21.	21.	21.	.04		
ROUTED TO	ROUTA1	91.	.63	21.	21.	21.	.04		
HYDROGRAPH AT	A2	26.	.62	6.	6.	6.	.01		
2 COMBINED AT	COMBINED	117.	.63	27.	27.	27.	.05		

2 COMBINED AT COMBINED		397.	.66	95.	95.	95.	.17
ROUTED TO ROUTE1		393.	.67	95.	95.	95.	.17
HYDROGRAPH AT H1		62.	.62	14.	14.	14.	.03
2 COMBINED AT COMBINED		448.	.67	109.	109.	109.	.19
HYDROGRAPH AT 12		52.	.62	12.	12.	12.	.02
HYDROGRAPH AT 11		8.	.62	2.	2.	2.	.00
2 COMBINED AT COMBINED		60.	.62	14.	14.	14.	.02
ROUTED TO ROUTE2		59.	.63	14.	14.	14.	.02
HYDROGRAPH AT J1		16.	.62	4.	4.	4.	.01
2 COMBINED AT COMBINED		75.	.63	18.	18.	18.	.03
HYDROGRAPH AT 13		33.	.62	8.	8.	8.	.01
ROUTED TO ROUTE3		31.	.70	8.	8.	8.	.01
HYDROGRAPH AT 102		69.	.62	16.	16.	16.	.03
2 COMBINED AT COMBINED		93.	.63	23.	23.	23.	.04
2 COMBINED AT COMBINED		160.	.63	41.	41.	41.	.07
ROUTED TO ROUTE2		166.	.67	41.	41.	41.	.07
HYDROGRAPH AT H2		62.	.62	10.	10.	10.	.02
2 COMBINED AT COMBINED		203.	.65	51.	51.	51.	.09
2 COMBINED AT COMBINED		632.	.67	160.	160.	160.	.29
ROUTED TO ROUTE1		650.	.67	159.	159.	159.	.29
HYDROGRAPH AT 1		105.	.62	24.	24.	24.	.04
ROUTED TO ROUTE		105.	.63	24.	24.	24.	.04
HYDROGRAPH AT B3		20.	.62	5.	5.	5.	.01
2 COMBINED AT COMBINED		125.	.63	29.	29.	29.	.05
2 COMBINED AT COMBINED		169.	.67	100.	100.	100.	.33
HYDROGRAPH AT 51		18.	.62	11.	11.	11.	.02
ROUTED TO ROUTE1		46.	.67	11.	11.	11.	.02
HYDROGRAPH AT 32		46.	.62	11.	11.	11.	.02
2 COMBINED AT COMBINED		80.	.63	21.	21.	21.	.04
2 COMBINED AT COMBINED		854.	.67	210.	210.	210.	.57
HYDROGRAPH AT 158		178.	.62	41.	41.	41.	.07

LINE ID.....1.....2.....3.....4.....5.....6.....7.....8.....9.....10

*** FREE ***

1	ID	SCOTTSDALE'S PERIMETER CENTRE							
2	ID								
3	ID	POST DEVELOPMENT DRAINAGE							
4	ID	25 YEAR - 1 HOUR							
5	ID								
6	ID	FILE NAME:PERM2							
7	ID								
	ID	DIAGRAM							
8	IT	1	28DEC88	1200	100				
9	IO	5							
10	KK	D1							
11	BA	0.0133							
12	PH		0.59	1.17	2.05				
13	LS	83	85						
14	UD	0.10							
15	KK	ROUTE1							
16	KK	900	0.012	0.027		TRAP	8	4	
17	FK	01							
18	BA	0.0038							
19	KK	COMBINED							
20	HC	2							
21	KK	02							
22	BA	0.0147							
23	KK	ROUTE2							
24	KK	650	0.010	0.027		TRAP	8	4	
25	KK	02							
26	BA	0.0120							
27	KK	COMBINED							
28	HC	2							
29	KK	COMBINED							
30	HC	2							
31	KK	ROUTE1							
32	KK	450	0.011	0.027		TRAP	8	4	
33	FK	01							
34	BA	0.0036							
35	KK	COMBINED							
36	HC	2							

LINE	ID	1	2	3	4	5	6	7	8	9	10
37	KK	DS									
38	BA	0.0067									
39	KK	F									
40	BA	0.0108									
41	KKCOMBINED										
42	HC	2									
43	KK	ROUTE									
44	RE	700	0.009	0.027		TRAP	8		4		
45	KK	F2									
46	BA	0.0108									
47	KKCOMBINED										
48	HC	2									
49	KKCOMBINED										
50	HC	2									
51	KK	ROUTE2									
52	RE	700	0.007	0.027		TRAP	8		4		
53	KL	A2									
54	BA	0.0152									
55	KKCOMBINED										
56	HC	2									
57	KK	B1									
58	BA	0.0242									
59	KKCOMBINED										
60	HC	2									
61	KK	A1									
62	BA	0.0373									
63	KK	ROUTE1									
64	RE	350	0.008	0.027		TRAP	8		4		
65	KK	A2									
66	BA	0.0108									
67	KKCOMBINED										
68	HC	2									
69	KK	ROUTE2									
70	RE	400	0.0045	0.027		TRAP	8		4		

[illegible]

SCHEMATIC DIAGRAM OF STREAM NETWORK

INPUT LINE	(V) ROUTING	(---) DIVERSION OR PUMP FLOW
NO.	(.) CONNECTOR	(---) RETURN OF DIVERTED OR PUMPED FLOW
10	D1	
	V	
	V	
15	ROUTD1	
	.	
17	.	G1
	.	.
	.	.
19	COMBINED.....	
	.	
21	.	B2
	.	V
	.	V
23	ROUTD2	
	.	
	.	
25	.	G2
	.	.
	.	.
27	COMBINED.....	
	.	
	.	
29	COMBINED.....	
	7	
	7	
31	ROUTC1	
	.	
	.	
33	.	F1
	.	.
	.	.
35	COMBINED.....	
	.	
37	.	D3
	.	.
	.	.
39	.	E
	.	.
	.	.
41	COMBINED.....	
	V	
	V	
43	ROUTE	
	.	
	.	
45	.	F2
	.	.
	.	.
47	COMBINED.....	
	.	
	.	
49	COMBINED.....	
	V	
	V	

53	.	02	.
55	COMBINED.....		
57	.	01	.
59	COMBINED.....		
61	.	A1	.
	.	V	.
	.	V	.
63	ROUTE1		
65	.		42
67	COMBINED.....		
	.	V	.
	.	V	.
69	ROUTE2		
71	COMBINED.....		
	.	V	.
	.	V	.
73	ROUTE1		
75	.	01	.
77	COMBINED.....		
79	.	12	.
81	.		11
83	COMBINED.....		
	.	V	.
	.	V	.
85	ROUTE12		
87	.		01
89	COMBINED.....		
91	.		15
	.	V	.
	.	V	.
93	ROUTE13		

97	.	.	COMBINED.....
	.	.	.
99	.	.	COMBINED.....
	.	.	V
	.	.	V
101	.	.	ROUT32
	.	.	.
103	.	.	R2
	.	.	.
105	.	.	COMBINED.....
	.	.	.
107	.	.	COMBINED.....
	.	.	V
	.	.	V
109	.	.	ROUT1
	.	.	.
111	.	.	V
	.	.	V
	.	.	V
113	.	.	ROUTE
	.	.	.
115	.	.	G3
	.	.	.
117	.	.	COMBINED.....
	.	.	.
119	.	.	COMBINED.....
	.	.	.
121	.	.	G1
	.	.	V
	.	.	V
123	.	.	ROUTE1
	.	.	.
125	.	.	G2
	.	.	.
127	.	.	COMBINED.....
	.	.	.
129	.	.	COMBINED.....
	.	.	.
131	.	.	138

(44*) RUNOFF ALSO COMPUTED AT THIS LOCATION

POST DEVELOPMENT DRAINAGE
25 YEAR - 1 HOUR

FILE NAME: PFR12

9 10 OUTPUT CONTROL VARIABLES

```

IPRNT      5  PRINT CONTROL
IPLOT      9  PLOT CONTROL
QSCAL      0.  HYDROGRAPH PLOT SCALE

```

II HYDROGRAPH TIME DATA

MIN	1	MINUTES IN COMPUTATION INTERVAL
IDATE	20DEC68	STARTING DATE
ITIME	1200	STARTING TIME
NO	100	NUMBER OF HYDROGRAPH ORDINATES
NDDATE	20JUN69	ENDING DATE
NRTIME	1439	ENDING TIME

COMPUTATION INTERVAL	.02 HOURS
TOTAL TIME BASE	1.65 HOURS

AUTHORS:

[illegible]

VALUE EXCEEDS TABLE IN LOGLOG	.01667	.01667	1.00000
VALUE EXCEEDS TABLE IN LOGLOG	.01667	.01667	1.00000
VALUE EXCEEDS TABLE IN LOGLOG	.01667	.01667	1.00000
VALUE EXCEEDS TABLE IN LOGLOG	.01667	.01667	1.00000
VALUE EXCEEDS TABLE IN LOGLOG	.01667	.01667	1.00000
VALUE _EXCEEDS TABLE IN LOGLOG	.01667	.01667	1.00000
VALUE EXCEEDS TABLE IN LOGLOG	.01667	.01667	1.00000
VALUE EXCEEDS TABLE IN LOGLOG	.01667	.01667	1.00000

RHHOFF SUMMARY
FLOW IN CUBIC FEET PER SECOND
TIME IN HOURS, AREA IN SQUARE MILES

OPERATION	STATION	PEAK FLOW	TIME OF PEAK	AVERAGE FLOW FOR MAXIMUM FLOOD			DRAIN AREA	MAXIMUM STAGE	TIME OF MAX. STAGE
HYDROGRAPH AT	D1	42.	.63	10.	10.	10.	.01		
ROUTED TO	ROUTE1	41.	.65	10.	10.	10.	.01		
HYDROGRAPH AT	C1	20.	.62	6.	6.	6.	.01		
2 COMBINED AT	COMBINED	60.	.65	16.	16.	16.	.02		
HYDROGRAPH AT	D2	46.	.62	11.	11.	11.	.01		
ROUTED TO	ROUTE2	45.	.63	11.	11.	11.	.01		
HYDROGRAPH AT	C2	38.	.62	9.	9.	9.	.01		
2 COMBINED AT	COMBINED	83.	.65	19.	19.	19.	.03		
2 COMBINED AT	COMBINED	151.	.65	35.	35.	35.	.05		
ROUTED TO	ROUTE1	150.	.65	35.	35.	35.	.05		
HYDROGRAPH AT	F1	11.	.62	3.	3.	3.	.00		
2 COMBINED AT	COMBINED	161.	.65	38.	38.	38.	.05		
HYDROGRAPH AT	D3	21.	.62	5.	5.	5.	.01		
HYDROGRAPH AT	E	34.	.62	8.	8.	8.	.01		
2 COMBINED AT	COMBINED	55.	.62	13.	13.	13.	.02		
ROUTED TO	ROUTE	54.	.65	13.	13.	13.	.02		
HYDROGRAPH AT	F2	34.	.62	8.	8.	8.	.01		
2 COMBINED AT	COMBINED	87.	.65	20.	20.	20.	.03		
2 COMBINED AT	COMBINED	248.	.65	58.	58.	58.	.08		
ROUTED TO	ROUTE2	246.	.65	58.	58.	58.	.08		
HYDROGRAPH AT	B2	48.	.62	11.	11.	11.	.02		
2 COMBINED AT	COMBINED	271.	.65	69.	69.	69.	.10		
HYDROGRAPH AT	G1	76.	.62	18.	18.	18.	.02		
2 COMBINED AT	COMBINED	364.	.65	87.	87.	87.	.12		
HYDROGRAPH AT	A1	117.	.62	27.	27.	27.	.04		
ROUTED TO	ROUTE1	117.	.65	27.	27.	27.	.04		
HYDROGRAPH AT	A2	34.	.62	8.	8.	8.	.01		
2 COMBINED AT	COMBINED	150.	.65	35.	35.	35.	.05		

2 COMBINED AT	COMBINED	512.	.65	122.	122.	122.	.17
ROUTED TO	ROUTE1	506.	.67	121.	121.	121.	.17
HYDROGRAPH AT	H1	60.	.62	18.	18.	18.	.03
2 COMBINED AT	COMBINED	537.	.67	140.	140.	140.	.19
HYDROGRAPH AT	12	66.	.62	15.	15.	15.	.02
HYDROGRAPH AT	11	11.	.62	2.	2.	2.	.00
2 COMBINED AT	COMBINED	77.	.62	18.	18.	18.	.02
ROUTED TO	ROUTE2	76.	.63	18.	18.	18.	.02
HYDROGRAPH AT	51	21.	.62	5.	5.	5.	.01
2 COMBINED AT	COMBINED	97.	.63	23.	23.	23.	.03
HYDROGRAPH AT	13	43.	.62	10.	10.	10.	.01
ROUTED TO	ROUTE3	40.	.68	10.	10.	10.	.01
HYDROGRAPH AT	12	88.	.62	20.	20.	20.	.03
2 COMBINED AT	COMBINED	121.	.63	30.	30.	30.	.04
2 COMBINED AT	COMBINED	218.	.63	53.	53.	53.	.17
ROUTED TO	ROUTE32	214.	.67	53.	53.	53.	.16
HYDROGRAPH AT	H2	54.	.62	12.	12.	12.	.02
2 COMBINED AT	COMBINED	263.	.65	65.	65.	65.	.19
2 COMBINED AT	COMBINED	839.	.67	205.	205.	205.	.28
ROUTED TO	ROUTE1	836.	.67	205.	205.	205.	.28
HYDROGRAPH AT	K	136.	.62	31.	31.	31.	.04
ROUTED TO	ROUTEK	134.	.65	31.	31.	31.	.04
HYDROGRAPH AT	G3	26.	.62	6.	6.	6.	.01
2 COMBINED AT	COMBINED	159.	.65	37.	37.	37.	.05
2 COMBINED AT	COMBINED	991.	.67	242.	242.	242.	.33
HYDROGRAPH AT	G1	61.	.62	14.	14.	14.	.02
ROUTED TO	ROUTE1	59.	.65	14.	14.	14.	.02
HYDROGRAPH AT	G2	59.	.62	13.	13.	13.	.02
2 COMBINED AT	COMBINED	115.	.63	28.	28.	28.	.04
2 COMBINED AT	COMBINED	1102.	.67	269.	269.	269.	.37
HYDROGRAPH AT	15S	229.	.62	53.	53.	53.	.07

LINE ID.....1.....2.....3.....4.....5.....6.....7.....8.....9.....10

*** FREE ***

4

1	ID	SCOTTSDALE'S PERIMETER CENTRE							
2	ID								
3	ID	POST DEVELOPMENT DRAINAGE							
4	ID	100 YEAR - 1 HOUR							
5	ID								
6	ID	FILE NAME:PERMS							
7	ID								
	ID	DIAGRAM							
8	IT	1	20DEC00	1200	100				
9	IO	5							
10	EX	01							
11	BA	0.0133							
12	PH		0.72	1.41	2.47				
13	LS		83	85					
14	UD	0.10							
15	KE	ROUTD1							
16	RE	400	0.012	0.027	TRAP	8	4		
17	EX	01							
18	BA	0.0088							
19	RECOMBINED								
20	HC	2							
21	EX	02							
22	BA	0.0147							
23	EX	ROUTD2							
24	RE	400	0.010	0.027	TRAP	8	4		
25	EX	02							
26	BA	0.0120							
27	RECOMBINED								
28	HC	2							
29	RECOMBINED								
30	HC	2							
31	EX	ROUTD1							
32	RE	450	0.011	0.027	TRAP	8	4		
33	EX	01							
34	BA	0.0076							
35	RECOMBINED								
36	HC	2							

LINE	ID	1	2	3	4	5	6	7	8	9	10
37	KK	D3									
38	BA	0.0067									
39	KK	E									
40	BA	0.0108									
41	KKCOMBINED										
42	HC	2									
43	KK	ROUTE									
44	KK	700	0.009	0.027		TRAP	0		4		
45	KK	F2									
46	BA	0.0108									
47	KKCOMBINED										
48	HC	2									
49	KKCOMBINED										
50	HC	2									
51	KK	ROUTE2									
52	KK	700	0.007	0.027		TRAP	0		4		
53	KK	R2									
54	BA	0.0152									
55	KKCOMBINED										
56	HC	2									
57	KK	E1									
58	BA	0.0242									
59	KKCOMBINED										
60	HC	2									
61	KK	A1									
62	BA	0.0373									
63	KK	ROUTE1									
64	KK	350	0.008	0.027		TRAP	0		4		
65	KK	A2									
66	BA	0.0108									
67	KKCOMBINED										
68	HC	2									
69	KK	ROUTE2									
70	KK	400	0.0045	0.027		TRAP	0		4		

LINE ID.....1.....2.....3.....4.....5.....6.....7.....8.....9.....10

105 UNCOMBINED

106 HC 2

107 RECOMBINED

108	90	2
-----	----	---

109 PM ROUTINE

110	RE	300	0.004	0.027	TRAP	16	4
-----	----	-----	-------	-------	------	----	---

111 112 113

111	FA	0.0431
112	FA	0.0431

113 KE ROUTE

114	RE	680	0.0036	0.027	TRAP	8	4
-----	----	-----	--------	-------	------	---	---

115 EK 63

115	KA	0.0037
116	EA	0.0037

117 EXCISE UNIT

118

119 RECORDED

120 4E 2

121 3 61

122 3A 0.0192

123 11 ROUTE

124	SE	870	0.004	0.027	TRAP	8
-----	----	-----	-------	-------	------	---

125 KF 64

126	5A	0.0104
-----	----	--------

127 ECONOMIC

128 HC

129 RECONCILE

150 HC

14 15 16

152 PA 0.072

میت	3
میت	3
میت	3

SCHEMATIC DIAGRAM OF STREAM NETWORK

INPUT LINE	(V) ROUTING	(---) DIVERSION OR PUMP FLOW
NO.	(.) CONNECTOR	(---) RETURN OF DIVERTED OR PUMPED FLOW
10	D1	
	V	
	V	
15	ROUTD1	
	.	
17	.	G1
	.	.
	.	.
19	COMBINED.....	
	.	
21	.	D2
	.	V
	.	V
23	ROUTD2	
	.	.
25	.	G2
	.	.
	.	.
27	COMBINED.....	
	.	
	.	
29	COMBINED.....	
	V	
	V	
31	ROUTC1	
	.	
33	.	F1
	.	.
	.	.
35	COMBINED.....	
	.	
37	.	D3
	.	.
39	.	E
	.	.
	.	.
41	COMBINED.....	
	V	
	V	
43	ROUTE	
	.	.
45	.	F2
	.	.
	.	.
47	COMBINED.....	
	.	
	.	
49	COMBINED.....	
	V	
	V	

53	.	B2	.
	.	.	.
55	COMBINED.....		.
	.		.
57	.	B1	.
	.	.	.
59	COMBINED.....		.
	.		.
61	.	A1	.
	.	V	.
	.	V	.
63	ROUTA1		.
	.		.
65	.	A2	.
	.	.	.
67	COMBINED.....		.
	.	V	.
	.	V	.
69	ROUTA2		.
	.		.
71	COMBINED.....		.
	.	V	.
	.	V	.
73	ROUTB1		.
	.		.
75	.	H1	.
	.	.	.
77	COMBINED.....		.
	.		.
79	.	I2	.
	.	.	.
81	.	I1	.
	.	.	.
83	COMBINED.....		.
	.	V	.
	.	V	.
85	ROUTI2		.
	.		.
87	.	J1	.
	.	.	.
89	COMBINED.....		.
	.		.
91	.	I3	.
	.	V	.
	.	V	.
93	ROUTI3		.
	.	.	.
	.	.	.

97	.	.	COMBINED.....
	.	.	.
99	.	COMBINED.....	
	.	V	
	.	V	
101	.	ROUTJ2	
	.	.	
103	.	.	H2
	.	.	.
105	.	COMBINED.....	
	.	.	
107	COMBINED.....	.	
	.	V	
	.	V	
109	ROUTH1	.	
	.	.	
111	.	K	
	.	V	
	.	V	
113	ROUTY	.	
	.	.	
115	.	.	G3
	.	.	.
117	COMBINED.....	.	
	.	.	
119	COMBINED.....	.	
	.	.	
121	.	G1	
	.	V	
	.	V	
123	ROUTG1	.	
	.	.	
125	.	.	G2
	.	.	.
127	COMBINED.....	.	
	.	.	
129	COMBINED.....	.	
	.	.	
131	.	138	

(***) RUNOFF ALSO COMPUTED AT THIS LOCATION

[illegible]

RUNOFF SUMMARY
 FLOW IN CUBIC FEET PER SECOND
 TIME IN HOURS, AREA IN SQUARE MILES

OPERATION	STATION	PEAK FLOW	TIME OF PEAK	AVERAGE FLOW FOR MAXIMUM PERIOD			BASIN AREA	MAXIMUM STAGE	TIME OF MAX STAGE
HYDROGRAPH AT	D1	51.	.62	12.	12.	12.	.01		
ROUTED TO	ROUTE1	51.	.65	12.	12.	12.	.01		
HYDROGRAPH AT	C1	34.	.62	8.	8.	8.	.01		
2 COMBINED AT	COMBINED	84.	.63	19.	19.	19.	.02		
HYDROGRAPH AT	D2	57.	.62	13.	13.	13.	.01		
ROUTED TO	ROUTE2	56.	.63	13.	13.	13.	.01		
HYDROGRAPH AT	C2	46.	.62	11.	11.	11.	.01		
2 COMBINED AT	COMBINED	102.	.63	24.	24.	24.	.03		
2 COMBINED AT	COMBINED	106.	.63	43.	43.	43.	.05		
ROUTED TO	ROUTE1	105.	.63	43.	43.	43.	.05		
HYDROGRAPH AT	F1	11.	.62	5.	5.	5.	.00		
2 COMBINED AT	COMBINED	108.	.63	46.	46.	46.	.05		
HYDROGRAPH AT	D7	16.	.62	6.	6.	6.	.01		
HYDROGRAPH AT	E	12.	.62	10.	10.	10.	.11		
2 COMBINED AT	COMBINED	68.	.62	15.	15.	15.	.02		
ROUTED TO	ROUTE	66.	.63	15.	15.	15.	.02		
HYDROGRAPH AT	F2	42.	.62	10.	10.	10.	.01		
2 COMBINED AT	COMBINED	107.	.63	25.	25.	25.	.03		
2 COMBINED AT	COMBINED	206.	.63	71.	71.	71.	.08		
ROUTED TO	ROUTE2	204.	.63	71.	71.	71.	.08		
HYDROGRAPH AT	G2	19.	.62	13.	13.	13.	.02		
2 COMBINED AT	COMBINED	139.	.63	31.	31.	31.	.10		
HYDROGRAPH AT	G1	91.	.62	21.	21.	21.	.02		
2 COMBINED AT	COMBINED	148.	.63	106.	106.	106.	.12		
HYDROGRAPH AT	A1	144.	.62	70.	33.	33.	.04		
ROUTED TO	ROUTE1	143.	.63	33.	77.	33.	.04		
HYDROGRAPH AT	A2	42.	.62	10.	10.	10.	.01		
2 COMBINED AT	COMBINED	184.	.62	42.	42.	42.	.05		

2 COMBINED AT	COMBINED	632.	.63	140.	140.	140.	.19
ROUTED TO	ROUTE1	617.	.62	140.	140.	140.	.17
HYDROGRAPH AT	H1	99.	.62	22.	22.	22.	.03
2 COMBINED AT	COMBINED	704.	.67	170.	170.	170.	.19
HYDROGRAPH AT	I2	82.	.62	17.	19.	19.	.02
HYDROGRAPH AT	I1	13.	.62	3.	3.	3.	.00
2 COMBINED AT	COMBINED	95.	.62	22.	22.	22.	.02
ROUTED TO	ROUTE2	94.	.63	22.	22.	22.	.02
HYDROGRAPH AT	J1	26.	.62	6.	6.	6.	.01
2 COMBINED AT	COMBINED	119.	.63	28.	28.	28.	.03
HYDROGRAPH AT	I5	53.	.62	12.	12.	12.	.01
ROUTED TO	ROUTE3	48.	.68	12.	12.	12.	.01
HYDROGRAPH AT	J2	108.	.62	25.	25.	25.	.03
2 COMBINED AT	COMBINED	149.	.63	37.	37.	37.	.04
2 COMBINED AT	COMBINED	268.	.63	64.	64.	64.	.10
ROUTED TO	ROUTE2	263.	.67	64.	64.	64.	.10
HYDROGRAPH AT	K2	86.	.62	15.	15.	15.	.02
2 COMBINED AT	COMBINED	325.	.65	79.	79.	79.	.09
2 COMBINED AT	COMBINED	1026.	.67	249.	249.	249.	.128
ROUTED TO	ROUTE1	1024.	.67	249.	249.	249.	.128
HYDROGRAPH AT	F	167.	.62	38.	38.	38.	.04
ROUTED TO	ROUTE1	163.	.65	38.	38.	38.	.04
HYDROGRAPH AT	G5	32.	.62	7.	7.	7.	.01
2 COMBINED AT	COMBINED	194.	.63	45.	45.	45.	.05
2 COMBINED AT	COMBINED	1212.	.67	294.	294.	294.	.133
HYDROGRAPH AT	G1	75.	.62	17.	17.	17.	.02
ROUTED TO	ROUTE1	73.	.65	17.	17.	17.	.02
HYDROGRAPH AT	G2	72.	.62	16.	16.	16.	.02
2 COMBINED AT	COMBINED	142.	.63	34.	34.	34.	.04
2 COMBINED AT	COMBINED	1348.	.67	328.	328.	328.	.137
HYDROGRAPH AT	I5S	281.	.62	64.	64.	64.	.07

APPENDIX I - PEAK DISCHARGE ASSUMPTIONS, HEC-1 INPUT DATA, AND OUTPUT SUMMARIES

1. Per the SCS Aguila-Carefree Soil Survey, April, 1986, the existing soils are classified in the "B" Hydrologic Soil Group.
2. The existing vegetative cover density is 20%. This is consistent with previous drainage reports by CWW and others.
3. Per Figure 2-15, COS Section 2 Design Manual, a "B" soil and 20% cover yields a runoff curve number of 83.
4. During a telephone discussion with Bill Erickson, COS, on 12/22/87, it was agreed upon to calculate post peak discharges based on additional impervious area, rather than adjusting (increasing) the runoff curve number. This approach is consistent with new State and Federal guidelines. The percent of impervious areas with development, both industrial and commercial, is assumed to be 85.
5. Due to the small drainage parcels, all on-site parcels were assumed to have a time of concentration of 10 minutes. Although several of the parcels would have a time of concentration less than 10 minutes, by using either Kirpich's formula or a time of travel relationship in parking areas and/or street, a 10 minute minimum was assumed. This is consistent with ADOT drainage procedures.

This is an conservative assumption, as the developer will impose sediment control on each development site, thereby increasing the time of concentration.

6. The individual drainage area boundaries are shown on Exhibit 1. The post development sub-basin drainage areas, within the individual parcels, were depicted to conform to the existing topography and landform as much as possible.
7. The 100 year, 1 hour precipitation value of 2.46 inches was obtained from Fig. 2-6, COS Section 2 Design Manual. The 5 minute and 15 minute values, 0.72 inches and 1.41 inches, respectively, were obtained by multiplying the 1 hour value of 0.29 and 0.57. This follows ADOT guidelines, per the "Hydrologic Design for Highway Drainage, Arizona". Likewise, the 2 year, 1 hour value was 0.99 inches, the 10 year, 1 hour value was 1.62 inches, and the 25 year, 1 hour value was 2.05 inches.
8. Channel routing was utilized by the Kinematic Wave channel routing method.

APPENDIX II CHANNEL CALCULATIONS

CHANNEL	SIDE SLOPE H:V	BOTTOM WIDTH (ft)	DEPTH (ft)	SLOPE (%)	VELOCITY (fps)	FLOW DEPTH	DESIGN Q (cfs)
A	1:1	10	4.4	0.80	15.6	3.4	722
B	1:1	10	4.8	1.18	19.8	3.8	1022
C	1:1	8	4.8	1.18	19.8	3.8	1022
D	4:1	8	2.1	0.75	4.0	1.1	51
E	4:1	8	2.0	1.00	4.6	1.0	57
F	4:1	8	2.4	1.00	5.4	1.4	102
G	4:1	8	2.2	1.21	5.5	1.2	84
H	4:1	8	2.9	1.08	6.7	1.9	199
I	4:1	8	2.5	0.90	5.3	1.5	107
J	4:1	8	3.8	0.69	6.6	2.8	359
K	4:1	8	2.4	0.80	4.9	1.4	94
L	4:1	8	3.3	0.45	4.7	2.3	184
M	4:1	8	1.4	1.20	3.1	0.4	13
N	4:1	8	2.3	0.90	4.9	1.3	82
O	4:1	8	2.1	0.58	3.7	1.1	53
P	4:1	8	3.1	0.4-1.0	4.3-6.0	2.1-1.7	149
Q	4:1	8	2.5	1.13	5.9	1.5	119
R	4:1	8	4.1	0.40	5.3	3.1	325
S	4:1	12	5.0	0.40	6.4	4.0	704
T	4:1	8	3.2	0.40	4.4	2.2	167
U	4:1	8	3.5	0.36	4.4	2.5	194
V	4:1	16	5.4	0.40	7.0	4.4	1026
W	4:1	8	3.1	0.35	4.0	2.1	142
X	4:1	8	2.8	0.8-1.2	5.5-6.3	1.8-1.6	144
Y	4:1	8	2.1	1.26	5.4	1.1	75
Z.1	3:1	24	6.0	0.30	6.9	5.0	1348
Z.2	8:1	24	4.5	0.30	5.2	3.5	925
Z.DIVER	3:1	24	3.4	0.50	5.9	2.4	435

1/3

ARIZONA DEPARTMENT OF TRANSPORTATION

CULVERT EXTENSION COMPUTATION RECORD

PROJECT: Redivider Center DESIGNER: 11/2
 STATION: _____ DATE: 12-28-88

HYDROLOGIC AND CHANNEL INFORMATION

DRAINAGE AREA: _____ $D_1 =$ _____ $TW_1 =$ _____
 $D_2 =$ _____ $TW_2 =$ _____

(Q_1 = DESIGN DISCHARGE, SAY Q_{25})
 (Q_2 = CHECK DISCHARGE, SAY Q_{50})

$S_{O1} =$ _____ $L_1 =$ _____
 $S_{OEXIST.} =$ _____ $L_{EXIST.} =$ _____
 $S_{O2} =$ _____ $L_2 =$ _____

MEAN STREAM VELOCITY = _____

SKETCH

CULVERT DESCRIPTION		Q	CAP. CHART HW	HEADWATER COMPUTATION												CONTROL HW	OUTLET VELOCITY	COST	DHW ELEV.	COMMENTS
				INLET CONT.		OUTLET CONTROL						HW = H + h _o - L ₅₀								
				HW D	HW	K _o	H	d _c	$\frac{d_c + O}{2}$	TW	H _o	L ₅₀	HW							
SIZE	ENTR. TYPE																			
QUAD CEC	10 x 4	1348	337 EA. 7/8 = 52.5	1.38	5.5															
QUAD CEC	10 x 4	632	158 EA. 7/8 = 15.8	0.76	3.0															
CEC	2 x 3	184	7/8 = 23	1.45	4.4															
QUAD CEC	6 x 3	306	15 EA. 7/8 = 55.5	1.60	4.8															
CEC	8 x 3	126	7/8 = 23	1.50	4.5															
RCP	30"	51	—	1.98	5.0															
RCP	30"	57	—	2.17	5.4															

SUMMARY AND RECOMMENDATIONS

* C.B.C's have 450 wingwalls and rounded ends ($K_e = 0.2$)

* RCP's have headwalls and rounded or grooved ends ($K_e = 0.2$)

2/3

ARIZONA DEPARTMENT OF TRANSPORTATION

CULVERT EXTENSION COMPUTATION RECORD

PROJECT: PRIMAVER CENTER DESIGNER: J. L. SKETCH

STATION: _____ DATE: 10-28-88

HYDROLOGIC AND CHANNEL INFORMATION

ORAINAGE AREA: _____ $Q_1 =$ _____ $TW_1 =$ _____

$Q_2 =$ _____ $TW_2 =$ _____

(Q_1 = DESIGN DISCHARGE, SAY Q_{25})
 (Q_2 = CHECK DISCHARGE, SAY Q_{50})

SO₁ = _____ L₁ = _____

SO_{EXIST.} = _____ L_{EXIST.} = _____

SO₂ = _____ L₂ = _____

MEAN STREAM VELOCITY = _____

EL. _____ SHLOR. _____

EL. _____ L₂ _____

EL. _____ L_{EXIST.} _____

EL. _____ L₁ _____

EL. _____ TW = _____

CULVERT DESCRIPTION		Q	CAP. CHART HW	HEADWATER COMPUTATION										CONTROL HW	OUTLET VELOCITY	COST	DHW ELEV.	COMMENTS
				INLET CONT.		OUTLET CONTROL						HW = H + h ₀ + L _{SO}						
				HW Q	HW	L₀	H	d _c	$\frac{d_c + 0}{2}$	TW	H ₀	L _{SO}	HW					
12	10 x 4	1026	348 EA 9/8 = 362	1.4	5.6													
	CBC	268	9/8 = 268	1.72	5.2													
	RCP	53	-	2.15	5.4													
	RCP	95	478 EA	1.9	4.8													
	CBC	167	9/8 = 272	1.72	5.3													

SUMMARY AND RECOMMENDATIONS

* CBC's have 45° wingwalls and rounded edges ($K_r = 0.2$)

* RCP's have headwalls and rounded or grooved ends ($K_r = 0.2$)

ARIZONA DEPARTMENT OF TRANSPORTATION

CULVERT EXTENSION COMPUTATION RECORD

PROJECT: Perimeter Center

DESIGNER: J.A.

STATION: _____

DATE: 12-28-88

HYDROLOGIC AND CHANNEL INFORMATION

DRAINAGE AREA: _____

$Q_1 =$ _____

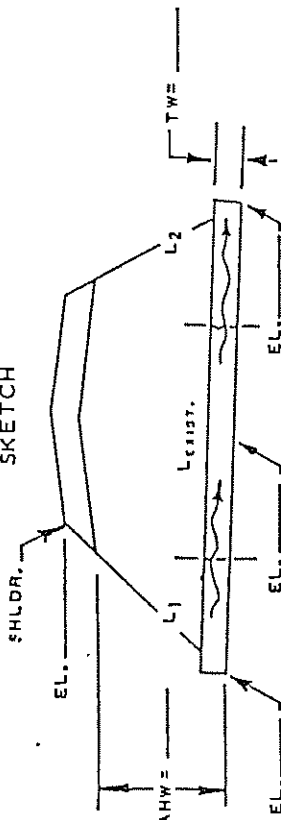
$TW_1 =$ _____

$Q_2 =$ _____

$TW_2 =$ _____

(Q_1 = DESIGN DISCHARGE, SAY Q_{25}
 Q_2 = CHECK DISCHARGE, SAY Q_{50})

SKETCH



$S_{O1} =$ _____ $L_1 =$ _____
 $S_{OEXIST.} =$ _____ $L_{EXIST.} =$ _____
 $S_{O2} =$ _____ $L_2 =$ _____

MEAN STREAM VELOCITY = _____
 $L/100 S_{O2} =$ _____

HEADWATER COMPUTATION

CULVERT DESCRIPTION	Q	CAP. CHART HW	INLET CONT.					OUTLET CONTROL					HW = H + h ₀ - L _{S0}				CONTROL HW	OUTLET VELOCITY	COST	OHK ELEV.	COMMENTS
			HW	HW	K ₀	H	d _c	d _c + 0 2	TW	H ₀	L _{S0}	HW									
14 CBC 10x5	818	BASED	ON	OPEN	CHANNEL	FLOW															
15 CBC 10x5	1022	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
16 CBC 10x5	1022	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"

SUMMARY AND RECOMMENDATIONS

COLLAR, WILLIAMS & WHITE ENGINEERING

2702 NORTH 44th STREET, SUITE 100A, PHOENIX, ARIZONA 85008

(602) 957-3350

January 24, 1989

Mr. Bill Erickson
Drainage Planner
City of Scottsdale
7447 East Indian School Road
Scottsdale, Arizona 85251

Subject: Scottsdale Perimeter Center
Downstream Drainage from Bell Road
C.W.W. Job No. 870514

Dear Mr. Erickson:

Our original conceptual plan that was submitted on December 9, 1989 has been revised to accommodate the landowners to the south and address Mr. Tom Beat's, (Special Projects Manager for the City of Scottsdale) comments to you in his letter of December 15, 1988.

The proposed channel would begin at the outlet of a 4 barrel 10'x4' R.C.B. at Bell Road. The grass lined channel would have 3:1 side slopes and a 24' wide bottom. A 3' wide concrete valley gutter would be centered in the channel for low flows. The channel slope would be 0.30% and an approximate velocity of 6.98 fps. The grass slopes and bottom would be additionally protected by using Miramat ECRM or Enkamat 7020.

As the flows enter the golf course the slopes could be laid back to accommodate the playability of the course. We have shown side slopes of 8:1 in our channel run for this section. Mr. Beat indicated his desire of D.G. and large rocks to create a natural wash. While this solution is aesthetically pleasing, engineering design could not accommodate this. The flows (1348 cfs - $Q_{1.00}$) and a velocity of approximately 5.7 fps ($Q_{1.00}$) would carry the D.G. downstream. If the natural wash theme is to be constructed, we would propose that the golf course design this portion and develop a maintenance plan.

The flow would then be directed to a stilling pond to allow the water to sheet flow over a 200' long weir. After crossing the weir, the flow would enter an existing channel. A pipe would provide positive drainage from the stilling pond.

Mr. Bill Erickson
January 24, 1989
Page Two

The Westcor Company II, as developers, are processing an agreement with Joyce Corrigan, C & H Development and the City of Scottsdale for the construction and maintenance of this contract.

If you have any further questions concerning this subject, please contact this office.

COLLAR, WILLIAMS & WHITE ENGINEERING



David Coble
Project Designer

DC:dl

Attachment
Exhibit

cc: Mr. David Scholl - The Westcor Company II
Mr. Bob Mohning - CWW
letters4/dl

NORMAL DEPTH FOR TRAPEZOIDAL CHANNELS
CHANNEL 2.1

DISCHARGE IS = 1348 CFS
BOTTOM WIDTH IS = 24 FT
SLOPE IS = .003 FT/FT
SIDE SLOPE (Z1) = 3
SIDE SLOPE (Z2) = 3
MANNING'S N = .027
TOP WIDTH IS = 53.9 FT
AREA IS = 194.13 SQ FT
WETTED PERIMETER IS = 55.52 FT
HYDRAULIC RADIUS IS = 3.5 FT
NORMAL DEPTH IS = 4.98 FT
VELOCITY IS = 6.94 FPS
FROUDE NUMBER IS = .64

NORMAL DEPTH FOR TRAPEZOIDAL CHANNELS
CHANNEL 2.2

DISCHARGE IS = 1348 CFS
BOTTOM WIDTH IS = 24 FT
SLOPE IS = .003 FT/FT
SIDE SLOPE (Z1) = 8
SIDE SLOPE (Z2) = 8
MANNING'S N = .027
TOP WIDTH IS = 90.2 FT
AREA IS = 236.25 SQ FT
WETTED PERIMETER IS = 90.72 FT
HYDRAULIC RADIUS IS = 2.6 FT
NORMAL DEPTH IS = 4.14 FT
VELOCITY IS = 5.71 FPS
FROUDE NUMBER IS = .62

NORMAL DEPTH FOR TRAPEZOIDAL CHANNELS
CHANNEL 1

DISCHARGE IS = 1348 CFS
BOTTOM WIDTH IS = 55 FT
SLOPE IS = .0167 FT/FT
SIDE SLOPE (Z1) = 8
SIDE SLOPE (Z2) = 8
MANNING'S N = .027
TOP WIDTH IS = 86.37 FT
AREA IS = 138.56 SQ FT
WETTED PERIMETER IS = 86.61 FT
HYDRAULIC RADIUS IS = 1.6 FT
NORMAL DEPTH IS = 1.96 FT
VELOCITY IS = 9.73 FPS
FROUDE NUMBER IS = 1.35

STILLING BASIN

WEIR LENGTH 200'

Q DESIGN = 1348 cfs
Weir Coefficient = 2.7

$$Q = 2.7 L H^{2/3}$$
$$h = \left(\frac{Q}{2.7L} \right)^{2/3}$$

$$H = 1.84'$$

Appendix F

Conditions, Covenants and Restrictions with Design Guidelines for The Perimeter Center Scottsdale

Conditions, Covenants and Restrictions
with
Design Guidelines
for



THE
PERIMETER
CENTER

Scottsdale

17207 N Perimeter Dr
Scottsdale, AZ 85255

800 528-1179 • 480 585-4500
Fax 480 585-2232

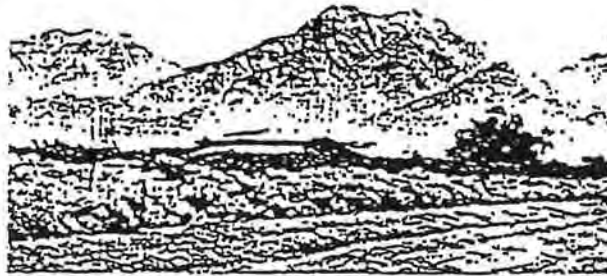


TABLE OF CONTENTS

INTRODUCTION	1
DESIGN OBJECTIVES	2
SITE DEVELOPMENT STANDARDS	4
Site Grading	
Setbacks and Landscape Easements	
Parking, Driveways and Sidewalks	
Loading and Service Areas	
Outside Storage and Equipment	
Plazas and Courtyards	
Site Furniture	
Site Walls	
Waste and Refuse	
Site Lighting	
Site Utilities	
Construction Phasing	
Public Art	
Landscape	
Plant Palette	
Energy Considerations	
Site Management	
Water Conservation	
ARCHITECTURAL STANDARDS	17
Site Development	
Massing and Scale	
Materials	
Building Details	
Energy Considerations	
SIGNAGE STANDARDS	20
Identification Signs	
Directional Signs	
Temporary Signs	
Design and Configuration	
Color	
Lettering Style	
Material	
Illumination	
Setbacks	
Setting	



DESIGN OBJECTIVES



Site Utilization should consider density and height relative to adjacent development. Other considerations should include views, a sense of arrival, service areas, pedestrian access and indoor/outdoor relationships. The impact of the automobile should be minimized.



The Architectural Vocabulary should be sensitive to the context of immediately adjacent development and the desert region relative to forms, materials, texture and colors. The use of glass should be minimized and the architecture should be unique to this site and its setting.



The Landscape Vocabulary should be harmonious with the existing streetscape and the arid desert landscape. The landscape should play a key role in the site aesthetics, the development of people spaces and energy and water conservation.



The Pedestrian Environment should be comfortable year round with places to sit, a shaded overhead plane, color, textures underfoot and fragrance. Pedestrian and vehicular conflicts should be minimized and effective lighting should foster a sense of security after dark.



Site Amenities and Special Features may include such elements as water features, thematic lighting and art. All such elements should be carefully designed within the context of the setting and the architecture. Key building components and details should be culled and incorporated into the site development thus minimizing a visual separation between site and building.



SITE DEVELOPEMENT STANDARDS

SITE GRADING

Set buildings and structures above flood limits.

Convey water to existing off-site drainage swales.

Small mounds should be avoided.

No retention is required.

Conveyance swales should be natural & riparian in character and aesthetically treated as an architectural feature.

Temporary berms may be removed.

Balance cut and fill.

Oil and grease traps and/or motor filters should be incorporated where feasible.

Grade planting areas and hardscape areas to take advantage of water harvesting potentials.

The following maximum gradients are recommended:

Landscape: 4:1

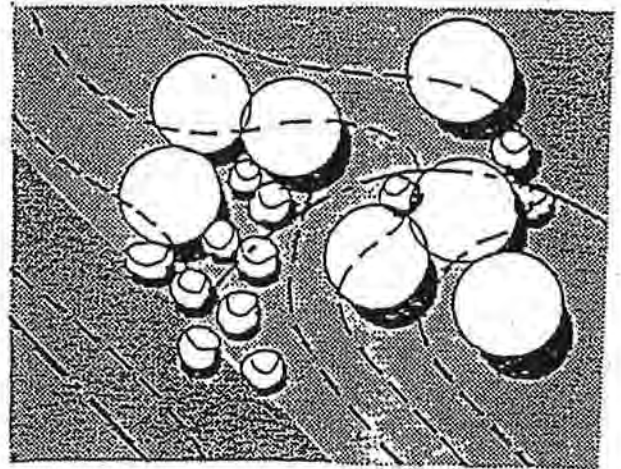
Drives: 3%

Walks: 3%

Plazas: 2%

The following berm heights are recommended:

3.5' maximum, to allow views.



Slope in landscaped areas:

Minimum slope: 2%

Maximum slope: (4:1)

Maximum slope in turf areas: 20% (5:1)

Pavement slopes:

Minimum slope: .5%

Maximum cross slope: 2%

Maximum longitudinal slope: 5% for pedestrian areas (limited application for handicapped access).

Parking slopes:

Minimum cross slope: .5%

Minimal concrete swale slope: .5%

Landscape swale slopes:

Minimum bottom and side slope: 1%

Maximum: 25% (4:1)

SETBACKS AND LANDSCAPE EASEMENTS

Landscape drainage improvements have been constructed as part of the infrastructure development. Such improvements have established an average setback of 40' throughout The Perimeter Center.

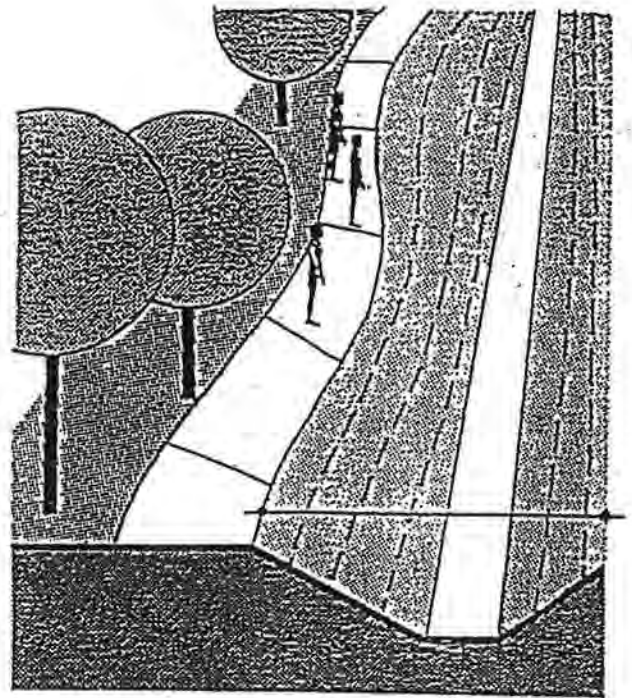
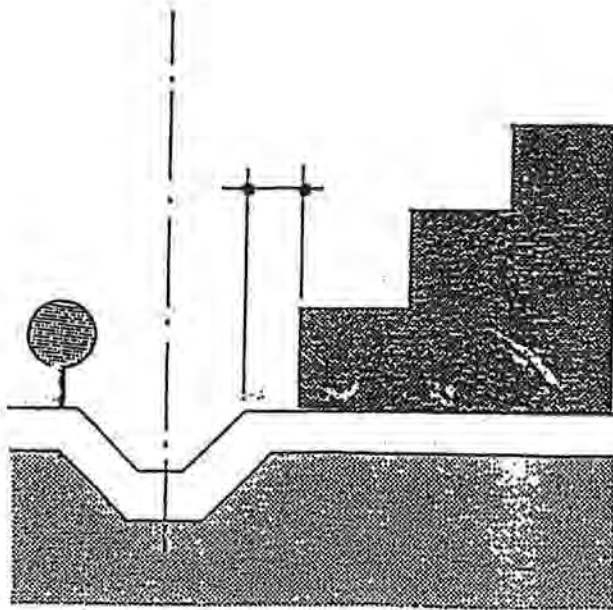
Setbacks

Between existing sodded drainage way and building or site improvements: 5 feet minimum (assumes removal of temporary berm).

Between existing streetscape and building or site improvements: Per City of Scottsdale Ordinance.

Side Yards and Rear Yards: Per City of Scottsdale Ordinance.

Landscape Area between Parking and Buildings: 10 feet minimum.



PARKING, DRIVEWAYS AND SIDEWALKS

Surface Parking

On street parking is not allowed

Paving should be either concrete, asphalt or stabilized decomposed granite (in limited areas)

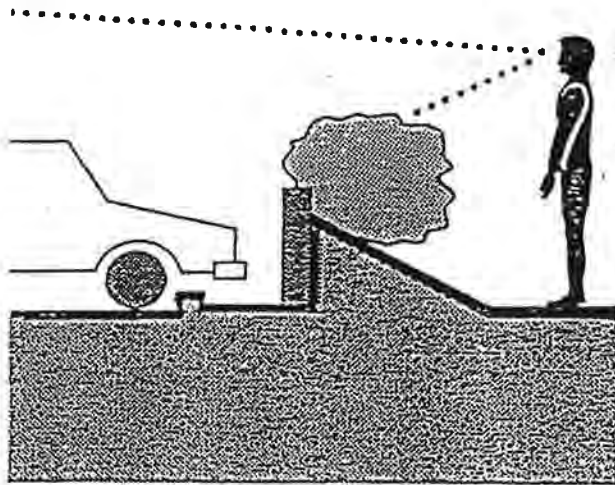
20% of the total parking area should be landscaped (area may include proposed perimeter landscape treatment)

Maximum distance from a parking space to a building entry should not exceed 500 feet

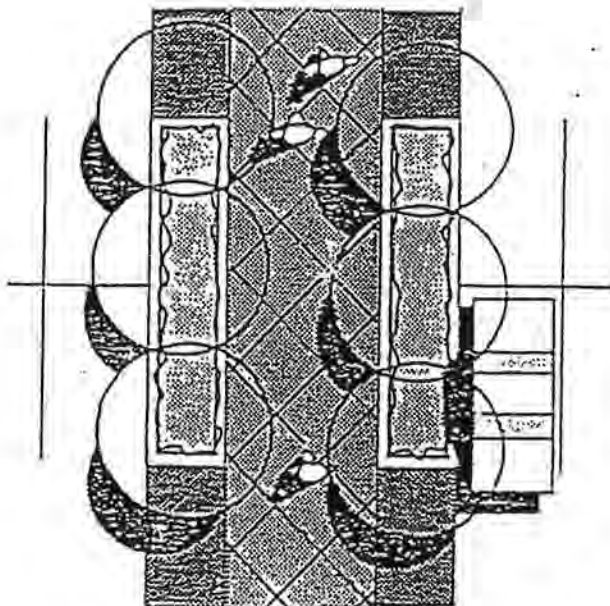
Parking stall and landscape island dimensions shall be per City of Scottsdale Ordinance

Parallel parking is discouraged.

Parking shall be screened with a combination of 3' high walls and berms. Walls shall be as described in the site wall section of these guidelines.



Parking "walks" should be developed to connect parking aisles with building entries, plazas or courtyards. When "walks" cannot be incorporated, the aisle layout should be conducive to efficient pedestrian circulation.



Visitor parking areas should be treated as parking "orchards" incorporating one canopy tree for every four spaces. The use of tree grates, tree guards, bollards and special paving is encouraged.

Parking canopies should incorporate a fascia and column surrounds which reflect the architectural character of the project.

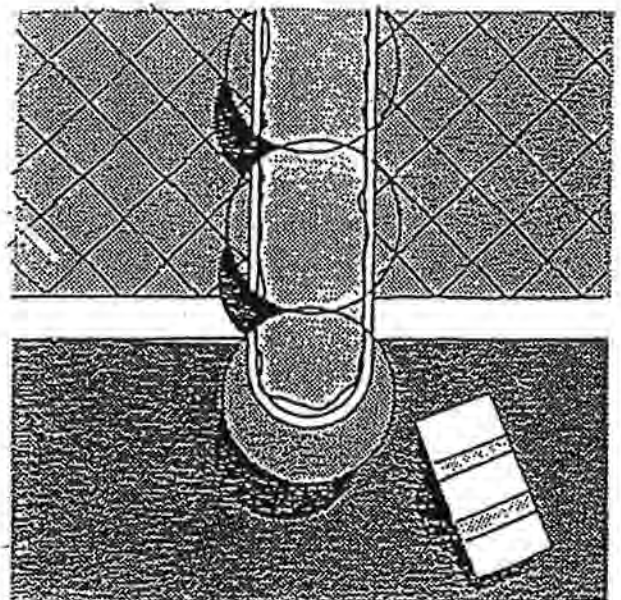
Driveways

Bridged driveway crossings shall match similar existing structures within the Center.

Adjacent sites should share driveways.

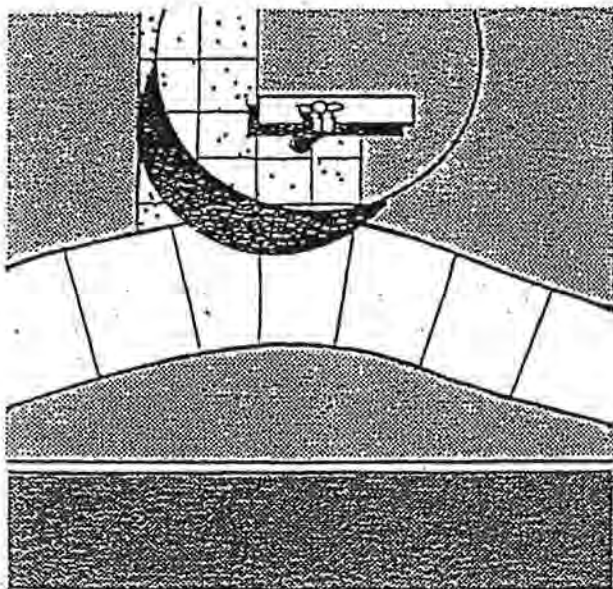
The use of geometrically patterned and textured concrete paving at driveways is encouraged. Natural materials such as brick and stone are also encouraged, while materials which imitate natural materials shall be avoided.

Landscaped medians at primary driveway entries are encouraged.



Sidewalks

Site sidewalks should connect to public sidewalks with minimal conflict with vehicular circulation. Horizontal connections should be as close to perpendicular as possible & should incorporate an oversized landing & seatwalls.



Site sidewalks should connect all site amenities, entry courts, building ingress/egress and parking areas.

Walks should be a minimum of five feet wide. The use of colored and/or textured paving is encouraged.

Walks should be partially shaded with either trees or architectural canopies.

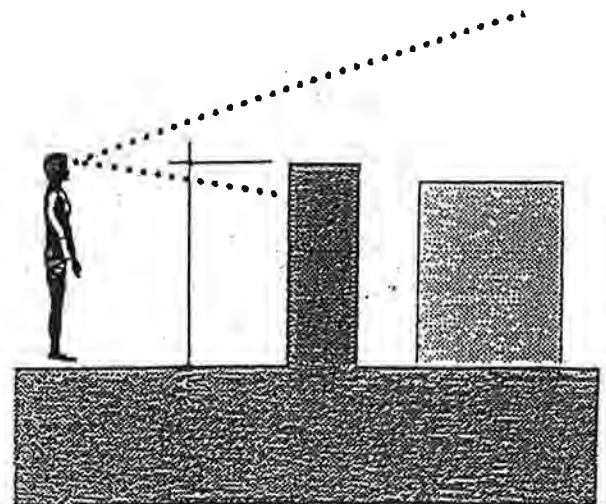
Walks should conform to current jurisdictional standards for handicapped access.

LOADING AND SERVICE AREAS

Loading and service areas should be located a minimum of 120' away from public streets and be screened with a site wall and opaquely gated.

Loading and service areas should be located away from view from upper floor windows.

Loading and service operations shall be accommodated within the confines of the specific site.



OUTSIDE STORAGE AND EQUIPMENT

Goods, materials, machinery, oversized vehicles or refuse shall be stored within the confines of six foot high site walls with opaque gates and located on the site away from public view and as far away from view from upper floor window view as possible.

Vehicles which are stored for 48 hours or more shall be stored in a screened enclosure away from view from public spaces, upper floor windows, public streets and sidewalks.

Methods of temporary storage of construction materials, supplies and associated construction waste necessary for permanent improvements within the site should be approved by the master developer.

PLAZAS AND COURTYARDS

The development of exterior people spaces is encouraged. The following list outlines recommended components and activities for these spaces:

Textured and patterned paving.

Water features.

Picnic and "brown bag" areas.

Sculpture gardens and strategically placed objects d'art.

Seasonally shaded area that allow the winter sun to penetrate through proper orientation and choice of overhead materials and elements.

Benches and seat walls.

Drinking fountains.

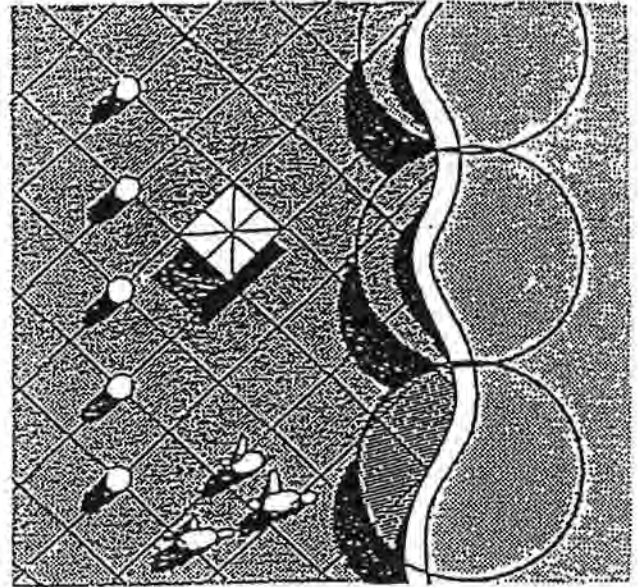
Waste receptacles.

Special lighting effects.

Cafe type chairs and other furniture amenities.

SITE FURNITURE

Site furniture includes flagpoles, entry monuments, fountains, benches, bollards, drinking fountains, waste receptacles and shade structures.



The forms and materials of the site furniture should be culled from the regional context of The Perimeter Center as well as from the site specific architectural theme.

The use of elements which are specifically designed for a particular site and building are greatly encouraged rather than "off-the-shelf" site elements.

SITE WALLS

The use of site walls to visually engage the buildings with the site is greatly encouraged. These walls should match the material and color of the building base.

Site walls which screen parking from public streets shall be 3 feet high and constructed of colored split face CMU or sand blasted CMU or an aesthetic combination thereof. The colors should complement The Perimeter Center entry monument walls.

The integration of walls and berms as well as the incorporation of upgraded materials and pilasters at site entries is encouraged.

SITE UTILITIES

All utility components including telephone pedestals, meters, backflow preventors, transformers, etc., shall be screened from public view with a combination of walls and plant materials.

Electrical panels should be mounted on the interior of buildings where possible. Where interior mounting is not possible, such equipment should be installed where it is out of the public view. In no case shall exterior electrical equipment be mounted on the street side or primary exposure side of any building.

Drain inlets should be located out of vehicular and pedestrian circulation patterns.

CONSTRUCTION PHASING

Undeveloped lots shall be maintained to be free of trash, debris and weeds.

Construction sites shall be maintained in a neat and orderly manner. All trash and construction debris shall be disposed of in enclosed containers and removed from the premises at regular intervals.

Existing improvements, plantings, desert trees and utilities shall be identified and protected during construction.

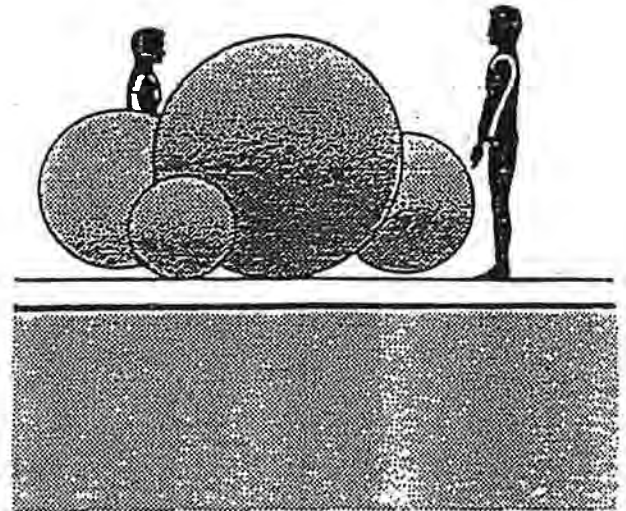
Adequate off-street parking for workers shall be provided during construction.

Adjacent sites, roadways and sidewalks shall be kept free of mud, soil, gravel and other debris during construction.

PUBLIC ART

The use of public art, primarily in exterior spaces is highly encouraged. The art may consist of sculpture, murals or water sculpture.

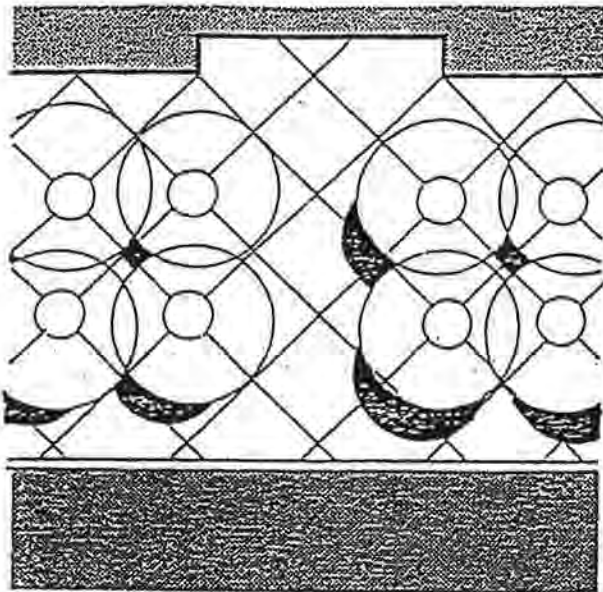
Sculpture should be of natural materials such as stone, bronze or copper. It should be sited in such a manner as to integrate with the geometry and patterns of the site and architectural design. The sculpture should invite participation. People should be able to walk up to it, touch it, walk through it or sit upon it. "Hands-off" pedestal placements should be avoided.



Murals should integrate with the site and/or building walls. Mural panels should conform to the geometric patterning of the wall. Mural images may be achieved by off-setting of masonry units, carving, sand blasting or tile inlays.

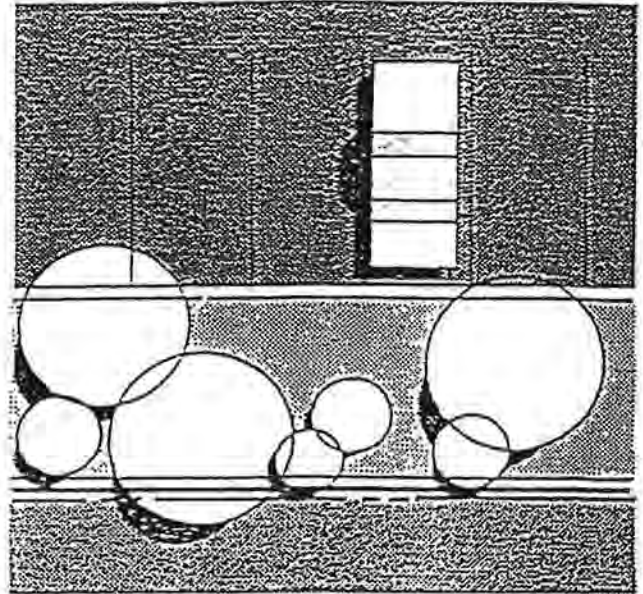
Murals and sculpture should be lit from concealed sources as located by the artist to best enhance the forms and textures of the piece.

Building entries, courtyards and plazas should be planted with closely spaced trees of unique characteristics such as form, flower or foliage. These trees should be underplanted with dense flowering and/or fragrant shrubs and ground covers, which are tolerant of the shady conditions created by the dense, overhead tree canopy. The use of pots, planted with flowering perennials or succulents is encouraged.

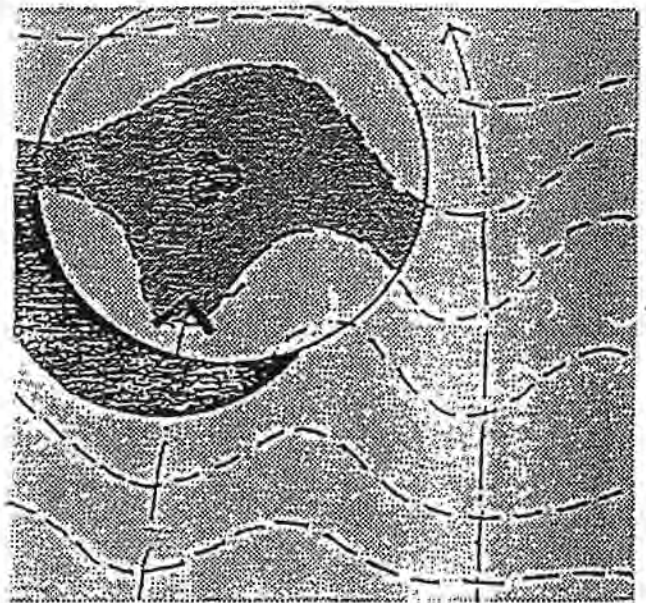


Site elements and spaces which are screened with walls should be screened with plant materials of approximate size and density.

Interior site boundaries should be treated with a combination of vertical trees and medium height shrubs which are randomly spaced to complement the six foot high property wall required under the ordinance. Trees should be spaced a minimum of ten feet and a maximum of thirty feet apart for an equivalent maximum spacing of twenty feet on center for the total lineal footage of the interior property boundaries.



Planting areas, particularly desert tree zones should be graded to capture surface run-off and thus decrease the dependency on potable water sources. Such sources shall be automatically controlled emitter systems, which are professionally designed and valved to separate exposures and plant types.





PLANT PALETTE

Trees

Acacia abyssinica
Acacia aneura
Acacia smallii
Acacia stenophylla
Cercidium floridum
Cercidium praecox
Chilopsis linearis
Prosopis species

Abyssinica Acacia
Mulga
Sweet Acacia
Shoestring Acacia
Blue Palo Verde
Palo Brea
Desert Willow
Mesquite

Shrubs

Acacia notabilis
Agave species
Aloe species
Ambrosia deltoidei
Baileya multiradiata
Baccharis sarothroides (male)
Calliandra californica
Calliandra eriophylla
Caesalpenia gilliesii
Caesalpenia pulcherrima
Ceanothus formosus
Dalea pulchra
Dasylirion aerotriche
Dasylirion wheeleri
Encelia farinosa
Hesperaloe parviflora
Justicia species
Larrea tridentata
Leucophyllum Species
Muhlenbergia cogens
Penstemon species
Ruellia peninsularis
Salvia species
Simmondsia chinensis
Tagetes lemmonii
Zauschneria californica latifolia

Acacia
Century Plant
Aloe
Bursage
Desert Marigold
Desert Broom
Baja Fairy Duster
Fairy Duster
Yellow Bird of Paradise
Red Bird of Paradise
Sturt Pea
Indigo Bush
Green Desert Spoon
Desert Spoon
Brittle Bush
Red Yucca
Mexican Honeysuckle
Creosote Bush
Texas Sage
Deer Grass
Penstemon
Ruellia
Sage
Jojoba
Mountain Marigold
Hummingbird Flower

Vines

Antigonon leptopus
Doxantha uguis-cati
Mascagnia species

Coral Wreath Vine
Cat Claw
Mascagnia

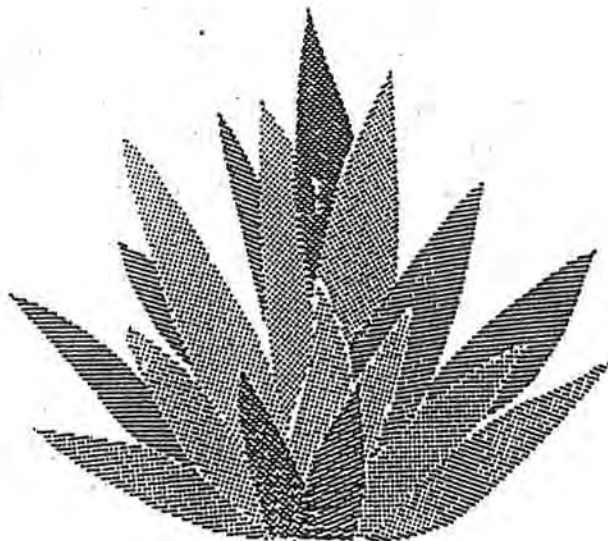
Ground Cover

Acacia r. 'Desert Carpet' tm
Baccharis "Centennial"
Dalea greggii
Lantana montevidensis
Oenothera species
Verbena pulchella

Trailing Acacia
Desert Broom
Trailing Indigo Bush
Trailing Purple Lantana
Evening Primrose
Desert Verbena

The use of turf is discouraged except as
a transition between the sodded drainage
ways and the site as previously described.

A native plant inventory and the reuse
of existing native trees on individual sites
is required by the City of Scottsdale.



SITE MANAGEMENT

Owners and occupants (including lessees) of any Lot in The Perimeter Center shall jointly have the duty and responsibility, at their sole cost and expense, to keep that part of the property so owned or occupied, including, but not limited to buildings, improvements, grounds and utilities easements, in a well maintained, safe, clean and attractive condition at all times. Such maintenance includes, but is not limited to, the following:

Prompt removal of all litter, trash, refuse and wastes.

Lawn mowing on a regular and frequent basis. The property shall be kept clean and neat in appearance at all times.

Provide 3" to 4" diameter by 6" lengths of black plastic drainage pipe as arbor guards at the base of all trees in lawn areas to prevent girdling by "line trimmers".

Tree and shrub pruning per sound horticultural practices and on a regular basis. Pruning should conform to the natural growth habit of the plant.

Watering by means of an automatic sprinkler system with supplemented hand watering as needed.

Keeping garden areas alive and any adjoining rights-of-way and drainage way edges attractive and free of weeds.

Removing and replacing any dead plant material in a timely manner.

Keeping vacant land well maintained to the curb line and the entire site free of trash.

Keeping exterior lighting and mechanical facilities in working order.

Keeping parking areas, driveways, and roads in good repair.

Complying with all governmental, health and police requirements.

Striping and resealing of parking areas and repainting of improvements on a regular schedule or as needed.

WATER CONSERVATION

Water usage should be consistent with the desert foothill setting of The Perimeter Center.

The exclusive use of arid region plants, especially Sonoran Desert materials, is highly encouraged. The use of lawn is allowed but should be sensitively incorporated in very limited areas as indicated in the landscape section of these guidelines.

Water features should respect the scarcity of water in the desert and should present a concept of minimal water with maximum effect.

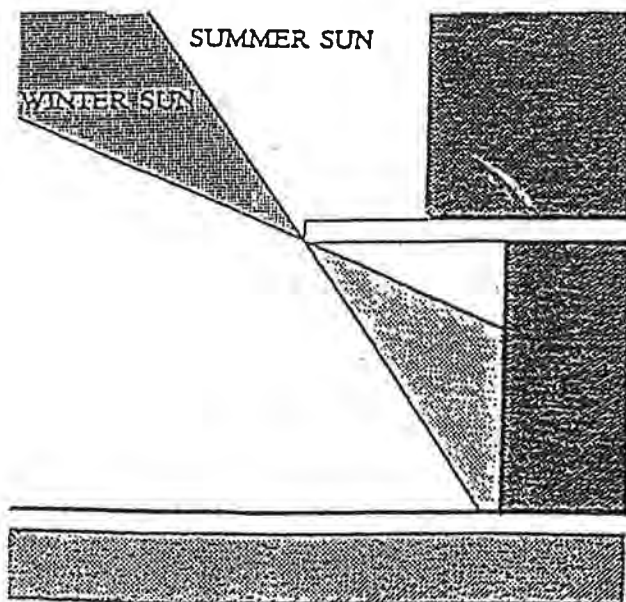
The concept of water harvesting is encouraged. Run-off from roofs and parking areas should be gathered and stored in underground cisterns until it is needed. It should be filtered and pumped into the site irrigation system. When water harvested is not feasible, planting areas should be graded to take maximum advantage of surface run-off to provide water for trees and shrubs.

ENERGY CONSIDERATIONS

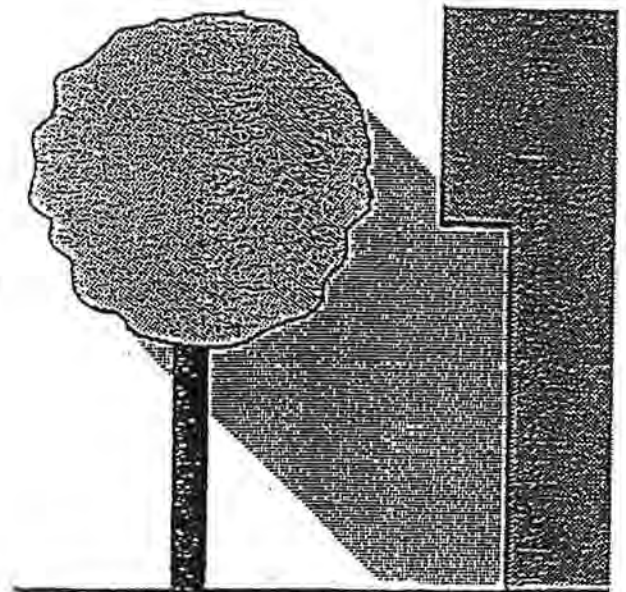
Three primary considerations address the common goal of energy conservation.

They are orientation, landscape/ground plane treatment and building treatment.

The optimum building orientation is north and south wherein vision glass can be located on both north and south elevations without direct summer solar penetration. The north side is always natural shaded except for very early and very late afternoon summer sun. The south side glass can be effectively shaded with overhangs or similar architectural devices from the summer sun while allowing the winter sun in, thus reducing the need for heating. East and west elevations should have a minimal amount of surface area exposed to the lower, more intense sun.



Landscape, particularly deciduous trees can help reduce summer solar penetration where architectural devices are not practical. They provide shading in the summer and allow the sun to enter in the winter. The ground plane immediately adjacent to buildings should be densely planted, and in cases where paving is required or desired, the paving should be shaded with a dense canopy of trees or vine covered trellises thus reducing reflected heat on the building surface.



The building surface treatment should be designed to complement building orientation and the location of site components such as parking and landscape zones. Glass which is not south facing should be shaded as much as possible with overhangs or louver elements which retain both distant and site views.



ARCHITECTURAL STANDARDS

SITE DEVELOPMENT

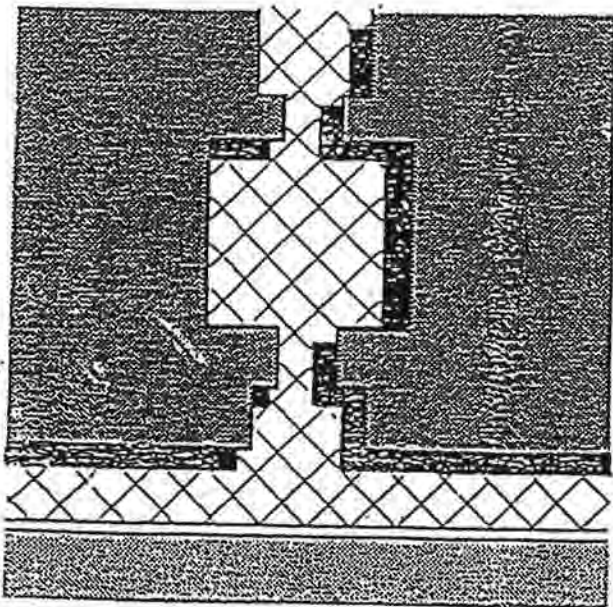
Site development should conform with the site development standards section of these guidelines.

Visually anchor the building to the site by gleaming forms, materials and colors, and incorporating them into the site development.

Create a balance between the size of the building foot print and open space.

Establish a foreground for the building from the street and from within the site proper.

Arrange buildings to create a hierarchy of spatial experiences and to define open spaces.



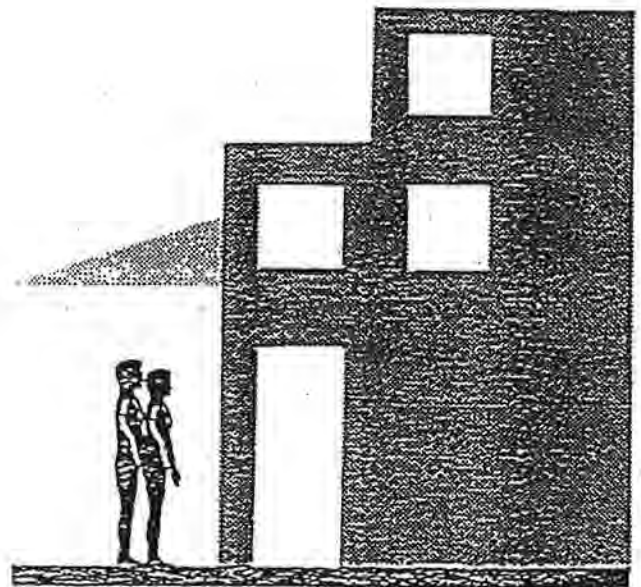
Locate buildings and open spaces with respect to solar aspect and mountain views.

Reflect the building geometry in the site ground plane development.

MASSING AND SCALE

Buildings should be clean, ordered and of a timeless expression. Trendy or vogue solutions should be avoided.

Buildings should reflect human scale and be sensitive to the scale of the streetscape and adjacent developments.



Floor area ratio (F.A.R.) should reflect what is appropriate for the building type, use and the setting as follows:

Corporate office buildings: .30 to .50
 R & D/Light Industrial: .35 to .60
 Hotels: .50 to 1.00
 Garden Office: .25 to .35
 Retail: .25 to .30

Building height and open space requirements shall conform to the City of Scottsdale Ordinance.

Building massing and roof forms should create a sense of order and be sensitive to the forms of the McDowell Mountains.

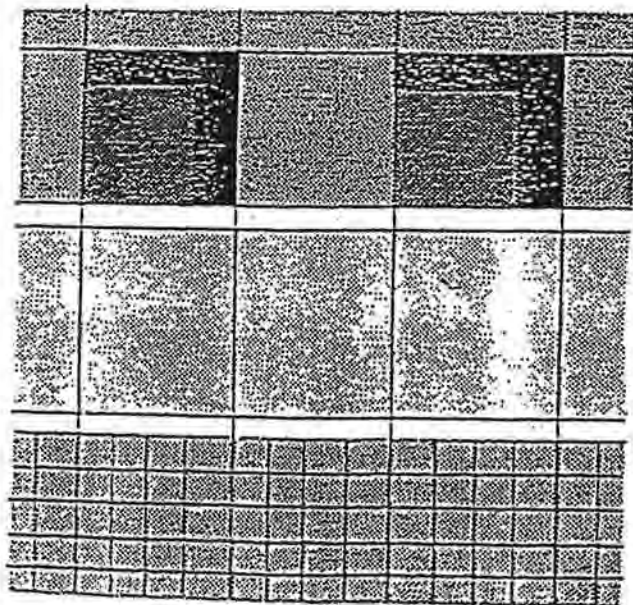
Massing should respect mountain views from adjacent building sites.

Rooftop equipment screening devices should integrate with the building massing and be constructed of like or complementary materials.

MATERIALS

Exterior materials should be of a high quality and reflect the desert setting. They should possess properties which allows them to age and weather while enhancing the aesthetic quality of the building.

Materials should include natural stone (sandstone, limestone, unpolished granite, polished granite, marble, etc.) precast concrete, cast in place concrete, brick, colored and textured CMU or smooth finish stucco. If stucco is used, a wainscot of a higher quality material should be incorporated for aesthetic and durability purposes.



Building elevations should be designed so that the use of glass should not constitute more than 60% of the total surface area of the combined elevations. Minimize the use of glass on east and west facing elevations unless adequate sun protection is provided. Avoid the use of highly reflective glass.

The use of natural patina copper with polished copper accents is encouraged for architectural metal work.

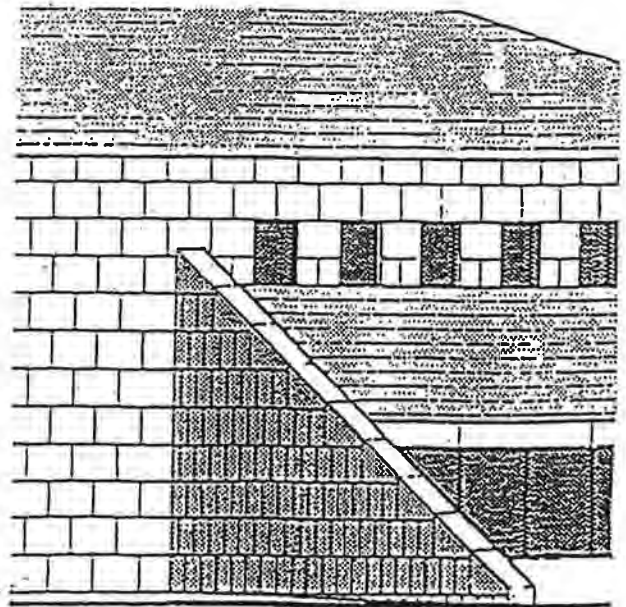
Exposed roofs should be metal, (with a factory applies finish), copper, (with a natural patina finish) or flat concrete tile.

Colors should reflect the desert environment including the desert floor, mountains and plants.

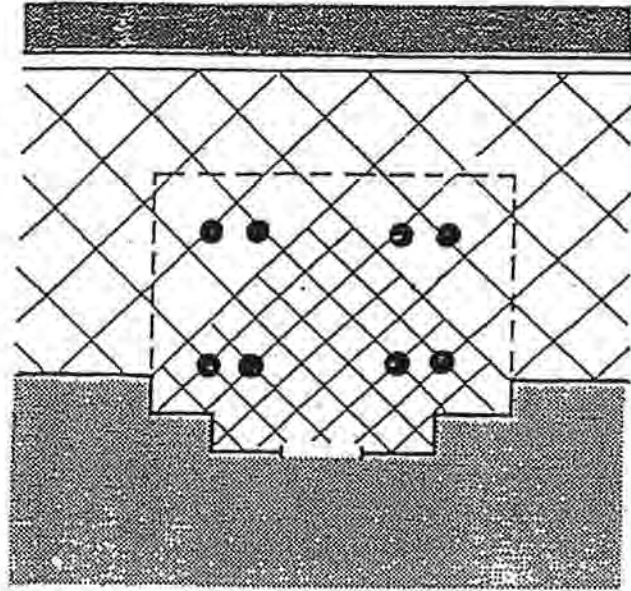
Material and color selections should complement adjacent developments.

BUILDING DETAILS

Details should reflect the clean contemporary order of the architecture.

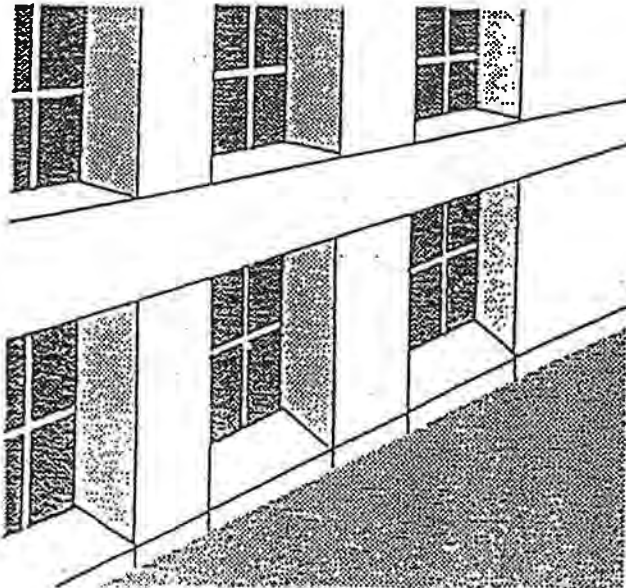


Main building entries should be deeply recessed and defined with special architectural features such as trellises, column groupings, and upgraded materials. The ground plane should be upgraded and articulated.



Building walls should extend beyond the footprint and become site walls to visually anchor the building to the site.

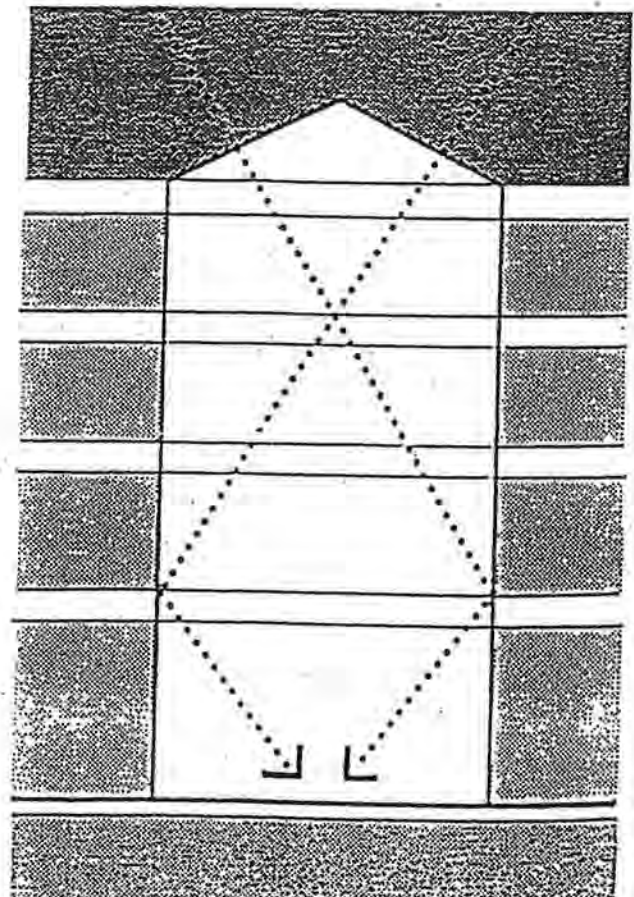
Incorporate deep recesses or architectural "eyebrows" to protect south facing glass and "punched" openings, architectural grill work or angled sun control "fins" to protect east and west facing glass.



ENERGY CONSIDERATIONS

Incorporate active solar energy design elements wherever feasible. Element should be incorporated into the building form and material vocabulary and shall not appear as after-thoughts or appendages.

Incorporate passive solar design elements into all buildings such as sun penetration control and natural day lighting features such as atriums and lightwells. Buildings should be oriented on the site to maximize the benefits of passive solar design.





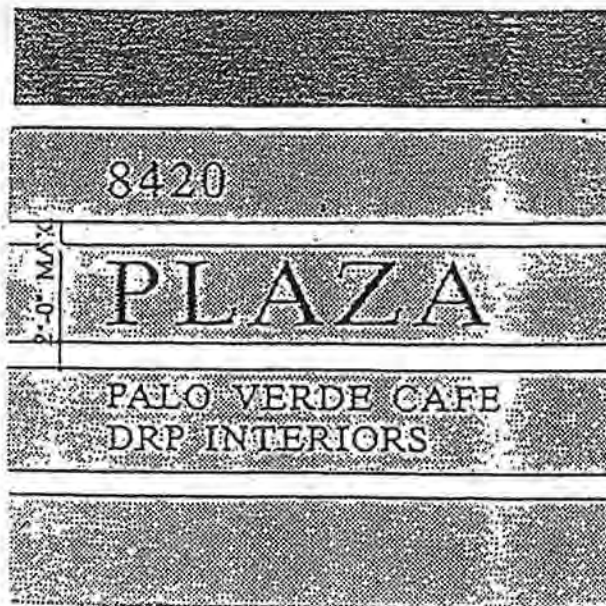
SIGNAGE STANDARDS

The Signage Standards address four types of signs; Identification, Directional, Vehicular Control and Temporary signs.

IDENTIFICATION SIGNS

Signs and graphics should assist the visitor in identifying businesses, building complexes and major tenants. The identification should be made from the adjacent roadway. Once the visitor is on site, identification signs become directional in nature.

A Building Mounted Sign is limited to the display of the building name and addresses or the name and/or symbol of the division, department, tenant or business occupying the building, and the building address. No more than the name of three tenants or the building name should be displayed, together with the building address. No message or advertising of any kind is permitted.



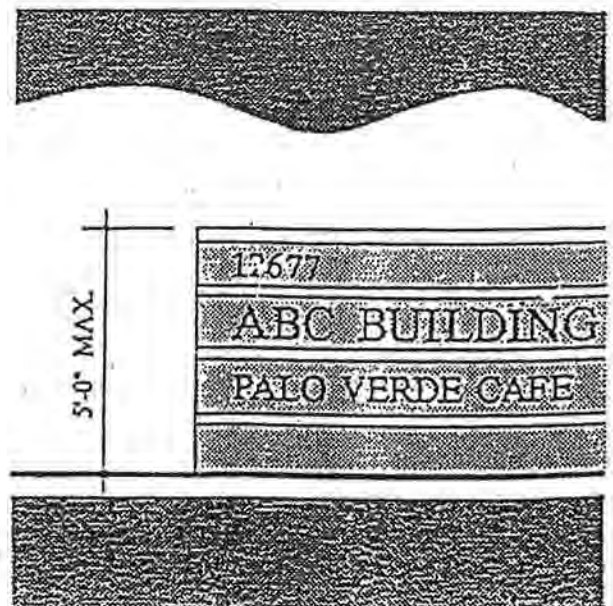
Wall signs for retail shopping and restaurant tenants should be located in accordance with the City of Scottsdale

Only the name and/or symbol in freestanding letters (no sign board) or letters integral with the architecture should be used.

Sign height and area shall be consistent with the City of Scottsdale Sign Ordinance and in no case may the top of the sign be closer to the roofline than 12". The maximum height of said sign shall be 2'-0".

Ground Mounted Building Or Complex Identification Sign identifies a single building or a cluster of buildings which is an entity. It should not identify more than three tenants in addition to the complex namesake (i.e., The ABC Building and the Palo Verde Cafe).

This sign should be set perpendicular to the street. It should not interfere with sight visibility triangles.

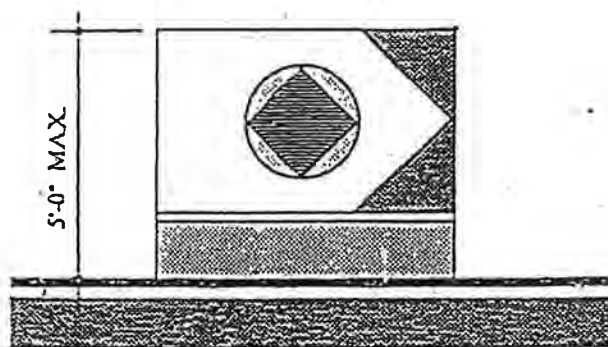
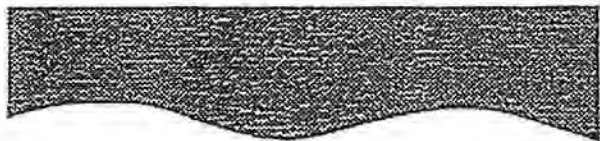


This sign is recommended for large, multi-building sites.

This is the largest sign that may be used, and may have a maximum face as allowed by the City of Scottsdale Sign Ordinance. Height of sign may not exceed 5 feet including base. There should be no more than one such sign per frontage street per project.

This sign may contain the names mentioned above, address numbers, and logo/symbol only.

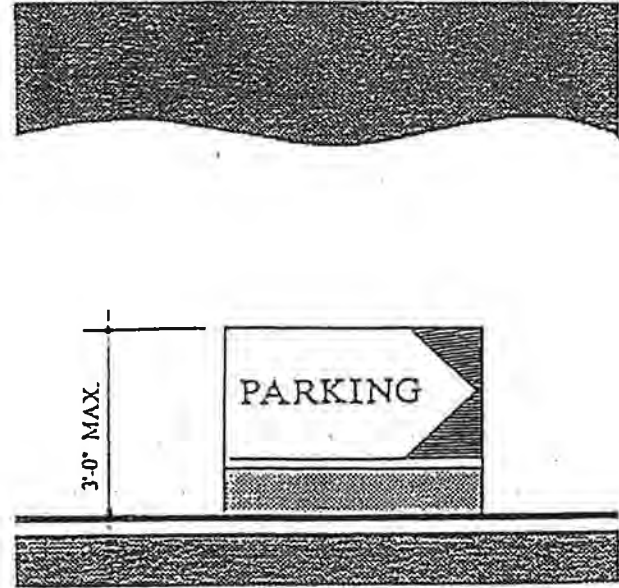
An Entry Identification Sign may identify entrances to a given site. It may have the word "Enter," a logo, and/or an arrow plus the name and/or address of the building and/or major tenants (up to three tenants). This sign should be located no closer than within 300 feet of a ground mounted single building identification sign.



An Entry Identification sign may have a maximum face of 4 square feet per side, excluding base, and with up to two sides. The sign should not exceed 5 feet in height, including base. Maximum of one sign per main entry into site.

DIRECTIONAL SIGNS

These signs may direct visitors to various on-site destinations. They may display the name of the destination, building names, Logo, building address, visitor parking, etc. and have arrows or other symbols.



These signs may have a maximum face of 8 square feet per sign, excluding base; on a maximum of two sides; and be a maximum of 3 feet in height, including base. The signs should not be readily visible from streets.

These signs may regulate traffic direction and speed, parking, etc. They include "STOP" signs, "ONE WAY" signs, speed limit signs, handicapped signs, etc.

These signs must meet City Standards for face size, face design, reflective qualities, height and other uniformities which make them instantly recognizable as traffic signs. However, it is encouraged that the signs have uniform material and background color and utilize the same base and mounting system utilized on other signage on the site.

No set limit on number of signs is recommended. The quantity of such signs will be determined on a case-by-case basis.

TEMPORARY SIGNS

The use of only one temporary development sign per street front ~~per~~ ~~one financing or leasing sign~~ (after obtaining building permits) per development parcel is allowed. Additional individual signs are discouraged except for small (maximum 4 square feet) directional signs for the direction of construction traffic.

Sign size to be consistent with City of ✓
Scottsdale ordinance.

The remaining sign area should be limited to identification of the following and no other information:

- Developer
- A concise identification of the development or building.

No temporary signs should be located as to reduce the safe flow of vehicles and pedestrians.

All temporary development signs should be removed prior to occupancy except on multi-building projects where temporary development signs may be placed near new construction after a completed phase is occupied.



DESIGN AND CONFIGURATION

All signs except traffic regulation signs for each parcel, should be designed as a single system, of like materials, construction methods, and should follow these guidelines.

Each sign should be square or rectilinear in shape and composed of one piece. It should have no moving parts.

COLOR

All signs except temporary signs, traffic regulating signs, traffic regulating signs and logos, should have a maximum of three colors, one for the base, another for the face and a third color for the lettering.

When a business name is in itself a business logo, it is exempt from these recommended color restrictions.

LETTERING STYLE

All lettering on all signs except business logo-types should be of the same lettering style on each parcel. The lettering style may be one of the following or very similar:

Optima

abcdefghijklmnopqrstuv
ABCDEFGHIJKLMN^{OP}Q

Times Roman

abcdefghijklmnopqrstuvw
ABCDEFGHIJKLMN^{OP}

Baskerville

abcdefghijklmnopqrstuvw
ABCDEFGHIJKLMN^{OP}

Souvenir

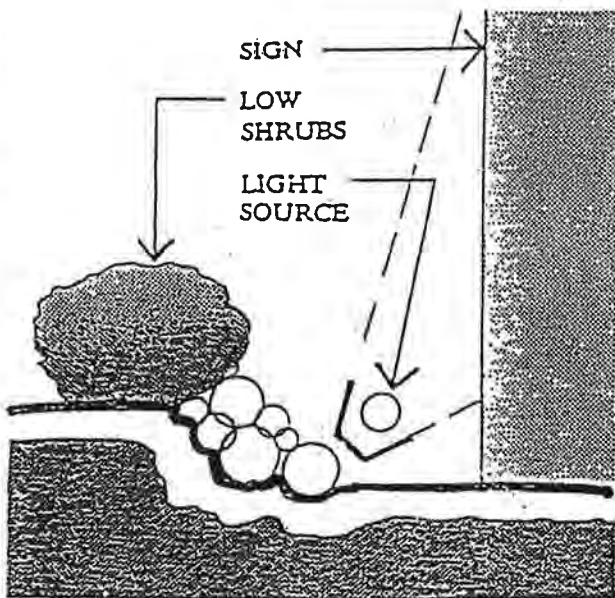
abcdefghijklmnopqrstuvw
ABCDEFGHIJKLMN^{OP}

MATERIAL

All sign posts and panels should be of a permanent, quality construction of: aluminum, copper or sheet metal; if painted, paint with a fade-resistant, durable, exterior finish; edges should be eased with no visible welds, natural stone or concrete with free-standing letters or letters cast into sign are also recommended. Panels should be plumb and square with no bows or warps. Exposed bases should have a smooth, durable, even architectural finish.

ILLUMINATION

Detached identification signs may be illuminated by continuous and uniform internal illumination, or ground lighting which meets the current Dark Sky and FAA Ordinance.



Identification Wall Signs may be illuminated by internal illumination or back lighting provided that the color and intensity of such lighting appears as an integral part of the overall architectural and site concept.

Exposed neon or argon signage should be limited to retail uses only, unless it is integral to the architectural design concept for a commercial building.

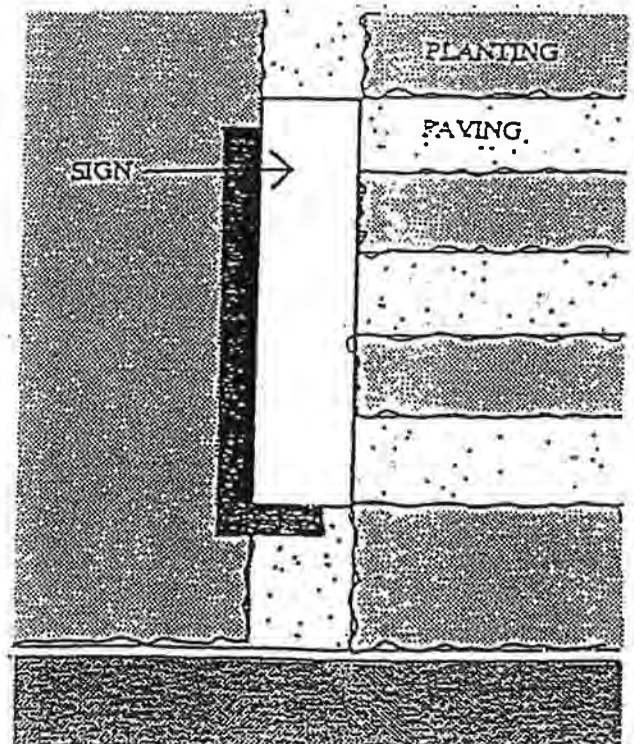
No sign illumination should cause a glare which will be visible from any street, access drive, or adjoining use.

SETBACKS

All sign setbacks, except on-site directional signs shall conform to the City of Scottsdale sign ordinance.

SETTING

Signs should be integrated into the landscape. For this reason, ground mounted signs are encouraged whenever practicable. Care must be taken to integrate free-standing signs into planting areas or to have clean base connections when in paved areas.



**FIRST AMENDED AND RESTATED
DECLARATION OF EASEMENTS, CONDITIONS,
COVENANTS AND RESTRICTIONS**

When Recorded, Return to:
Lawyers Title Insurance Corporation
Irma Hickman
40 E. Mitchell Dr., #100
Phoenix, AZ 85012

**THE PERIMETER CENTER
SCOTTSDALE, ARIZONA**

2/4 600 728 DO
LAWYERS TITLE INSURANCE CORPORATION

This DECLARATION is made as of this 16th day of January, 1996 by Scottsdale Land Trust Limited Partnership, a Delaware limited partnership (hereinafter "Declarant").

WITNESSETH:

WHEREAS, Declarant is the Owner of certain property in the County of Maricopa, State of Arizona, which is more particularly described on Exhibit "A" attached hereto and incorporated herein by this reference (the "Property");

WHEREAS, Declarant wishes to establish general and, at its discretion, specific guidelines for the development and improvement of the Property;

WHEREAS, Declarant caused to be recorded against the Property that Declaration of Conditions, Covenants and Restrictions, dated February 5, 1991 and recorded on February 8, 1991 in Document Number 91-056418 records of Maricopa County; and

WHEREAS, Declarant desires and intends to amend and restate the Declaration in its entirety by this First Amended and Restated Declaration of Easements, Conditions, Covenants and Restrictions (the "Declaration").

NOW, THEREFORE, Declarant hereby declares that all of the Property shall be held, sold and conveyed subject to the following covenants, conditions and restrictions, all of which are for the purpose of enhancing and protecting the value, desirability and attractiveness of the Property, as Declarant deems appropriate. All of the foregoing is hereby declared to be for the benefit of all of the Property and the Owners thereof, their heirs, successors, grantees and assigns. These covenants, conditions and restrictions shall run with the Property and shall be binding on all persons having or acquiring any right, title or interest in the Property, or any part thereof, and shall inure to the benefit of each Owner thereof.

ARTICLE I DEFINITIONS

1.1 "Association" shall mean the Arizona nonprofit corporation to be organized by Declarant to administer and enforce the Declaration and to exercise the rights, powers and duties set forth in this Declaration, its successors and assigns. Declarant hereby reserves the exclusive right to cause such Association to be incorporated. It is the present intent of the Declarant that the Association shall be referred to as the "The Perimeter Center Owners' Association." Declarant, however, shall be entitled to name the Association as it deems appropriate.

1.2 "Board" shall mean the Board of Directors of the Association.

1.3 "Bylaws" shall mean the Bylaws of the Association as the same may from time to time be amended or supplemented.

1.4 "Boundary" means the legal perimeter of a Site which marks the line of division of two contiguous Sites or between a Site and a dedicated street.

1.5 "Declarant" means Scottsdale Land Trust Limited Partnership, a Delaware limited partnership, and its successors or assigns. The rights of Declarant shall not be assigned except by express written instrument specifically referring to the Declarant's rights hereunder. Declarant shall be entitled, at any time, to assign part or all of its rights hereunder to the Association. The party making an assignment of Declarant's rights shall be under no obligation or liability thereafter for the exercise of or failure to exercise such rights; and the party to whom such Assignment has been made shall defend, indemnify and hold the assignor harmless therefrom.

1.6 "Common Elements" shall include any and all real property maintained by Declarant or the Association for the common use and enjoyment of the Owner as shown on the plat of the Property and all improvements placed thereon, including, without limitation, any real property as might be conveyed from time to time to the Association for such purpose.

1.7 "Improvements" means all buildings, parking areas, loading areas, refuse, storage or collection areas, fences, walks, hedges, landscaping, poles, signs and any other types of structures or improvements.

1.8 "Master Drainage Plan" means that certain drainage plan dated April 12, 1989 and prepared by Collier, Williams & White Engineering on file with Declarant.

1.9 "Owner" means the record owner of the equitable fee title (or legal fee title if equitable fee title has merged therewith) to any portion of the Property.

1.10 "Person" means any individual, corporation, partnership, or other legal entity.

1.11 "Property" means the real property described on Exhibit "A" and such other real property adjoining the property described on Exhibit "A" as Declarant may designate by written instrument specifically referring to this Declaration recorded with the Maricopa County Recorder's Office.

1.12 "Site" means all contiguous portions of the Property under one ownership from time to time.

ARTICLE II CONSTRUCTION OF IMPROVEMENTS

2.1 Minimum Setback Lines.

A. **General.** No portion of any Improvement shall be placed on any Site closer to the boundary lines of said Site than as herein provided. The following Improvements are specifically excluded from these setback provisions:

1. Steps and walks.
2. Except as provided in Section 2.7B, paving and associated curbing, except that vehicle parking areas shall not be permitted within twenty (20) feet of a street right-of-way line or lines.
3. Fences, except that no fence shall be placed within a street setback area unless specific approval is given by Declarant in writing.
4. Landscaping.
5. Planters or Site walls, not to exceed three (3) feet in height.
6. Roof overhangs, subject to specific approval of Declarant in writing.
7. Except as provided in Section 2.7B, covered parking structures.

B. Setback from interior boundary lines shall not be less than fifteen (15) feet from each side of boundary line and not less than twenty (20) feet from each rear boundary line. The interior boundary lines for a corner Site shall be considered side boundary lines.

C. Setback from street right-of-way lines shall not be less than fifty (50) feet from each street right-of-way line.

D. The setback requirements set forth in B and C above shall be increased by six (6) inches for each one (1) foot in the height of any Improvement above 25 feet from finished grade.

2.2 Completion of Construction.

After commencement of construction of any Improvement, the Owner shall diligently prosecute the work thereon, to the end that the Improvement shall not remain in a partly finished condition any longer than reasonably necessary for completion thereof. During construction, the Owner shall ensure that the construction materials shall not obstruct any public or private right-of-way and shall be stored in a neat and safe fashion.

2.3 Excavation.

No excavation shall be made except in connection with construction of an Improvement, and upon completion thereof exposed openings shall be backfilled and disturbed ground shall be graded and leveled.

2.4 Drainage Channels.

Each Site shall be required to provide for the drainage of runoff into approved drainage channels as shown on and in conformance with the Master Drainage Plan. No Owner or other person shall make any alteration to or modification of the approved drainage channels without the prior written approval of Declarant.

2.5 Landscaping.

Every Site on which Improvements shall have been placed shall be landscaped with lower Sonoran zone desert flora to the extent reasonably possible and according to plans approved as specified herein and maintained thereafter in a healthy and well-kept condition. A minimum of fifteen percent (15%) of each Site in its entirety and a minimum of fifteen percent (15%) of all parking areas shall be devoted to landscaping which meets or exceeds the following requirements:

1. At least one tree located on the perimeter of the Site for each 30 lineal feet of said perimeter (clustering of a single species encouraged);
2. At least one tree located on the perimeter of any building for each 20 lineal feet of said perimeter (clustering encouraged);

3. At least one tree located in parking areas for each 20 parking spaces;

4. At least 25% of all trees must initially be 36" boxes, and no trees will be smaller than 24" boxes when planted;

5. Berms and screen walls will be utilized to buffer parking areas from adjacent properties;

6. All areas between Site Boundary and the setback lines shall be landscaped, other than as used for walks, driveways and parking areas. The portion of the Site to a depth of twenty (20) feet which is immediately adjacent to any street shall be used exclusively for landscaping except for walks and driveways traversing the required landscaped area. Berms or screen walls shall be utilized to shield parking areas from the streets.

7. The Owner shall provide automatic irrigation systems, hose bibs and maintenance facilities in the vicinity of the landscaped areas for purposes of servicing said areas.

2.6 Signs.

A. No billboard or advertising signs shall be permitted, provided, however, that signs shall be allowed for the following limited purposes:

1. Those identifying the Site or the Owners thereof or the name, business, address or location of the person or firm occupying the Site; and

2. Those offering the Site for sale or lease when specifically approved by Declarant in writing.

B. Signs shall conform to setback lines unless specific approval to the contrary is granted by Declarant in writing.

C. Signs shall only be of such size, design and color as is specifically approved by Declarant in writing.

2.7 Parking Areas.

A. General. Each Owner shall be responsible for providing adequate off-street parking to accommodate all parking needs for employees, visitors and company vehicles

on that Owner's Site. The intent of this provision is to comply with applicable zoning and eliminate the need for any on-street parking. If parking requirements increase as a result of a change in use or number of employees, additional off-street parking by the respective Owner on its Site shall be provided to satisfy the intent of this section. Unless such requirement is specifically modified by Declarant as to a particular Site, each Owner shall provide on its Site a minimum of one parking space per each 600 square feet of gross leasable area within any building being used for industrial, fabricating, assembly or manufacturing purposes, and one parking space per each 250 square feet of gross leasable area within any building being used for office, clerical or administrative purposes or such other ratios as Declarant may specifically approve in writing.

B. Parking shall not be permitted:

1. Between public street pavement and Site boundary liens.
2. Closer than twenty (20) feet to any front Site Boundary, ten (10) feet to any side Site Boundary, and five (5) feet of the rear Boundary; provided, however, no covered parking shall be permitted in the area between the front of any building and the street.
3. In required loading areas.
4. Blocking access to refuse collection areas.

C. The parking requirements may be modified by Declarant as to any particular Site as Declarant deems appropriate, in its sole discretion, in writing specifically referring to these provisions. Declarant reserves the right to make an independent determination of the parking requirements for each Site and no determination for any other Site shall limit Declarant's discretion as to any other Site. No modification by Declarant as to a particular Site shall be deemed approval as to any other.

2.8 Storage, Loading and Trash Areas.

A. No materials, supplies or equipment, including company-owned or operated trucks, shall be stored in any area on a Site except inside a closed building, or behind a visual barrier screening such areas from the view of adjoining properties and/or public streets. All such screening shall be masonry walls or, at Declarant's option, appropriate landscaping.

B. Loading areas shall not encroach into setback areas unless specifically approved by Declarant in writing; nor shall loading be permitted to occur in the front of any building.

C. Loading docks shall be set back and screened to minimize the effect from the street. Docks shall not be closer than seventy (70) feet to the street right-of-way line or one hundred ten (110) feet from the street center line (whichever is greater), unless specifically approved by Declarant in writing. Loading will be permitted behind the rear setback line from that portion of a structure not fronting a street and from the areas to the rear of any building located on a corner Site.

D. All trash storage and pick-up areas shall be screened from the view of adjoining property and/or all public streets. Such screening shall be masonry walls, gates and appropriate landscaping. All refuse will be accumulated in an approved container provided by a licensed refuse service or a public agency.

2.9 Building Regulations.

Any building erected on a Site shall conform to the Design Guidelines promulgated by Declarant and to the following construction practices:

1. Exterior walls, of brick, granite, natural stone, masonry, glass, stucco or tilt-up panels of exposed aggregate shall be permitted. Exterior walls of traditional sheet or corrugated iron, steel, aluminum or asbestos will not be permitted, although use of high grade metal finishes such as "Aluco-Bond" will be considered by Declarant.

2. Exterior walls shall be painted, stuccoed or otherwise suitably treated in a manner approved by Declarant.

3. All roof mounted mechanical equipment and/or ductwork or any other protruding element which projects vertically above the roof or roof parapet is to be screened by an enclosure (which is consistent with the architectural detail of the building).

4. No mechanical equipment is to be exposed on the wall surface of a building.

5. Gutters and downspouts are to be painted to match the surface to which attached unless used as a major design element, in which case the color is to be aesthetically consistent with the color scheme of the building.

6. Vents, louvers, exposed flashing, tanks, stacks, overhead doors, rolling and "man" service doors, and the like, are to be designed and painted aesthetically consistent with the color scheme of the building.

7. No part of the roof may project above the parapet.

8. All exterior wall elevations of each building are to have architectural treatment.

9. Covered parking structures may be constructed of steel support elements, corrugated metal decking or other appropriate materials, provided that the canopy shall have an attractive fascia of suitable material.

A. Declarant shall be entitled, but under no obligation, to promulgate such guidelines for the design and construction of improvements and landscaping as it may deem appropriate (the "Design Guidelines"). Declarant reserves the right, however, to approve any Improvements and landscaping on a case-by-case basis and to modify the Design Guidelines or grant variances therefrom as Declarant deems appropriate. In no event shall Declarant's approval of a particular Improvement in any way limit or affect Declarant's discretion with respect to any other Improvement. No approval or waiver as to a particular Improvement or landscaping shall be deemed approved with respect to any other.

2.10 Utilities.

All utilities shall be enclosed in buildings or otherwise screened from public view in a fashion which is architecturally and aesthetically compatible with the Improvements constructed for the Site, or shall be placed underground. Said services are to include electrical, gas, water, sewer, telephone and any other special piping, conduit, containers, or other equipment and appurtenances as may be required.

2.11 Exterior Lighting.

All lighting will be designed and installed in such a manner so as to be shielded from adjoining Sites and public streets and shall be architecturally and aesthetically compatible with the Improvements constructed on the Site. Parking lot lighting fixtures are to have a maximum height of 16 feet and must be of a design approved in writing by Declarant. Walkway lighting fixtures are to have a maximum height of 8 feet. Security lighting fixtures are not to project above the fascia or roof line of any building and are to be shielded from direct street view. All shields shall be painted.

2.12 Conformance with City Code.

All planning or design criteria set forth herein is intended to supplement the requirements of the City of Scottsdale. All development of the Property shall also conform to all City of Scottsdale requirements.

2.13 Public Art.

Each Site on which Improvements shall be placed shall, at the Owner's sole cost and expense, be reasonably provided with such works of art as shall enhance the attractiveness of the Site and any Improvements thereon. The selection and placement of all such works of art shall be approved in writing by Declarant.

ARTICLE III APPROVAL OF PLANS

3.1 Plan Approval.

A. No Improvement shall be erected, placed, constructed, altered, maintained or permitted to remain on any Site until plans and specifications showing plot layout (with loading areas, parking stalls, and landscaped areas and drainage plans clearly indicated thereon) and all exterior elevations, with materials and colors therefor and structural design, signs and landscaping (including the size of plants and types of plants and planting material), shall have been submitted to and approved in writing by Declarant. All plans and specifications for proposed Improvements shall be prepared by a duly licensed and qualified architect. Such plans and specifications shall be submitted in writing over the signature of the Owner and Owner's architect with a specific written request for approval pursuant to this Agreement. Each submission shall include a certification from Owner's architect that the proposed Improvements comply with the provisions of this Declaration.

B. Approval shall be based, among other things, on adequacy of Site dimensions; adequacy of structural design; conformity and harmony of external design with neighboring structures; effect of location and use of Improvements on neighboring Sites, Improvements, operations and uses; relation of topography, grade and finished ground elevation of the Site being improved to that of neighboring Sites; proper orientation of main elevation with respect to nearby streets; compliance with the public art provisions of Section 2.13 and conformity of the plans and specifications to the purpose and general plan and intent of these restrictions. Declarant shall not arbitrarily or unreasonably withhold its approval of such plans and specifications.

C. If Declarant fails either to approve or to disapprove such plans and specifications within thirty (30) days after the same have been submitted to it in accordance with Section 3.1 above, it shall be conclusively presumed that Declarant has disapproved said plans and specifications, subject, however, to the restrictions contained in Article II hereof.

D. Neither Declarant nor its successors or assigns shall be liable in damages to anyone submitting plans to them for approval, or to any other Person affected by this Declaration, by reason of mistake in judgment, negligence or nonfeasance arising out of or in

connection with the approval or disapproval or failure to approve any such plans or any defect in design or workmanship of any such improvements or the failure of such plans or specifications to comply with any applicable government rule, regulation or ordinance. Every Person who submits plans to Declarant for approval agrees, by submission of such plans, and every Owner, and all Persons who may claim ownership or an interest through said Owner, of any Site agrees, by acquiring title thereto or interest therein, that he will not bring any action or suit against Declarant to recover any such damages.

E. Declarant reserves the right to evaluate each submission on a case-by-case basis and no approval or disapproval of a submission shall in any way limit or affect Declarant's discretion for any other submission. All determinations, including determinations of compatibility of use or design shall be made by Declarant in its sole discretion.

3.2 Estoppel Certificates and Presumption of Compliance.

A. Notwithstanding anything to the contrary herein contained, after the expiration of thirty (30) days from the date of issuance of a certificate of occupancy by a municipal or other governmental authority for any Improvement, said Improvement shall be deemed to be in compliance with all provisions of this Article III as to all purchasers and encumbrances in good faith and for value, unless actual notice of such noncompliance or noncompletion, executed by Declarant, shall appear of record in the office of the County Recorder of Maricopa County, Arizona, or unless legal proceedings shall have been instituted to enforce compliance or completion.

B. Within sixty days after written demand is delivered to Declarant, Declarant shall furnish an Estoppel Certificate certifying that as of the date thereof either: (a) all Improvements made/or other work done on or within a Site comply with the requirements of this Declaration; or (b) such Improvements or work do not so comply in which event the Certificate will identify the noncomplying Improvements or work and set forth with particularity the cause or causes for such noncompliance. Any Owner, lessee, purchaser or encumbrancer in good faith for value will be entitled to rely on such Certificate with respect to the matters set forth therein, such matters being conclusive as between Declarant and all such persons.

3.3 Architectural Review Fee.

An architectural review fee of \$250.00 which may be periodically adjusted, shall be paid to Declarant at the time plans are submitted for approval. The fee shall be non-refundable regardless of Declarant's determination.

ARTICLE IV REGULATION OF OPERATIONS AND USES

4.1 Permitted Operations and Uses.

A. No Site shall be used for any purpose except as specifically authorized by Declarant in a Site Use Declaration recorded with the Maricopa County Recorder. The Site Use Declaration may be amended from time to time with the consent of the Owner of the Site and Declarant. No other Owner shall have any right to approve another Site Use Declaration or modification or termination thereof. Declarant reserves the right to determine the use of the Site, in its sole discretion, and no determination regarding a particular Site shall limit or affect Declarant's discretion with respect to any other Site. In no event shall Declarant's approval or disapproval of a particular use be deemed a representative or covenant to approve or disapprove the same or other uses for other Sites. Declarant shall be entitled to approve any use consistent with the zoning for the Property, as such zoning may from time to time be modified. In no event shall Declarant be liable to any Owner or occupant of a Site for any use of another Site approved pursuant hereto.

4.2 Prohibited Operations and Uses.

A. The operation and use of drilling for and/or removal of oil, gas or other hydrocarbon substances on any Site shall not be permitted.

B. The following operations and uses shall not be permitted on any Site:

1. Residential;
2. Junk yards;
3. Commercial excavation of building or construction materials;
4. Dumping, disposal, incineration or reduction of garbage, sewage, offal, dead animals or refuse;
5. Stockyard or slaughter of animals;
6. Automobile service stations or the refining or sale of petroleum or of its products;
7. Smelting of iron, tin, zinc or other ores;
8. Livestock raising.

9. Any operation which discharges or emits into the atmosphere or onto the Site any contaminant producing a nuisance or hazard beyond Site boundary lines.

10. Any operation which discharges any toxic or noxious material in such concentrations as to be detrimental to or endanger the public health, safety, or welfare.

11. Any other use or operation that the Declarant determines is inconsistent with the established aesthetics of the Property or which might adversely affect the value, desirability or attractiveness of the Property.

4.3 Other Operations and Uses.

A. Operations and uses which are neither specifically prohibited nor specifically authorized by these restrictions may be permitted in a specific case if operational plans and specifications are submitted to and approved in writing by Declarant. If Declarant fails either to approve or to disapprove such operational plans and specifications within thirty (30) days after the same have been submitted to it, it shall be conclusively presumed that Declarant has disapproved said plans and specifications.

B. Neither Declarant, nor its successors or assigns, shall be liable in damages to anyone submitting operational plans and specifications to them for approval, or to any other Persons affected by this Declaration, by reason of mistake in judgment, negligence or nonfeasance arising out of or in connection with the approval or disapproval or failure to approve any such operational plans and specifications or with the operation or use of the Property or individual Sites. Every person who submits operational plans and specifications to Declarant for specifications, and every owner, and all Persons who may claim ownership or an interest through said Owner of any Site agrees, by acquiring title thereto or interest therein, that he will not bring any action or suit against Declarant to recover any such damages.

4.4 Maintenance.

A. All vacant or unimproved portions of any Site shall be kept and maintained in a healthy, neat, clean and weed-free condition.

B. The exterior of all Improvements and all walks, driveways and parking areas and landscaping on each Site shall be maintained in good order, repair and condition. All

exterior painted surfaces shall be maintained in first-class condition and if painted shall be repainted at least once in every four (4) years.

C. In the event of a failure to comply with this provision, Declarant shall enjoy the right but not the obligation to maintain the Site and charge the Owner the costs of such maintenance. All costs so incurred shall be a lien upon the Site and said lien may be foreclosed by Declarant in the same manner as provided in Section 6.3H below.

ARTICLE V
TERM, TERMINATION, MODIFICATION AND ASSIGNMENTS
OF DECLARANT'S RIGHTS AND DUTIES

5.1 Term.

This Declaration, every provision hereof and every covenant, condition and restriction contained herein shall continue in full force and effect until fifty (50) years from the date hereof, and shall be automatically renewed each ten (10) years thereafter unless terminated as provided in Section 5.2 below.

5.2 Termination and Modification.

Except as provided in Section 7.6, this Declaration, or any provision hereof, or any covenant, condition or restriction contained herein, may only be terminated, extended, modified or amended, as to the whole of said Property or as to any portion thereof, by the written consent of the Owners of seventy-five percent (75%) of the Property, based on the number of acres, or portions thereof exceeding one-half, owned as compared to the total number of acres subject to these restrictions; provided, however, that so long as Declarant owns any portion of the Property, no such termination, extension, modification or amendment shall be effective without the written approval of Declarant thereto. No such termination, extension, modification or amendment shall be effective until the same has been reduced to writing, executed, acknowledged and recorded in the office of the Recorder of Maricopa County, Arizona.

5.3 Assignments of Declarant's Rights and Duties.

Any and all of the rights, powers and reservations of Declarant herein contained may be assigned to any Person, which will assume the duties of Declarant pertaining to the particular rights, powers and reservations assigned, and upon any such Person evidencing its consent in writing to accept such assignment and assume such duties he or it shall, to the extent of such assignment, have the same rights and powers and be subject to the same obligations and duties as are given to and assumed by Declarant herein. No Person shall be a successor or assignee of Declarant as to any such rights, powers and reservations, including the right to amend its document, except by a written assignment of such rights which refers specifically to this

Declaration and this Section 5.3 and which has been executed by Declarant and recorded in Maricopa County, Arizona. Any such successor or assignee of Declarant shall have the right to further assign such rights to its successors and assigns. The term "Declarant" as used herein includes all such specific assignees. If at any time Declarant ceases to exist and has not made such an assignment, the Board shall exercise all rights of Declarant.

ARTICLE VI ASSOCIATION

6.1 Establishment of Association.

The Declarant may in the manner hereafter set forth, establish an Owners' Association which, if established, shall be a non-profit Arizona corporation charged with the duties and invested with the powers prescribed by law and set forth in the Articles of Incorporation, the Bylaws for the Association and the Declaration. Neither the Articles nor the Bylaws shall for any reason be amended or otherwise changed or interpreted so as to be inconsistent with this Declaration. The Association, if established, shall be named "The Perimeter Center Owners' Association, Inc.". At the Declarant's option, the Association may be established on the fifth anniversary of this Declaration, at such time as 80% of the Property is developed, or at such other time as the Declarant in its sole discretion shall determine. A Declaration of Establishment signed by the Declarant shall be recorded with the Maricopa County Recorder, which Declaration of Establishment shall refer to this Declaration and shall incorporate it therein by reference. Simultaneously with the recording of the Declaration of Establishment, Declarant shall file with the Corporation Commission of the State of Arizona the Articles of Incorporation of the Association in conformance with all applicable laws and regulations.

When established as set forth above, the Association shall be governed by and in accordance with the requirements set forth herein and shall be organized for the purposes and with the powers set forth herein. By acceptance of a deed or by acquiring any ownership interest in any of the real property included within this Declaration, each Owner for himself or itself, his heirs, personal representatives, successors, transferees and assigns binds himself, his heirs, personal representatives, successors, transferees and assigns to automatically become members of the Association upon its establishment, subject to the rights and obligations set forth herein.

6.2 Membership and Voting Rights.

Every Owner shall be a member of the Association. Membership shall be appurtenant to and may not be separated from ownership of a portion of the Property.

A. The Association shall have two (2) classes of voting membership.

Class A. Class A Owners shall be all Owners, with the exception of Declarant. Class A Owners shall be entitled to one (1) vote for each acre, or fraction thereof exceed one-half, contained in the Site owned by each such person or legal entity as of the date of the notice of the meeting at which the vote is to be cast. When two (2) or more persons or entities hold undivided interests in any Site, all such persons or entities shall be members of the Association; provided, however, that the vote for such Site shall be exercised as they, among themselves, determine, and in no event shall more than one (1) vote be cast with respect to each Acre, or fraction thereof exceeding one-half, contained in a Site in which such Owners own undivided interests.

Class B. The Class B Owner shall be the Declarant. The Class B Owner shall be entitled to seven and one-half (7.5) votes for each acre of land, or fraction thereof exceeding one-half, of the Property owned by Declarant as of the date of the notice of the meeting at which the vote is to be cast. Notwithstanding anything contained herein to the contrary, the Class B membership shall cease and be converted to Class A membership when Declarant shall have sold and conveyed all of the land area within the Property (including any additions or annexations thereto) to third-party purchasers of Sites.

B. Each member shall have such other rights, duties and obligations as set forth in the Articles of Incorporation and the Bylaws of the Association as they may be amended from time to time.

C. The Association membership of each Owner of a Site within the Property shall be appurtenant to said Site. The rights and obligations of an Owner and membership in the Association shall not be assigned, transferred, pledged, conveyed or alienated in any way except upon transfer of ownership of the Owner's interest in the Property and then only to the transferee of ownership to such interest or by interest succession, testamentary disposition, foreclosure of a mortgage of record, exercise of a power of sale under a deed of trust or such other legal process as is now in effect or as may hereinafter be established under or pursuant to the laws of the State of Arizona. Any attempt to make a prohibited transfer shall be void.

D. The affairs of the Association shall be conducted by a Board of no less than three (3) nor more than seven (7) Directors, with the assistance of such officers as are designated in the Articles or Bylaws.

6.3 Funds and Assessments.

A. Creation of the Lien and Personal Obligation of Assessments. Each Owner by acceptance of a deed therefor, whether or not it shall be expressed in such deed, is deemed to covenant and agree to pay to the Association:

1. Annual assessments or charges; and

2. Special assessments for capital improvements or unexpected requirements, such assessments to be established and collected as hereinafter provided. The annual and special assessments, together with interest, costs and reasonable attorneys' fees shall be a charge on the land and shall be a continuing lien upon the Property against which each such assessment is made. Each such assessment, together with interest, costs and reasonable attorneys' fees shall also be the personal obligation of the Person who is the Owner of such Site at the time when the assessments fell due. The personal obligation for delinquent assessments shall not pass to the successors in title of the person owning the Property at the time the assessment fell due, unless expressly assumed by them.

B. Purpose of Assessments. The assessments levied by the Association shall be used to promote the health, safety and welfare of the Owners for the maintenance of the Property and of the common areas and improvements situated or to be situated thereon, including, without limitation, the maintenance and repair of the drainage channels specified in Section 2.4, and to comply with all of the obligations of the Association hereunder, all in accordance with this Declaration and the Articles of Incorporation of the Association.

C. Annual Assessments. The annual assessments will be determined by the initial Board of Directors of the Association for the calendar year in which the Association is established. All assessments shall be based upon the respective size of each Site held by an Owner and shall be determined by dividing the total square footage of all of the land subject to this Declaration into the number of square feet of the Site held by each Owner and then multiplying such dividend by the amount of the assessment. Such assessments shall be calculated by the Board based upon its operating budget for the ensuing year which budget may include water and electric charges, maintenance and landscaping expenses, and the cost of professional management, as provided for in Section 6.4, and such other matters as the Board may deem advisable.

D. Special Assessments for Capital Improvements. The Association may choose to construct, reconstruct, repair or replace landscaping or improvements upon the Property. In addition to the annual assessments authorized above, the Association may levy, in any assessment year, a special assessment payable in not more than the next succeeding ten (10) years for the purpose of defraying in whole or in part the cost of any construction, reconstruction, repair or replacement of any landscaping or other improvement which may be installed by the Association upon the Property, provided that any such assessment shall have the consent of two-thirds (2/3) of the votes of the members who are voting in person or by proxy at a meeting duly called for that purpose.

E. Meetings - Notice and Quorum for Action Authorized Under Sections C and D above. Written notice of any meeting called for the purpose of taking any action authorized under Sections C and D above shall be sent to all members of the Association not less than fifteen (15) days nor more than sixty (60) days in advance of the meeting. At the first such meeting called, the presence of members or proxies entitled to cast more than fifty percent (50%) of all votes of the membership shall constitute a quorum. If the required quorum is not present, another meeting may be called subject to the same written notice requirement and the required quorum at the subsequent meeting shall be one-half (1/2) of the required quorum at the preceding meeting. No such subsequent meeting shall be held more than sixty (60) days following the preceding meeting.

F. Uniform Rate of Assessment. Both annual and special assessments must be fixed at a uniform rate based upon the number of acres or part thereof owned by each Owner and may be collected on a monthly, quarterly, or annual basis as determined by the Board of Directors of the Association. The Board of Directors shall determine when an assessment is due and payable and when the payment of an assessment shall be deemed delinquent. In addition to its obligation to pay assessments as an Owner, Declarant shall be entitled, but under no obligation, to subsidize Association activities. In no event, however, shall Declarant be under any obligation to continue such subsidies even if the affect of discontinuing such subsidies is to substantially increase assessments.

G. Date of Commencement of Annual Assessments-Due Dates. The annual assessments provided for herein shall commence as to all Sites on the first day of the month following the meeting of the Board of Directors of the Association at which the initial annual assessment is established. The first annual assessment shall be adjusted according to the number of months remaining in the calendar year. The Association shall, upon demand, and for a reasonable charge, furnish a certificate signed by an officer of the Association setting forth whether the assessments on a specified Site have been paid.

H. Enforcement of Assessments. Each assessment levied hereunder shall be a separate, distinct and personal debt and obligation of the Owner or Owners against whom the same is assessed, and shall constitute a lien and charge upon the Site held by that Owner to which the assessment relates. The Declarant and all other Owners by acceptance of a deed or by acceptance of any other document or instrument conveying an ownership interest therein, whether or not it shall be so expressed in any such deed or other document or instrument, are and shall be deemed to covenant and agree to pay to the Association the assessments, both annual and special, provided for herein, and agree to the enforcement of the assessments in the manner herein specified. In the event the Association employs an attorney or attorneys for collection of an assessment, whether by suit or otherwise, or to enforce compliance with or specific performance of the terms and conditions of this Declaration, or for any other purpose in connection with the breach of this Declaration, each Owner agrees to pay reasonable attorneys' fees, accountants' fees, and costs thereby incurred in addition to any other amounts due from the

Owner or any other relief or remedy obtained against said Owner. In the event of a default in the payment of any such assessment when due, in which case the assessment shall be deemed delinquent, and in addition to any other remedies herein or by law provided, the Association may enforce each such obligation in any manner provided by law or in equity, or without any limitation of the foregoing, by any one or all of the following procedures:

1. Enforcement by Suit. The Association may bring a suit at law against each Owner or Owners to enforce each such assessment obligation. Each Owner agrees that any judgment rendered in any such action shall include a sum for reasonable attorneys' fees in such amount as the Court may adjudge against the defaulting Owner, plus all court costs and necessary expenses and accounting fees incurred by the Association, plus interest on the amount of said assessment at the maximum lawful rate from the date the assessment becomes delinquent until paid in full.

2. Enforcement by Lien. The Association may give notice to each Owner whose assessment is due and unpaid by mailing to said Owner a copy of a Notice and Claim of Lien which shall state the following: (i) the last known name of the delinquent Owners; (ii) the legal description and street address of the Site against which Claim of Lien is made; (iii) the amount claimed to be due and owing (with any proper offset allowed); (iv) that the Claim of Lien is made by the Association pursuant to the terms of the Declaration; and (v) that a lien is claimed against the Site in an amount equal to the amount of the stated delinquency. The Association may record a duly executed original or copy of such Notice and Claim of Lien and the lien claimed therein shall immediately attach and become effective as a lien upon the Site against which such assessment was levied. Each default in payment of an assessment shall constitute a separate basis for a Claim of Lien or a lien, but any number of defaults may be included within a single Notice and Claim of Lien. The amount of the lien shall include the amount of all unpaid assessment, plus interest at 24% per annum or such other default rate as the Board may from time to time establish, plus a lien charge to cover recording, legal and accounting expenses incident thereto. The amount of said lien charge may be increased or decreased by the Board of Directors in its sole discretion. Any such lien may be foreclosed by appropriate action in Court, or in the manner provided by law for the foreclosure of a realty mortgage, or the exercise of a power of sale in a trust deed, as elected by the Association, as set forth by the laws of the State of Arizona, as the same may be changed or amended. The lien provided for herein shall be in favor of the Association and shall be for the benefit of all other Owners. The Association shall have the power to bid in its interest at any foreclosure sale and to purchase, acquire, hold, lease, mortgage and convey any Site. In the event such foreclosure is by action in court, reasonable attorneys' fees, court costs, title search fees, interest and all other costs and expenses shall

be allowed to the extent permitted by law. Each Owner hereby expressly waives any objection to the enforcement and foreclosure of this lien in this matter and also hereby expressly waives the defense of the statute of limitations applicable to the bringing of any suit or action thereon.

3. Notification. The Association may notify all Owners of the names of all persons who have defaulted in the payment of any assessment when due and the amount thereof in the discretion of the Board of Directors.

4. Subordination of the Lien to Mortgages. The lien of the assessments provided for herein shall be subordinate to the lien of any first mortgage or deed of trust. The sale or transfer of any Site shall not affect the assessment lien. However, the sale or transfer of any Site pursuant to mortgage foreclosure, exercise of a power of sale pursuant to a deed of trust, or any proceeding in lieu thereof, shall extinguish the lien of such assessments as to payments which became due prior to such sale or transfer. No sale or transfer shall relieve an Owner from liability for any assessments thereafter becoming due or from the lien thereof.

6.4 Professional Management.

The Association may engage the services of a professional manager or a professional management company or otherwise contract to maintain the Property, to collect assessments, to attend to the Security within the Property and to perform such other duties and to fulfill such other functions as may be determined by the Board of Directors of the Association.

ARTICLE VII
ENFORCEMENT

7.1 Abatement and Suit.

Violation or breach of any restriction herein contained shall give to Declarant and every Owner, upon ten (10) days' prior written notice and failure of the other Owner to commence curing said violation or breach within said ten (10) day period and to thereafter diligently pursue the same, the right to enter upon the Site as to which said violation or breach exists and to summarily abate and remove, at the expense of the Owner thereof, any structure, thing or condition that may be or exist thereon contrary to the intent and meaning of the provisions hereof, or to prosecute a proceeding at law or in equity against the Person or Persons who have violated or are attempting to violate any of these restrictions to enjoin or prevent them from doing so, to cause said violation to be remedied or to recover damages for said violation.

Declarant shall be entitled, but under no obligation, to prosecute any action or seek any remedy hereunder. No Owner shall be entitled to take any action unless it notifies Declarant in writing of its intent to do so and Declarant fails to institute an action or pursue appropriate remedies within 60 days after receipt of such notice or after commencing such action fails or refuses to prosecute such action to completion. The prosecuting and defending Owners shall defend, indemnify and hold Declarant harmless from all liability, obligations, costs, expenses, fees (including reasonable attorney's fees) arising from such action.

7.2 Deemed to Constitute a Nuisance.

The result of every action or omission whereby any restriction herein contained is violated in whole or in part is hereby declared to be and to constitute a nuisance, and every remedy allowed by law or equity against an Owner, either public or private, shall be applicable against every such result and may be exercised by Declarant or by any Owner of any Site.

7.3 Attorneys' Fees.

In any legal or equitable proceeding for the enforcement or to restrain the violation of this Declaration or any provision hereof, the losing party or parties shall pay the attorneys' fees of the prevailing party or parties, in such amount as may be fixed by the court in such proceedings. All remedies provided herein or at law or in equity shall be cumulative and not exclusive.

7.4 Inspection.

Declarant may from time to time at any reasonable hour or hours, enter and inspect any Site and/or Improvements thereon to ascertain compliance therewith.

7.5 Failure to Enforce Not a Waiver of Rights.

With the exception of the time limit for action by Declarant contained in Section 3.1 of Article III hereof, the failure of Declarant or any Owner to enforce any restriction herein contained shall in no event be deemed to be a waiver of the right to do so thereafter nor of the right to enforce any other restriction.

7.6 Lien Against Site.

If any Improvements or landscaping are not constructed or maintained in accordance with the requirements of this Declaration, Declarant shall have the right, after ten (10) days' written notice to Owner, or to the occupant if Owner's address is unknown, to enter upon the Site and perform any alteration, removal, maintenance, or repairs, to cause conformity with the requirements hereof. The expenses of such work, together with a fee equal to fifteen percent (15%) of the amount thereof, shall be the joint and several liability of Owner and any occupant

who has been provided with a notice of violation, and shall be payable upon demand of Declarant. If such expense is not paid within five (5) days of demand, such obligation shall bear interest at the rate of twenty-four percent (24%) per annum or such other default rate as the Association may establish. Should the demand of Declarant for payment for corrective work undertaken not be paid within thirty (30) days of the date of mailing such demand, a lien upon the affected Site shall arise in favor of Declarant to secure payment of the cost of such maintenance and repair together with the 15% fee referred to above, interest, and reasonable costs, expenses, and attorneys' fees incurred in imposing and foreclosing the lien. The lien shall be established or recorded upon the recording of a notice of lien with the Office of the County Recorder of Maricopa County, setting forth the name of the record owner, a description of the Property subject to the lien, and the amount secured thereby. The lien may be foreclosed by an appropriate proceeding in a court of competent jurisdiction. Entry upon a Site by Declarant or its agents for the purpose of maintenance or repair shall not be deemed a trespass and the Owner and all occupants shall be deemed to have consented thereto. The obligation to pay for such corrective work shall not pass to successors in interest unless they have actual notice thereof. In the event such successors in interest have actual notice prior to acquisition, they shall be jointly and severally liable with the former Owner and/or occupant of the Site.

7.7 Subordination of the Lien to Mortgages.

The lien described herein shall be subordinate to the lien of any first mortgage or deed of trust. The sale or transfer of any Property shall not, however, affect the lien. However, the sale or transfer of any Property pursuant to exercise of a power of sale in a deed of trust or foreclosure or any proceeding in lieu thereof, shall extinguish the lien as to payments which became due prior to such sale or transfer. No sale or transfer shall relieve such Property from liability for any obligation thereafter becoming due or from the lien thereof.

ARTICLE VIII EASEMENTS

8.1 Reservation of Easements.

Declarant shall retain the following perpetual easements with respect to the Property:

A. A perpetual and exclusive easement for the maintenance of any Common Elements, which may presently or hereafter encroach upon a Site; and

B. The Declarant shall have the perpetual and nonexclusive right of access to each Site (i) to inspect same for the purpose of verifying conformance with this Declaration, and following the establishment of the Association, the Bylaws, the Articles of Incorporation, and any Rules and Regulations of the Association, (ii) to remedy any violations set forth in this Declaration, the Bylaws, or in any Rules or Regulations of the Association, and (iii) to perform

any operations required in connection with the maintenance, repairs, or replacements of or to the Common Elements or any equipment, facilities, or fixtures affecting or serving other Sites or the Common Elements; provided that requests for entry are made in advance and that any such entry is at a time reasonably convenient to the Owner. In case of an emergency, such right of entry shall be immediate, whether the Owner is present at the time or not; and

C. A perpetual, blanket, and nonexclusive easement in, upon, over, under, across, and through the Property and the Common Elements for surface water runoff and drainage caused by natural forces and elements, grading, and/or the improvements located upon the Property and for the maintenance and repair of the drainage channels specified in Section 2.4. No individual Owner shall directly or indirectly interfere with or alter the drainage and runoff patterns and systems within the Property; and

D. A blanket, perpetual, and nonexclusive easement in, upon, over, across and through the Common Elements for the purpose of the installation, maintenance, repair, service, and replacement of all sewer, water, power, and telephone pipes, lines, mains, conduits, water, poles, transformers, meters, and any and all other equipment or machinery necessary or incidental to the proper functioning of any utility systems serving the Property and the Sites, which easement shall be for the benefit of any governmental agency or utility company or other entity that requires same for the purpose of furnishing one or more of the foregoing services.

8.2 Conveyance of Easements.

At such time as the Association is established, the Declarant may convey and assign all of the easements reserved herein to the Association.

ARTICLE IX MISCELLANEOUS PROVISIONS

9.1 Constructive Notice and Acceptance.

Every Person who now or hereafter owns or acquires any right, title or interest in or to any portion of said Property is and shall be conclusively deemed to have consented and agreed to every covenant, condition and restriction contained herein, whether or not any reference to this Declaration is contained in the instrument by which such Person acquired an interest in said Property. Whenever this Declaration creates or imposes an obligation with respect to a Site, the Owner of the Site shall be responsible for the timely and proper performance of the obligation, notwithstanding any delegation of such responsibility of lease, contract, or otherwise to another party.

9.2 Rights of Mortgagees.

The breach of any provision hereof shall not in any way reduce the security or affect the validity of any mortgage or deed of trust now or hereafter executed upon any Site; provided, however, that any Person who acquires ownership of any Site by reason of a sale under a foreclosure of any mortgage or under the provisions of any deed of trust, such purchaser and his successors and assigns shall hold any and all Property so purchased subject to all of the restrictions and other provisions of this Declaration.

9.3 Mutuality Reciprocity: Runs with Land.

All restrictions, conditions, covenants and agreements contained herein are made for the direct, mutual and reciprocal benefit of each and every part and portion of said Property; shall create mutual, equitable servitude upon each portion in favor of every other portion; shall create reciprocal rights and obligations between the respective Owners of all portions and privity of contract and estate between all grantees of said portions, their heirs, successors and assigns; and shall, as to the Owner of each portion, his heirs, successors and assigns, operate as covenants running with the land, for the benefit of all other portions.

9.4 Paragraph Headings.

Paragraph headings, where used herein, are inserted for convenience only and are not intended to be part of this Declaration or in any way to define, limit or describe the scope and intent of the particular paragraphs to which they refer.

9.5 Effect of Invalidation.

If any provision of this Declaration is held to be invalid by any court, the invalidity of such provision shall not affect the validity of the remaining provisions hereof.

9.6 Waiver and Variance.

The Declarant hereunder shall enjoy the right to waive or vary any of the requirements of this Declaration in particular respects whenever in its sole discretion it determines such waiver or variance will not adversely affect the purposes of this Declaration, provided that such waiver or variance must be contained in a writing signed by Declarant.

9.7 No Personal Liability.

Notwithstanding anything to the contrary provided in this Declaration, there shall be absolutely no personal liability on the part of the Declarant (including any successor) with respect to any of the terms, conditions and covenants of this Declaration, and any Owner or other Person shall look solely to the interest of Declarant in the Property for the satisfaction of

each and every remedy of such Owner or other Person in the event of any breach by Declarant, Owner or any other Person of any terms, conditions and covenants of this Declaration.

9.8 Joinder by Adjoining Property Owner.

Franchise Finance Corporation of America does hereby join in the terms and conditions of this Declaration and, as the owner thereof, does subject that certain real property described on Exhibit B to all of the covenants, conditions and restrictions contained herein.

9.9 Approval of Declarant.

Notwithstanding anything to the contrary provided in this Declaration, whenever approval of the Declarant is required herein such approval shall be required to be obtained in writing and prior to the performance of the act for which approval is sought.

9.10 Disclaimer.

Nothing contained herein shall be deemed a representation or agreement by Declarant to develop the Property in any general or specific manner, and Declarant reserves the right to amend or abandon any plans it may from time to time pursue, in its sole discretion.

9.11 Name.

Declarant retains all rights in the name "Perimeter Center" and Owners shall be entitled to use the name only with reference to the location of the Site. Each Owner shall execute such instrument as Declarant may request to evidence Declarant's full ownership thereof.

IN WITNESS WHEREOF, the undersigned has executed this Declaration on the date set forth above.

SCOTTSDALE LAND TRUST LIMITED
PARTNERSHIP, a Delaware limited partnership

BY: FFCA Management Company Limited
Partnership, a Delaware limited partnership,
General Partner

BY: Perimeter Center Management
Company, a Delaware corporation,
Managing General Partner

By: 
M. H. Fleischer
President

By: 
M. H. Fleischer, Individual
General Partner of FFCA
Management Company, L.P.

ADJOINING LAND OWNER OF THAT REAL
PROPERTY DESCRIBED ON EXHIBIT "B"
ATTACHED HERETO HEREBY CONSENTS TO
THE FOREGOING DECLARATION AND
SUBMITS SUCH REAL PROPERTY TO THE
PROVISIONS HEREOF:

FRANCHISE FINANCE CORPORATION OF
AMERICA, a Delaware corporation

By: 
M. H. Fleischer, President

STATE OF ARIZONA)
) ss.
COUNTY OF MARICOPA)

The foregoing instrument was acknowledged before me on this 12th day of January, 1996 by M.H. Fleischer, President of Franchise Finance Corporation of America, General Partner of FFCA Management Company, L.P., which is the general partner of the Partnership.

Mary E. Leeland
Notary Public

My Commission Expires:
12-9-96



STATE OF ARIZONA)
) ss.
COUNTY OF MARICOPA)

The foregoing instrument was acknowledged before me on this 12th day of January, 1996 by M.H. Fleischer, Individual General Partner of FFCA Management Company, L.P., which is the general partner of the Partnership.

Mary E. Leeland
Notary Public

My Commission Expires:
12-9-96



STATE OF ARIZONA)

) ss.

COUNTY OF MARICOPA)

The foregoing instrument was acknowledged before me on this 12th day of January, 1996 by M.H. Fleischer, President of FRANCHISE FINANCE CORPORATION OF AMERICA, an adjoining land owner.

Mary E. Leeland
Notary Public

My Commission Expires:

12-9-96



760032/25

2702 N. 44th Street

Suite 100A

Phoenix AZ 85008-1578

(602) 957-3350

FAX: (602) 956-5488

COLLAR, WILLIAMS & WHITE

Description

For

CWW No. 901202

Scottsdale Land Trust

December 10, 1990

Limited Partnership

SCOTTSDALE PERIMETER CENTER

Being a portion of Section 36, Township 4 North, Range 4 East, G. & S. R. B. & M., Maricopa County, Arizona, more particularly described as follows:

BEGINNING at the south quarter Corner of said Section 36;
THENCE N89°13'33"W, along the south line of said Section, 925.00 feet;
THENCE N00°46'27"E, 455.35 feet to a point marking the beginning of a tangent curve, having a radius of 1800.00 feet to the right;
THENCE northerly, along the arc of said curve, through a central angle of 06°58'18", having an arc distance of 219.03 feet;
THENCE N07°44'46"E, 564.82 feet to a point marking the beginning of a tangent curve, having a radius of 1350.00 feet to the left;
THENCE northerly, along the arc of said curve, through a central angle of 31°50'58", having an arc distance of 750.44 feet;
THENCE N24°06'13"W, 373.03 feet to a point marking the beginning of a tangent curve, having a radius of 2522.95 feet to the right;
THENCE northerly, along the arc of said curve, through a central angle

SCB/jm



760032/25
Description For
Scottsdale Land Trust
Limited Partnership
SCOTTSDALE PERIMETER CENTER
CWW No. 901202
December 10, 1990

of 24°08'05", having an arc distance of 1062.74 feet;
THENCE N00°01'52"E, 65.00 feet;
THENCE S89°58'08"E, 156.69 feet to a point marking the beginning of a
tangent curve, having a radius of 2200.00 feet to the left;
THENCE easterly, along the arc of said curve, through a central angle
of 45°11'27", having an arc distance of 1735.20 feet to a point on a
curve on the Centerline of The Outer Loop (State Route 117), from which
the central point, of said curve bears S36°25'05"W, 5729.58 feet;
THENCE southeasterly, along the arc of said curve, through a central
angle of 42°57'41", having an arc distance of 4296.14 feet to a point
on the east line of the aforementioned Section 36;
THENCE S00°00'04"W, along the east line of said Section, 516.12 feet to
the southeast Corner of said Section 36;
THENCE N89°52'20"W, along the south line of said Section, 2643.72 feet
to the POINT OF BEGINNING.

Comprising 261.360 Acres.

Excepting therefrom the Rights-Of-Way as shown on the MAP OF
DEDICATION OF RIGHT-OF-WAY AND EASEMENTS FOR PERIMETER CENTER PHASE
ONE, as recorded in Book 331, Page 25, Maricopa County records, and the
Rights-Of-Way as shown on the MAP OF DEDICATION OF RIGHT-OF-WAY AND
EASEMENTS FOR PERIMETER CENTER PHASE TWO, as recorded in Book 332, Page
49, Maricopa County records.

Total Rights-Of-Way Comprising 39.472 Acres.

Description For
Scottsdale Land Trust
Limited Partnership
SCOTTSDALE PERIMETER CENTER
CWW No. 901202
December 10, 1990

Also Excepting therefrom the following described parcel:

COMMENCING at the aforementioned south quarter Corner of Section 36;

THENCE S89°52'20"E, along the south line of said Section, as shown on the aforementioned Map Of Dedication Phase One, 787.50 feet to the Centerline of Perimeter Drive;

THENCE N00°07'40"E, along the Centerline of said Perimeter Drive, 557.45 feet to a point marking the beginning of a tangent curve, having a radius of 1275.00 feet to the left;

THENCE northerly, continuing along said Centerline and along the arc of said curve, through a central angle of 14°17'59", having an arc distance of 318.21 feet to a point hereinafter referred to as point "A";

THENCE N75°49'41"E, 45.00 feet to a point on the easterly Right-Of-Way line of said Perimeter Drive marking the POINT OF BEGINNING;

THENCE S89°52'20"E, 480.00 feet;

THENCE S00°07'40"W, 445.00 feet to a point on a curve on the northerly Right-Of-Way line of Hartford Drive, from which the central point of said curve bears N06°14'10"W, 565.00 feet;

THENCE westerly, along said northerly Right-Of-Way line, and along the arc of said curve, through a central angle of 06°21'50", having an arc distance of 62.76 feet;

Description For
Scottsdale Land Trust
Limited Partnership
SCOTTSDALE PERIMETER CENTER
CWW No. 901202
December 10, 1990

THENCE N89°52'20"W, continuing along said northerly Right-Of-Way line, 356.47 feet to a point marking the beginning of a tangent curve, having a radius of 20.00 feet to the right;

THENCE northwesterly, along the arc of said curve, through a central angle of 90°00'00", having an arc distance of 31.42 feet to a point on the aforementioned easterly Right-Of-Way line of Perimeter Drive;

THENCE N00°07'40"E, along said easterly Right-Of-Way line, 102.45 feet to a point marking the beginning of a tangent curve, having a radius of 1320.00 feet to the left;

THENCE northerly, continuing along said easterly Right-Of-Way line, and along the arc of said curve, through a central angle of 14°17'59", having an arc distance of 329.44 feet to the POINT OF BEGINNING.

Comprising 4.619 Acres.

Also Excepting therefrom the following described parcel:

COMMENCING at the aforementioned point "A", said point being on a curve, and on the Centerline of the aforementioned Perimeter Drive, from which the central point of said curve bears S75°49'41"W, 1275.00 feet;

THENCE northerly, along said Centerline, and along the arc of said curve, as shown on the aforementioned Map Of Dedication, Phase Two, through a central angle of 05°14'14", having an arc distance of 116.55 feet;

Description For
Scottsdale Land Trust
Limited Partnership
SCOTTSDALE PERIMETER CENTER
CWW No. 901202
December 10, 1990

THENCE N19°24'34"W, continuing along said Centerline, 808.48 feet;
THENCE N70°35'26"E, 51.00 feet to a point on the aforementioned
easterly Right-Of-Way line of Perimeter Drive marking the POINT OF
BEGINNING;

THENCE N19°24'34"W, along said easterly Right-Of-Way line, 300.00 feet;
THENCE N21°45'02"W, continuing along said Right-Of-Way line, 126.04
feet to a point marking the beginning of a tangent curve, having a
radius of 20.00 feet to the right;

THENCE northerly, along the arc of said curve through a central angle
of 92°20'29", having an arc distance of 32.23 feet to a point on the
southerly Right-Of-Way line of St. John Road;

THENCE N70°35'26"E, along said southerly Right-Of-Way line, 84.71 feet
to a point marking the beginning of a tangent curve, having a radius of
1365.00 feet to the left;

THENCE easterly, continuing along said Right-Of-Way line, and along the
arc of said curve, through a central angle of 17°33'26", having an arc
distance of 418.28 feet;

THENCE N53°02'00"E, continuing along said Right-Of-Way line, 92.94
feet;

THENCE S31°36'24"E, 474.06 feet to a point on the northerly
Right-Of-Way line of Princess Drive;

THENCE S58°23'36"W, along said northerly Right-Of-Way line, 169.81 feet
to a point marking the beginning of a tangent curve, having a radius of
2635.00 feet to the right;

Description For
Scottsdale Land Trust
Limited Partnership
SCOTTSDALE PERIMETER CENTER
CWW No. 901202
December 10, 1990

THENCE southwesterly, along said Right-Of-Way line, and along the arc of said curve, through a central angle of $11^{\circ}15'49''$, having an arc distance of 518.01 feet to a point of compound curvature marking the beginning of a tangent curve, the central point of which bears $N20^{\circ}20'34''W$, 20.00 feet;

THENCE westerly, along the arc of said curve, through a central angle of $90^{\circ}56'01''$, having an arc distance of 31.74 feet to the POINT OF BEGINNING.

Comprising 7.00000 Acres.

Subject Parcel, after deducting above Exceptions, Comprising 210.269 Acres, more or less, subject to all easements of record.



3702 N. 44th Street
Suite 100A
Phoenix AZ 85008-1578
(602) 957-3356
FAX: (602) 956-3488

COLLAR, WILLIAMS & WHITE

Description

For

CWW No. 901202 Franchise Finance Corporation Revised December 7, 1990
June 15, 1988
of America

SCOTTSDALE'S PERIMETER CENTER PARCEL

Being a portion of the southeast quarter of Section 36, Township 4 North, Range 4 East, G. & S. R. B. & M., Maricopa County, Arizona, more particularly described as follows:

COMMENCING at the south quarter Corner of Section 36;
THENCE S89°52'20"E, along the south line of said Section, as shown on the MAP OF DEDICATION OF RIGHT-OF-WAY AND EASEMENTS FOR PERIMETER CENTER PHASE ONE, as recorded in Book 331, Page 25, Maricopa County records, a distance of 787.50 feet to the Centerline of Perimeter Drive;
THENCE N00°07'40"E, along the Centerline of said Perimeter Drive, 557.45 feet to a point marking the beginning of a tangent curve, having a radius of 1275.00 feet to the left;
THENCE northerly, continuing along said Centerline and along the arc of said curve, through a central angle of 14°17'59", having an arc distance of 318.21 feet;
THENCE N75°49'41"E, 45.00 feet to a point on the easterly Right-Of-Way

SCB/jm

EXHIBIT "B"

PAGE 1 OF 2



Description For
Franchise Finance Corporation
of America
SCOTTSDALE'S PERIMETER CENTER PARCEL
CWW No. 880107
June 15, 1990
Revised December 7, 1990

line of said Perimeter Drive, marking the POINT OF BEGINNING;
THENCE S89°52'20"E, 480.00 feet;
THENCE S00°07'40"W, 445.00 feet to a point on a curve on the northerly
Right-Of-Way line of Hartford Drive, from which the central point of
said curve bears N06°14'10"W, 565.00 feet;
THENCE westerly, along said northerly Right-Of-Way line, and along the
arc of said curve, through a central angle of 06°21'50", having an arc
distance of 62.76 feet;
THENCE N89°52'20"W, continuing along said northerly Right-Of-Way line,
356.47 feet to a point marking the beginning of a tangent curve, having
a radius of 20.00 feet to the right;
THENCE northwesterly, along the arc of said curve, through a central
angle of 90°00'00", having an arc distance of 31.42 feet to a point on
the aforementioned easterly Right-Of-Way line of Perimeter Drive;
THENCE N00°07'40"E, along said easterly Right-Of-Way line, 102.45 feet
to a point marking the beginning of a tangent curve, having a radius of
1320.00 feet to the left;
THENCE northerly, continuing along said easterly Right-Of-Way line, and
along the arc of said curve, through a central angle of 14°17'59",
having an arc distance of 329.44 feet to the POINT OF BEGINNING.

Comprising 4.619 Acres, more or less, subject to all easements of
record.